Lecture notes. Lecture 7

Time series, dynamical systems, chaos and delay embedding

Marcin Płodzień
Deep Learning in Classical and Quantum Physics
Institute of Theoretical Physics, Jagiellonian University in Kraków

1 Records of a hidden motion

To forecast a measured record we need some idea of what produced its numbers. We start from one record and work back to the physical system behind it. We describe its state and law of motion and the instrument that turns the state into stored numbers.

1.1 A laser that pulses irregularly

The running example of these notes is a measured record that nobody can forecast by eye. A laser consists of an amplifying medium between two mirrors and a source of energy, the pump. The pump keeps more molecules of the medium in an excited state than in the lower state. Light travelling back and forth between the mirrors is amplified by stimulated emission, and a small fraction of it leaves through one partially transmitting mirror. A detector behind this mirror records the intensity of the outgoing light (Figure 1). For most lasers and most pump strengths the intensity settles to a constant value. In the far-infrared laser of Ref. [1], whose medium is ammonia gas (NH_3), the intensity never settles. It rises in a sequence of pulses whose heights grow over many pulses, then drops abruptly to a low level, and the growth starts again. The number of pulses before a collapse and the height of the largest pulse change from one cycle to the next without repeating.

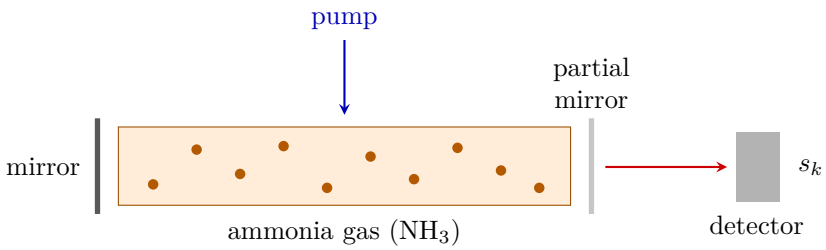


Figure 1: The far-infrared ammonia laser. The pump keeps the gas between the mirrors excited, and light is amplified while it travels back and forth. The part that leaves through the partial mirror is recorded by the detector as the number s_k at the sampling step k.

The detector output was digitised with 8 bits at equal time intervals. The stored record is a sequence of 10093 integers between 0 and 255, of which 243 different values occur. The rounding to these levels is a part of the measurement error. The record is known as data set A of the Santa Fe time-series competition [2], and we call it the Santa Fe laser record. We use its first 4000 values, shown in Figure 2, and measure time in samples. The fast oscillation in the lower panel is the sequence of pulses, which recur about every 7.26 samples. The slow saw-tooth envelope in the upper panel is the cycle of growth and collapse.

A sequence of numbers measured at equal time intervals is called a time series. Besides the laser, weather stations produce one by recording the temperature every hour, solar observatories by

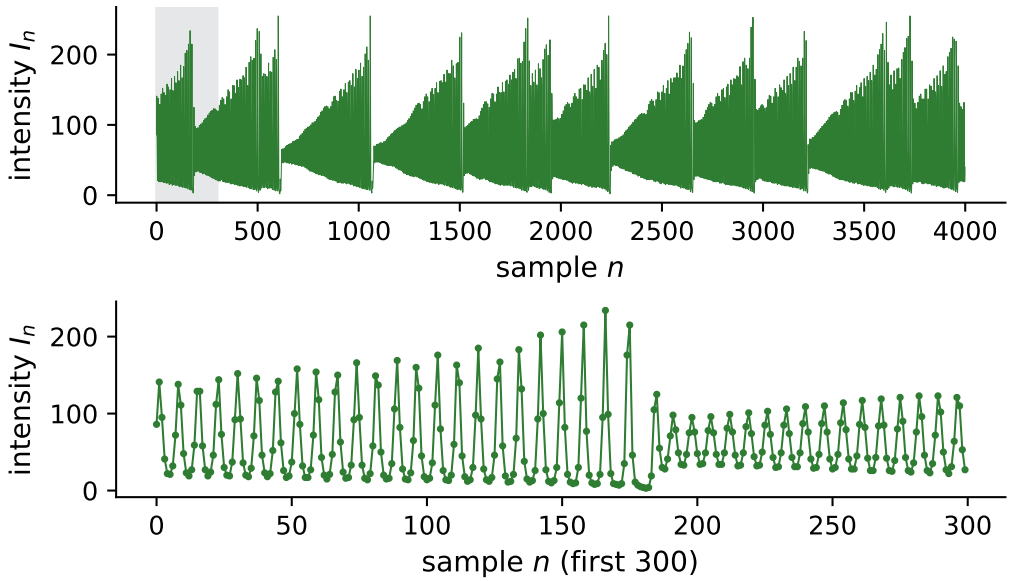


Figure 2: The Santa Fe laser record. The vertical axis shows the measured intensity, an integer between 0 and 255, written I_n on the axis and s_k in the text. The horizontal axis shows the sample index, written n on the axis and k in the text. Top, the 4000 samples used in these notes, with the first 300 samples shaded. Bottom, the shaded part enlarged, which resolves the individual pulses (dots are the stored samples) and one collapse near sample 180.

counting sunspots every day, and electrodes by recording the electrical activity of the heart. Each record comes from a system that changes according to some law. We ask whether the record can be forecast, how far ahead and how well, and which law produced it, a deterministic law with a few variables or many unobserved influences. If the law is deterministic, we also want the number of variables it needs and the rate at which it amplifies small errors. Both questions require a precise language for the system behind the record.

1.2 State and law of motion

In classical physics the next value of the laser record is fixed by the state of the system. At every moment this state is a list of d numbers

$$\mathbf{x}(t) \coloneqq (x_1(t), \dots, x_d(t)) \in \mathbb{R}^d \tag{1.1}$$

that, together with the law of motion, fixes everything about the future of the system. The state of a pendulum consists of the angle and the angular velocity, so $d = 2$. The angle alone does not suffice, since a pendulum passing the lowest point may be moving to the left or to the right. The state of the atmosphere holds the temperature, the pressure and the wind at every point of a grid, and d is very large. The set of all possible states is the phase space, here $\mathbb{R}^d$ or a part of it, and every state is one point in it.

The law of motion is a system of first-order ordinary differential equations,

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \qquad \text{in components} \quad \dot{x}_i = F_i(x_1, \dots, x_d), \quad i = 1, \dots, d, \tag{1.2}$$

where the dot denotes the time derivative. The vector field $\mathbf{F}\colon \mathbb{R}^d \to \mathbb{R}^d$ assigns to every point of phase space a velocity, the direction and speed in which the state moves when it is at that point. It can be pictured as an arrow attached to every point, and the motion of the system as a point that always moves along the local arrow. The curve $\mathbf{x}(t)$ traced in this way is a trajectory. The field $\mathbf{F}$ contains the forces, the couplings between the components, friction, external driving and all parameters, such as a viscosity or a pump strength. Parameters are constants inside $\mathbf{F}$ and do not belong to the state. We assume throughout that $\mathbf{F}$ is smooth, in general nonlinear, and independent of time. A law independent of time is called autonomous.

Equations of higher order are brought to the form (1.2) by introducing derivatives as new variables. Newton's equation $m_0\ddot{q} = f(q)$ for a particle of mass m_0 at the coordinate q becomes $\dot{q} = p/m_0$, $\dot{p} = f(q)$ with the momentum p, so one mechanical degree of freedom contributes two state variables. For the pendulum with the angle θ, the angular velocity $\omega = \dot{\theta}$, the damping rate γ and time measured in units of the inverse small-oscillation frequency, the law reads

$$\dot{\theta} = \omega, \qquad \dot{\omega} = -\sin\theta - \gamma\,\omega. \tag{1.3}$$

The first equation says that the angle changes at the rate ω. The second contains the torque of gravity and the friction torque, which opposes the motion.

In some systems the law is known well and its parameters can be measured, as for a pendulum, an electrical circuit or a planetary orbit. In others it is known only approximately, with missing terms, uncertain parameters and unknown couplings, as for a living cell, an ecosystem or the climate. Where the law is imperfect, the measured record is the main source of knowledge about the system.

1.3 Determinism and the absence of crossings

A law of the form (1.2) fixes the future, and this property has a geometric consequence for every record. It rests on the existence and uniqueness theorem for ordinary differential equations applied to equation (1.2). If $\mathbf{F}$ is continuously differentiable, then through every initial state passes one and only one solution, at least for a finite time interval. From a given state the system can move in one way only, so two trajectories never cross in phase space (Figure 3). At a crossing point one state would continue along two different curves, and the present state would leave the future open. Determinism and the absence of crossings are therefore the same statement. Running the argument backwards in time shows that the present state also fixes the whole past.

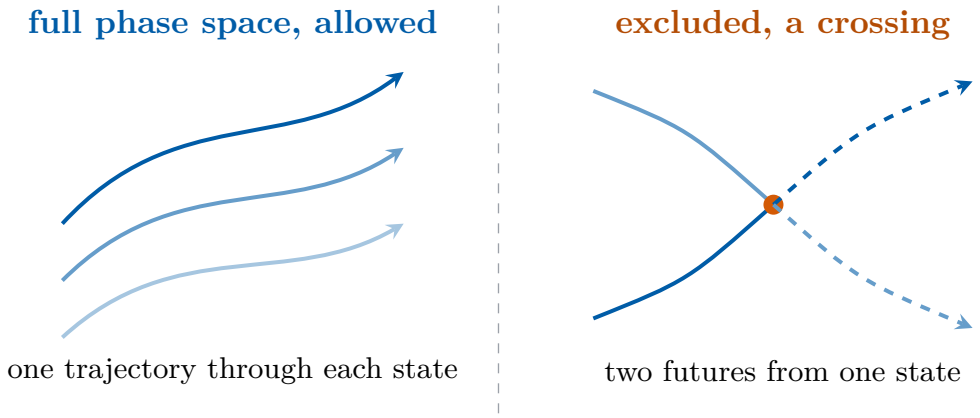


Figure 3: Left, trajectories of a smooth autonomous law in the full phase space, one through each state. Right, a crossing (orange dot), which would give one state two futures (dashed) and is excluded by the uniqueness of solutions.

A plotted curve that crosses itself cannot be a trajectory in the full phase space. Such crossings occur in a projection of a trajectory onto two of its coordinates, or in a measured coordinate plotted against an earlier value of itself. They signal that some state variables are missing from the plot.

Laplace drew the extreme conclusion from this property of the laws of mechanics [3]. He imagined an intelligence that knows at one instant all the forces acting in nature and the positions of all bodies, and that can analyse these data. For such an intelligence, he wrote, nothing would be uncertain, and the future, like the past, would be present before its eyes. In our notation the positions and velocities of all bodies form the state $\mathbf{x}$ and the forces form the vector field $\mathbf{F}$. The analysis is the solution of equation (1.2). Laplace's statement holds in the classical description of nature, in which a state is a point and the law of motion is a deterministic differential equation. In quantum mechanics a completely known state fixes only the probabilities of measurement outcomes. For the classical

systems considered here, two practical limitations remain. No observer knows the full state, and every observer knows it only to finite precision.

Randomness in a classical system therefore describes the information that the observer lacks, and it is called epistemic. The toss of a coin is the standard example. Its motion follows Newton's equations, and the outcome is fixed by the upward velocity and the rate of spin at the moment the coin leaves the hand [4]. In the plane of these two initial values, the starts that give heads and those that give tails form alternating bands. The bands become narrower as the velocity and the spin increase. A real toss is launched with an imprecision much larger than the width of a band. Since the observer cannot tell into which band it falls, the outcome is described by the probability one half. A machine that launched the coin with sufficient precision would produce the same outcome every time. An irregular record may thus come from a deterministic law with few variables, as the coin does, or from many unobserved influences that are best treated as random.

1.4 The measured record

An experiment rarely sees the state that fixes the future. It measures one quantity at successive times separated by the sampling interval Δt, and the measured value is a function of the state,

$$s_k = h(\mathbf{x}(k\Delta t)) + \eta_k, \qquad k = 1, \dots, N. \tag{1.4}$$

The measurement function $h\colon \mathbb{R}^d \to \mathbb{R}$ describes the instrument, which turns the d numbers of the state into one number, and η_k is the measurement noise. The record, or time series, is the sequence $s_1, \dots, s_N$ of N stored numbers. A camera that films a pendulum and stores only its angle measures $h(\theta, \omega) = \theta$ and loses the velocity. Figure 4 shows the two reductions contained in equation (1.4). The state moves continuously along its trajectory, and only at the sampling times $t = k\Delta t$ does the instrument evaluate one function of the current state. Time becomes discrete, because the motion between the sampling times is not stored, and the d coordinates of the state shrink to one number. Since two different states can give one and the same number, a measured value does not fix the state. The record is a one-dimensional image of a motion in d dimensions.

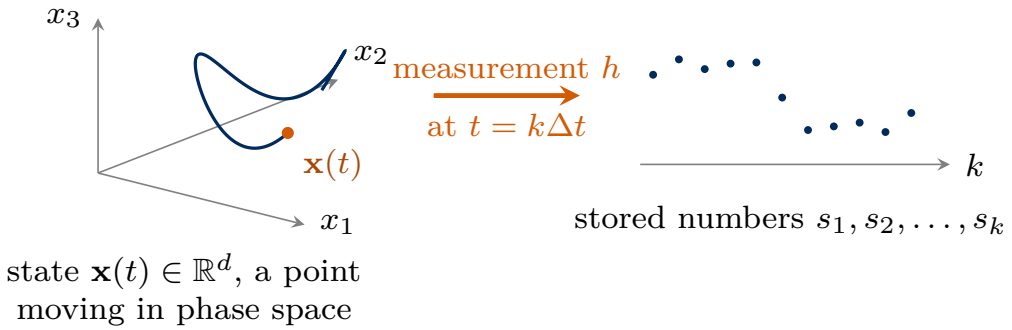


Figure 4: The state $\mathbf{x}(t)$ (orange dot) moves along a trajectory (blue) in a phase space drawn here with three coordinates x_1, x_2, x_3. At the times $k\Delta t$ the measurement function h records one number. The stored record $s_1, s_2, \dots$ (dots on the right, plotted against the step k) is all that remains of the motion.

Randomness enters a record in two ways. The measurement noise η_k corrupts the stored numbers and leaves the motion untouched, whereas dynamical noise acts on the motion itself. A pollen grain in water is pushed by molecules whose positions nobody records, and a law written for the grain alone must contain a random force $\boldsymbol{\xi}(t)$, $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) + \boldsymbol{\xi}(t)$. Such a force changes the future of the system, while measurement noise changes only what we know about it.

1.5 The laser as a dynamical system

With state, law and measurement defined, we describe the laser of Figure 2 as a dynamical system and locate the origin of the difficulties of its record. The physics of a single-mode laser is known in

principle and involves three variables. The cavity field E is the light in the laser. The polarisation P is the electric dipole moment per unit volume that the field induces in the molecules. The population inversion W is the excess of molecules in the upper energy level over the lower one. In a semiclassical description the three variables obey the Maxwell–Bloch equations,

$$\dot{E} = -\gamma_c\,(E - gP), \qquad \dot{P} = -\gamma_\perp\,(P - EW), \qquad \dot{W} = -\gamma_\parallel\,(W - W_0 + EP), \tag{1.5}$$

where γ_c, $\gamma_\perp$ and $\gamma_\parallel$ are the decay rates of the field, of the polarisation and of the inversion, and g is a coupling constant. The inversion that the pump maintains in the absence of light is W_0. The state is $\mathbf{x} = (E, P, W)$ and $d = 3$. According to the first equation, the field decays through the mirrors and is fed by the polarisation. In the second, the polarisation decays and is driven by the field in proportion to the inversion. In the third, the pump restores the inversion while stimulated emission, the product EP, depletes it.

The detector measures the intensity, proportional to E^2, so $h(E, P, W) \propto E^2$. This description reveals three difficulties that every record of a partly observed system shares. The system is observed only partially. The polarisation and the inversion are never measured, and the intensity loses even the sign of E. Equations (1.5) and the intensity E^2 are unchanged under $(E, P, W) \mapsto (-E, -P, W)$, and two states related by this symmetry produce one intensity record forever. The decay rates and the pump level of the experiment are not known to the precision that a quantitative prediction would need, which makes the law itself imprecise. In most lasers the three rates differ by orders of magnitude, the fast variables follow the slow ones, and the motion is effectively one- or two-dimensional. In the ammonia laser the three rates are comparable and all three variables remain active [1]. Finally, for suitable parameters the solutions of equations (1.5) never repeat and depend sensitively on the initial state. Even the correct law would then predict the state only for a limited time.

2 The forecasting problem

We wish to forecast the laser record despite these difficulties, and we first fix the information a model may use and the way its forecasts are judged.

2.1 Recall and forecast

At the present step k we know the stored values $s_1, \ldots, s_k$, and nothing else. The state $\mathbf{x}$, the law $\mathbf{F}$ and even the measurement function h are unknown. A task is defined by its target y_k, the value that has to be produced at step k. In a forecast the target is a future value,

$$y_k = s_{k+\tau}, \qquad \tau \geq 1, \tag{2.1}$$

where the integer τ is the horizon, also called the lead, and the step k is the forecast origin. In a recall task the target is a past value, $y_k = s_{k-\tau}$, which tests what a model keeps of the past. The estimate of a target is written with a tilde, $\tilde{y}_k$, for example $\tilde{s}_{k+\tau}$ for a forecast. Figure 5 marks both targets on a stretch of the laser record.

Figure 6 contrasts the forecast with the situation of full knowledge. With the initial state and the law, the solution of equation (1.2) gives the future state, limited only by the precision of the initial state. With the record alone, a forecasting model has to replace the hidden dynamics, and it sees the system only through the compressed numbers s_k.

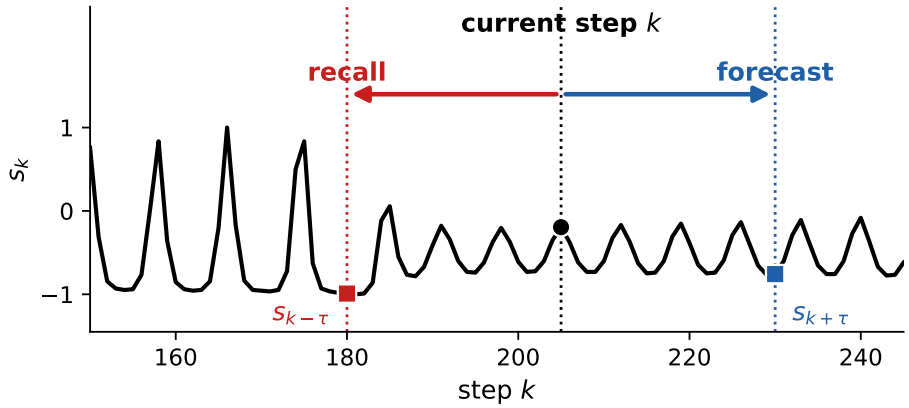


Figure 5: Recall and forecast on a stretch of the Santa Fe laser record, rescaled to the interval $[-1, 1]$ for the drawing and plotted against the step k. The black dot is the present step k. The red square is the recall target $s_{k-\tau}$ and the blue square the forecast target $s_{k+\tau}$, here for $\tau = 25$.

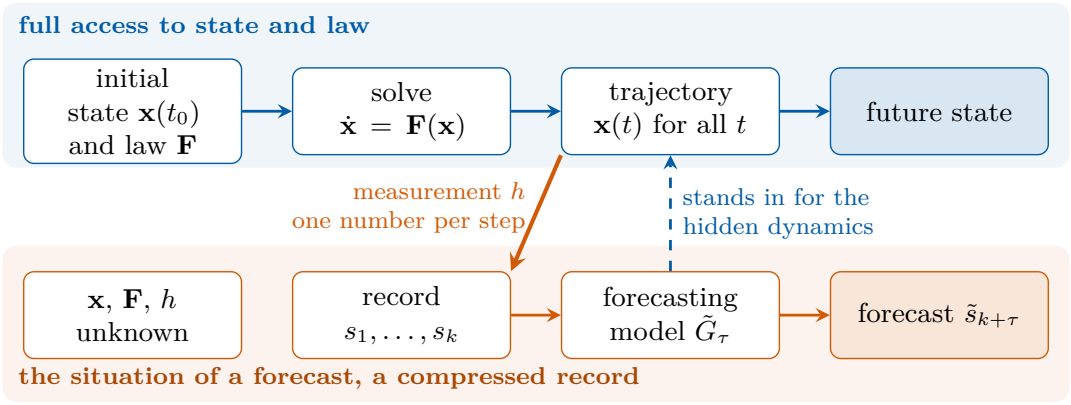


Figure 6: Top, with the initial state and the law the equations of motion give the future state. Bottom, the measurement compresses the state into the record $s_1, \ldots, s_k$, and a forecasting model $\tilde{G}_\tau$ built from the record returns the forecast $\tilde{s}_{k+\tau}$.

2.2 A window of the past

A forecasting model is a function of the known past. One sample gives one number, while the hidden state of the laser has three variables, so the model needs several past values. It cannot read an infinite past and reads the window of the last L values,

$$\mathbf{u}_k \coloneqq (s_k,\, s_{k-1},\, \ldots,\, s_{k-L+1}) \in \mathbb{R}^L, \qquad \tilde{s}_{k+\tau} = \tilde{G}_\tau(\mathbf{u}_k), \tag{2.2}$$

where the integer L is the window length and $\tilde{G}_\tau$ is the model for the horizon τ, the tilde marking a quantity estimated from data. The window takes the place of the state in equation (1.2) as the information from which the future is computed. A model of this kind needs a prescription for the window, namely how many past values, at what spacing, suffice to fix the present state. It also needs the function $\tilde{G}_\tau$ itself, learned from the record. Forecasting a measured record by a function fitted to the record goes back to Ref. [5].

A forecast for many horizons can be produced in three ways, sketched in Figure 7. A one-step model predicts only the next value. A direct forecast uses a separate map $\tilde{G}_\tau$ for every horizon, each fed with the measured window $\mathbf{u}_k$. A recursive forecast applies the one-step model repeatedly and appends each prediction to the window as if it were a measurement. The error of one step then enters the input of the next.

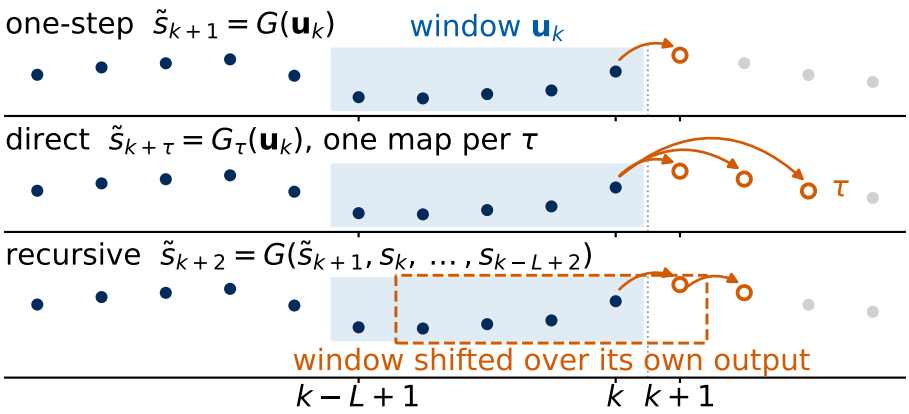


Figure 7: Three ways to forecast. Filled dots are samples of the record against the step index, and the shaded box is the window $\mathbf{u}_k$ of the last L values. Grey dots are future values, and open circles are forecasts. Top, one-step forecast. Middle, direct forecast with one map $\tilde{G}_\tau$ for each horizon τ, written G in the drawing. Bottom, recursive forecast, in which the window (dashed) shifts over the model's own outputs.

2.3 Training and testing in time order

A model is fitted on one part of the record and judged on another. For a record the choice of the two parts decides whether the score measures forecasting at all. In many learning problems the samples are independent and can be shuffled before they are divided. The pairs $(\mathbf{u}_k, s_{k+\tau})$ of a record are far from independent, since neighbouring windows share $L-1$ values. A window that starts after a test origin contains samples that follow the origin, for some windows including the value to be forecast. A forecast is a statement about the future, and its evaluation follows the order of time.

The laser record used here has $N = 4000$ samples, numbered from 0 to 3999, and we divide it into three consecutive blocks (Figure 8). The training block, samples 0 to 2499, is used to fit a model. The validation block, samples 2500 to 2999, is used to choose between variants of a model, such as the window length. The test block, samples 3000 to 3999, is used once at the end to measure the quality of the chosen model on values it has never seen. The input of a training pair and its target must both lie inside the training block. Any preprocessing estimated from data, and any statistic that chooses a setting of a model, is computed on the training block alone and then frozen. The record is rescaled linearly so that its minimum and maximum over samples 0 to 2499 become 0 and 1, and validation and test values can lie slightly outside $[0, 1]$.

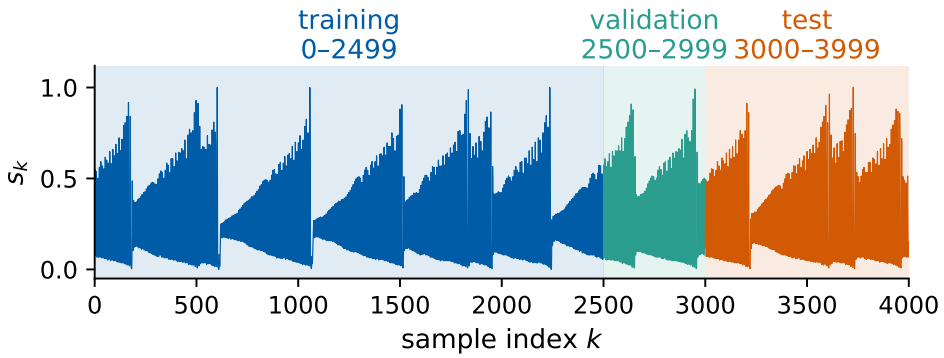


Figure 8: Rescaled Santa Fe record s_k against the sample index k. It is divided into the training block (samples 0 to 2499), the validation block (2500 to 2999) and the test block (3000 to 3999).

Consider $L = 3$, the horizons $\tau = 1$ and 2, a training block that ends at sample 5 and a test block that starts at sample 8. The origin $k = 3$ gives the training input (s_3, s_2, s_1) and the targets (s_4, s_5), both inside the training block. The origin $k = 4$ is excluded from training, since its second target s_6 lies outside the training block. The origin $k = 7$ is a valid test origin. Its input (s_7, s_6, s_5) is known at that time, and its targets (s_8, s_9) lie in the test block. A random division violates the rule on a

large scale. For windows of $L = 8$ samples, the horizon $\tau = 25$ and 950 test windows, it makes a fraction 0.997 of all test targets appear as inputs of training windows. The division in time makes none. A score measured after shuffling describes how well a model fills gaps in a record it has partly seen, which is a much easier task than forecasting.

2.4 Scoring a forecast

On the test block, a forecast at a fixed horizon is judged by comparing the predictions $\tilde{y}_i$ with the targets y_i over the test origins $i = 1, \dots, n$. An exact forecast puts every point $(y_i, \tilde{y}_i)$ on the diagonal $\tilde{y} = y$. The tightness of the cloud of points around a straight line is measured by the Pearson correlation coefficient. For any two sets of n numbers a_i and b_i with means $\bar{a}$ and $\bar{b}$ it is

$$r(a, b) \coloneqq \frac{\sum_{i=1}^{n} (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_{i=1}^{n} (a_i - \bar{a})^2 \; \sum_{i=1}^{n} (b_i - \bar{b})^2}}. \tag{2.3}$$

The numerator is n times the covariance of the pairs and the denominator n times the product of the two standard deviations. The coefficient lies between -1 and 1. It equals 1 when the pairs lie on a rising straight line and -1 when they lie on a falling one. It is close to 0 when the cloud shows no linear trend. Shifting a or b, or multiplying either by a positive factor, leaves it unchanged.

Definition 2.1 (Capacity)**.** The capacity of a forecast at the horizon τ is the squared Pearson correlation between predictions and targets over the test origins. The total capacity is its sum over the horizons,

$$r(\tau) \coloneqq r(\tilde{y}, y), \qquad C(\tau) \coloneqq r(\tau)^2, \qquad C_{\text{tot}} \coloneqq \sum_{\tau=1}^{50} C(\tau), \tag{2.4}$$

with $\tilde{y}_i = \tilde{s}_{k_i+\tau}$ and $y_i = s_{k_i+\tau}$ at the test origins k_i.

The capacity is the fraction of the variance of the targets that the best affine function $a\tilde{y} + b$ of the forecasts accounts for. It equals 1 when the targets are an affine function of the forecasts and 0 when the forecasts carry no linear information about them. C_{tot} therefore lies between 0 and 50. A constant forecast has an undefined correlation and is assigned $C = 0$ by convention. This squared correlation also measures how well a dynamical system reconstructs a function of its past inputs [6]. With the test origins $k = 3000, \dots, 3949$ every target up to $\tau = 50$ lies in the test block, and each $C(\tau)$ is computed from these 950 origins.

Since the Pearson coefficient is invariant under shifts and positive rescalings and the capacity squares it, the capacity ignores the amplitude, the offset and the sign of a forecast. Figure 9 shows this for three artificial predictions of the laser test values y. The prediction $\tilde{y} = 0.5\, y + 0.3$ has half the amplitude of the signal and a wrong offset, and yet $r = +1.00$ and $C = 1.00$. The prediction $\tilde{y} = 1 - y$ moves against the target, with $r = -1.00$, and again $C = 1$ because squaring removes the sign. The prediction $\tilde{y} = y + \xi$, with independent Gaussian noise ξ of standard deviation 0.1, has the correct level and sign and the lowest capacity of the three, $C = 0.81$. We always read the capacity together with the sign of r and a scatter plot of forecasts against targets.

The simplest forecast, persistence, needs no fitting and repeats the last known value,

$$\tilde{s}_{k+\tau} \coloneqq s_k, \tag{2.5}$$

for every horizon. It is a good forecast for a slowly varying signal at a short horizon, and a forecasting model is useful only where its capacity exceeds that of persistence.

The window of equation (2.2) is still a guess. Its length, and more generally the set of past values that carry the information about the present state, depend on how the hidden state moves.

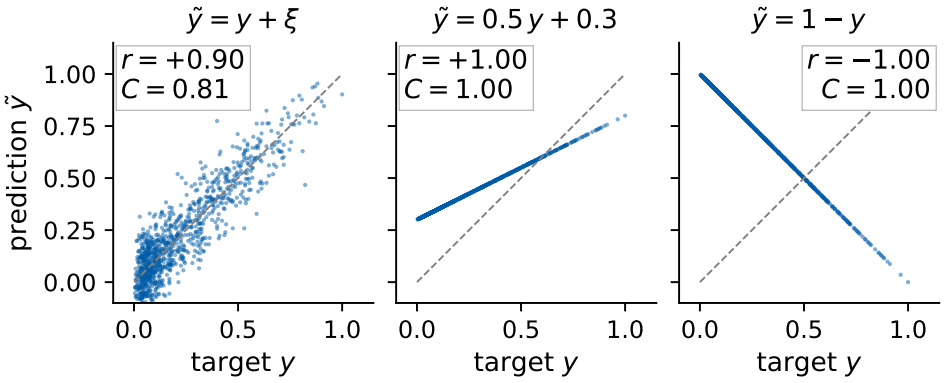


Figure 9: Three artificial predictions $\tilde{y}$ of the rescaled Santa Fe test values y, each point one test value. Left, the target plus independent Gaussian noise ξ of standard deviation 0.1. Middle, the target rescaled and shifted, $0.5\,y + 0.3$. Right, the reversed target, $1 - y$. The dashed line is the diagonal $\tilde{y} = y$, and the boxes give the Pearson correlation r and the capacity $C = r^2$.

3 Flows, attractors and phase-space volume

Three notions of the theory of dynamical systems describe the motion of a hidden state, the flow map, the attractor and the phase-space volume.

3.1 The flow map

A record samples the motion at discrete times, while the law (1.2) gives only the instantaneous velocity at every state. Following the arrows from a given initial state for a finite time defines a map on phase space, which is the object a record sees.

Definition 3.1 (Flow map)**.** Let $\mathbf{x}(t; \mathbf{x}_0)$ be the solution of $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$ with $\mathbf{x}(0) = \mathbf{x}_0$. The flow map $\varphi_t \colon \mathbb{R}^d \to \mathbb{R}^d$ sends every initial state to the state reached after the time t,

$$\varphi_t(\mathbf{x}_0) \coloneqq \mathbf{x}(t; \mathbf{x}_0). \tag{3.1}$$

The flow map moves every point of phase space along its own trajectory for the time t. For the one-dimensional law $\dot{x} = ax$ with a constant a it is $\varphi_t(x_0) = e^{at} x_0$. The uniqueness of solutions gives every flow map three properties,

$$\varphi_0(\mathbf{x}_0) = \mathbf{x}_0, \qquad \varphi_{t+s} = \varphi_t \circ \varphi_s, \qquad \varphi_{-t} = \varphi_t^{-1}. \tag{3.2}$$

Evolving for zero time leaves every state unchanged. Since the law does not change in time and only one trajectory passes through the intermediate state, evolving for s and then for t ends where evolving for $t + s$ ends. Evolving for t and then for $-t$ returns to the initial state, so the flow map is invertible. For the one-dimensional example, $e^{at} e^{as} x_0 = e^{a(t+s)} x_0$ and $e^{-at} e^{at} x_0 = x_0$.

A record sampled at the interval Δt sees the system only through the map $\varphi_{\Delta t}$ (Figure 10). With $\mathbf{x}_k \coloneqq \mathbf{x}(k\Delta t)$, the composition rule in equation (3.2) gives

$$\mathbf{x}_{k+1} = \varphi_{\Delta t}(\mathbf{x}_k), \qquad \mathbf{x}_{k+n} = \underbrace{\varphi_{\Delta t} \circ \cdots \circ \varphi_{\Delta t}}_{n \text{ times}}(\mathbf{x}_k) = \varphi_{n\Delta t}(\mathbf{x}_k), \tag{3.3}$$

so a continuous-time system observed at discrete times is a discrete-time map. Systems defined directly by a map $\mathbf{x}_{n+1} = \mathbf{f}(\mathbf{x}_n)$ also occur. A system defined by a differential equation is called a flow, one defined by a map is called a map, and a sampled flow is a map with $\mathbf{f} = \varphi_{\Delta t}$ [23].

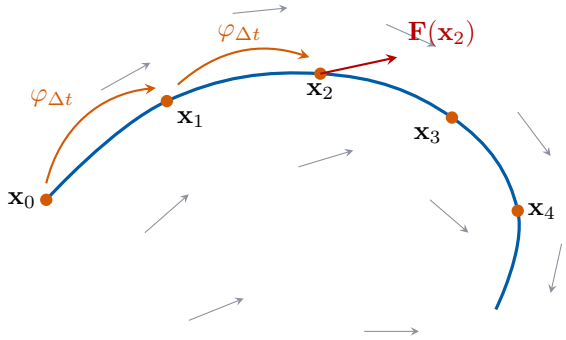


Figure 10: The vector field $\mathbf{F}$ attaches a velocity to every state (grey arrows), and a trajectory (blue) is tangent to the arrows at every point, as the arrow $\mathbf{F}(\mathbf{x}_2)$ shows. Sampling at the interval Δt picks the states $\mathbf{x}_k = \mathbf{x}(k\Delta t)$ (orange), and the flow map $\varphi_{\Delta t}$ carries each of them to the next one.

3.2 Attractors

A pendulum with friction loses energy and comes to rest, whatever its initial swing. Many systems share this long-time behaviour, in which after a transient the motion approaches a subset of phase space and stays close to it.

Definition 3.2 (Attractor)**.** An attractor $A \subset \mathbb{R}^d$ is a set on which the motion settles after the transient for all initial states in a region of phase space, called its basin. A trajectory that starts on A stays on A.

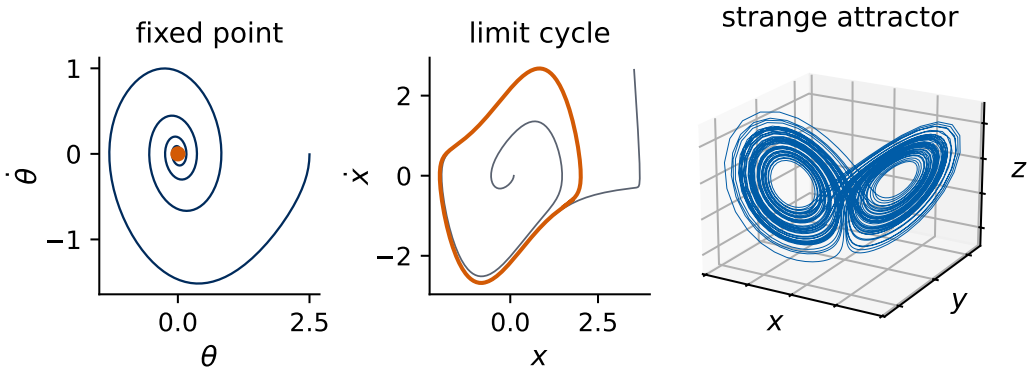


Figure 11: Three kinds of attractor. Left, the damped pendulum in the plane of angle θ and angular velocity $\dot\theta$, whose trajectory spirals into the fixed point at rest (orange dot). Middle, the Van der Pol oscillator in the plane of x and $\dot x$, with two orbits (grey) started inside and outside that end on the limit cycle (orange). Right, one trajectory of the Lorenz model in its phase space (x, y, z), which settles on an attractor on which the orbit never closes.

Figure 11 shows the three kinds of attractor met in these notes. The damped pendulum (1.3) settles at rest, so its attractor is a fixed point, a state with $\mathbf{F}(\mathbf{x}^*) = 0$. The Van der Pol oscillator $\ddot x = (1 - x^2)\dot x - x$ pumps energy in at small amplitude, where the factor $1 - x^2$ is positive, and removes it at large amplitude. Orbits started inside and outside end on the same closed curve, a limit cycle, which the motion traces once per period. The third panel shows the attractor of a model of convection, the Lorenz model. The orbit on it never closes, and the set it fills consists of infinitely many thin layers, whereas the fixed point and the limit cycle are smooth sets. An attractor with such a layered structure is called strange.

3.3 Phase-space volume

A frictionless pendulum settles on no attractor and swings forever with the amplitude it was given. Whether a system has an attractor depends on what the flow does to sets of states. Let A_0 be a set

of initial states, for example a small ball, and let

$$A_t \coloneqq \varphi_t(A_0) = \{\varphi_t(\mathbf{x}_0) \colon \mathbf{x}_0 \in A_0\}, \qquad V(t) \coloneqq \int_{A_t} d^d y \tag{3.4}$$

be the set of the states reached at time t from all starts in A_0 and its volume, where $\mathbf{y}$ denotes a point of A_t. The volume is computed by a change of variables from $\mathbf{y}$ to the initial state $\mathbf{x}_0$. Every $\mathbf{y} \in A_t$ comes from one $\mathbf{x}_0 \in A_0$, $\mathbf{y} = \varphi_t(\mathbf{x}_0)$, and the volume elements are related by the Jacobian matrix of the flow map,

$$(M_t)_{ij}(\mathbf{x}_0) \coloneqq \frac{\partial\, \varphi_{t,i}(\mathbf{x}_0)}{\partial x_{0,j}}, \qquad M_0 = I, \qquad d^d y = \det M_t(\mathbf{x}_0)\, d^d x_0, \qquad V(t) = \int_{A_0} \det M_t(\mathbf{x}_0)\, d^d x_0, \tag{3.5}$$

where $\varphi_{t,i}$ is the i-th component of the flow map and I the identity matrix. The entry $(M_t)_{ij}$ measures how much the i-th coordinate at time t changes per unit change of the j-th initial coordinate. The determinant $\det M_t(\mathbf{x}_0)$ is the ratio of the volume of a small element around $\mathbf{x}_0$ at time t to its initial volume.

Theorem 3.3 (Rate of change of phase-space volume)**.** For every set A_0 of initial states the volume changes at the rate

$$\frac{dV}{dt} = \int_{A_t} \nabla \cdot \mathbf{F}\; d^d y, \qquad \nabla \cdot \mathbf{F} \coloneqq \sum_{i=1}^{d} \frac{\partial F_i}{\partial x_i}. \tag{3.6}$$

To derive the theorem, we first obtain an equation of motion for M_t. The trajectory obeys $\frac{d}{dt}\varphi_{t,i}(\mathbf{x}_0) = F_i(\varphi_t(\mathbf{x}_0))$. Differentiating both sides with respect to $x_{0,j}$, exchanging the order of the derivatives on the left and using the chain rule on the right, gives

$$\frac{d}{dt}(M_t)_{ij} = \sum_{l=1}^{d} \frac{\partial F_i}{\partial x_l}\Big|_{\varphi_t(\mathbf{x}_0)} (M_t)_{lj}, \tag{3.7}$$

which in matrix form reads

$$\frac{d}{dt} M_t = J\big(\varphi_t(\mathbf{x}_0)\big)\, M_t, \qquad M_0 = I, \qquad J_{ij} \coloneqq \frac{\partial F_i}{\partial x_j}, \tag{3.8}$$

where J is the Jacobian matrix of the vector field, evaluated along the trajectory. Jacobi's formula for the derivative of the determinant of an invertible matrix, $\frac{d}{dt}\det M = \det M\, \mathrm{tr}\big(M^{-1}\frac{dM}{dt}\big)$, together with the invariance of the trace under cyclic permutations, $\mathrm{tr}(M_t^{-1} J M_t) = \mathrm{tr}\, J = \nabla \cdot \mathbf{F}$, gives

$$\frac{d}{dt}\det M_t = \big(\nabla \cdot \mathbf{F}\big)\big(\varphi_t(\mathbf{x}_0)\big)\, \det M_t, \qquad \det M_0 = 1. \tag{3.9}$$

Jacobi's formula needs M_t to be invertible. At $t = 0$ the determinant is 1, and by equation (3.9) a positive determinant stays positive, so $\det M_t > 0$ for all t. The absolute value that a change of variables normally requires is therefore absent from equation (3.5). Differentiating $V(t) = \int_{A_0} \det M_t\, d^d x_0$ under the integral and inserting equation (3.9) gives

$$\frac{dV}{dt} = \int_{A_0} \big(\nabla \cdot \mathbf{F}\big)\big(\varphi_t(\mathbf{x}_0)\big)\, \det M_t(\mathbf{x}_0)\, d^d x_0 = \int_{A_t} \nabla \cdot \mathbf{F}\; d^d y, \tag{3.10}$$

where the second step is the change of variables $\mathbf{y} = \varphi_t(\mathbf{x}_0)$ with $d^d y = \det M_t\, d^d x_0$, which proves equation (3.6).

Equation (3.9) is a linear equation $\dot{u} = a(t)\, u$ for the number $u = \det M_t$, and its solution is

$$\det M_t(\mathbf{x}_0) = \exp\Big(\int_0^t \big(\nabla \cdot \mathbf{F}\big)\big(\mathbf{x}(s)\big)\, ds \Big), \tag{3.11}$$

with $\mathbf{x}(s) = \varphi_s(\mathbf{x}_0)$. The divergence is therefore the rate at which small volumes grow, $\frac{1}{\delta V}\frac{d\,\delta V}{dt} = (\nabla \cdot \mathbf{F})(\mathbf{x}(t))$ for a small element of volume δV around the trajectory. For a small set A_0 the divergence is nearly constant on A_t and equation (3.6) reduces to $dV/dt \approx (\nabla \cdot \mathbf{F})(\mathbf{x}(t))\, V$. When the divergence is the same constant at every state, $V(t) = V(0)\, e^{(\nabla \cdot \mathbf{F})t}$ holds for any set A_0.

3.4 Conservative and dissipative flows

Equation (3.6) turns the question whether volumes shrink into the sign of a divergence, and two classes of systems follow from it. A system without friction is described by a Hamiltonian $H(\mathbf{q}, \mathbf{p})$ of the coordinates $\mathbf{q} = (q_1, \ldots, q_n)$ and the conjugate momenta $\mathbf{p} = (p_1, \ldots, p_n)$, and its law of motion is given by Hamilton's equations,

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, n, \tag{3.12}$$

with the state $\mathbf{x} = (\mathbf{q}, \mathbf{p})$ and $d = 2n$. The divergence of this vector field is

$$\nabla \cdot \mathbf{F} = \sum_{i=1}^{n} \Big(\frac{\partial^2 H}{\partial q_i \, \partial p_i} - \frac{\partial^2 H}{\partial p_i \, \partial q_i} \Big) = 0, \tag{3.13}$$

since the mixed second derivatives of a smooth function are equal, and equation (3.6) gives $dV/dt = 0$. The flow of a Hamiltonian system conserves the volume of every set of states, a result known from statistical mechanics as Liouville's theorem, and such flows are called conservative. The motion may stretch and fold a cloud of initial states, as it would an incompressible fluid, but the volume of the cloud stays fixed. A conservative system therefore has no attractor.

Friction makes the divergence negative. For the damped pendulum (1.3) the vector field is $\mathbf{F}(\theta, \omega) = (\omega, \, -\sin\theta - \gamma\omega)$, and

$$\nabla \cdot \mathbf{F} = \frac{\partial \omega}{\partial \theta} + \frac{\partial(-\sin\theta - \gamma\omega)}{\partial \omega} = -\gamma. \tag{3.14}$$

The divergence is constant, so an area $\mathcal{A}$ of initial states in the phase plane shrinks as $\mathcal{A}(t) = \mathcal{A}(0)\, e^{-\gamma t}$. Every set of initial states contracts towards a set of zero area, the fixed point at rest. A flow whose divergence is negative, everywhere or on average along its trajectories, is called dissipative (Figure 12). Its volumes shrink, and the motion settles on an attractor of lower dimension than the phase space, where it can nevertheless remain complicated.

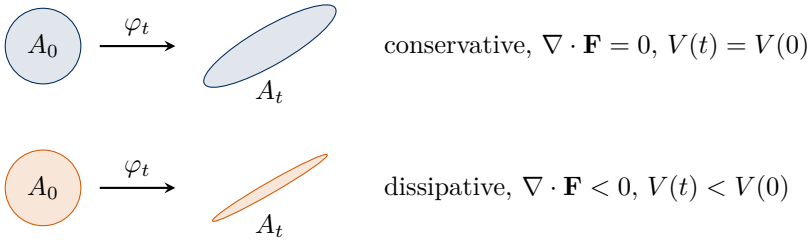


Figure 12: A set A_0 of initial states and its image $A_t = \varphi_t(A_0)$. Top, a conservative flow deforms the set and keeps its volume. Bottom, a dissipative flow deforms it and shrinks its volume, so the states crowd onto a set of lower dimension.

The energy H of the damped pendulum shows in detail how friction creates an attractor. It is the kinetic plus potential energy, equal to the Hamiltonian of the frictionless pendulum, and counted from the lowest point in the units of equation (1.3) it reads

$$H(\theta, \omega) = \tfrac{1}{2}\omega^2 + 1 - \cos\theta, \qquad \frac{dH}{dt} = \omega\dot{\omega} + \sin\theta\,\dot{\theta} = -\gamma\,\omega^2, \tag{3.15}$$

where the second equation follows by inserting equation (1.3). Without friction, $\gamma = 0$, the energy is conserved and every trajectory runs along a curve of constant H in the phase plane. For $H < 2$ these curves are closed and the pendulum swings. For $H > 2$ the bob passes over the top and the angle grows without bound. The curve $H = 2$, called the separatrix, divides the two kinds of motion. With friction the energy decreases whenever the pendulum moves, so every trajectory crosses the curves of constant energy inwards. The fixed points, where $\mathbf{F} = 0$, are the hanging position $(\theta, \omega) = (0, 0)$

and the upright position $(\pi, 0)$, each repeated every 2π in the angle. A small deviation from a fixed point $(\theta^*, 0)$ obeys $\dot{\boldsymbol{\delta}} = J\boldsymbol{\delta}$ with the Jacobian

$$J = \begin{pmatrix} 0 & 1 \\ -\cos\theta^* & -\gamma \end{pmatrix}, \qquad \mu^2 + \gamma\mu + \cos\theta^* = 0, \qquad \mu_\pm = -\frac{\gamma}{2} \pm \sqrt{\frac{\gamma^2}{4} - \cos\theta^*}, \tag{3.16}$$

and the deviation grows or decays as $e^{\mu t}$, with μ an eigenvalue of J, a root of the characteristic equation in the middle. At the hanging position $\cos\theta^* = 1$, and for $\gamma < 2$ the roots are complex, $\mu_\pm = -\gamma/2 \pm \mathrm{i}\sqrt{1 - \gamma^2/4}$. The imaginary part is the frequency of the swing and the negative real part its decay. A deviation circles the hanging position with an amplitude falling as $e^{-\gamma t/2}$, a stable spiral. At the upright position $\cos\theta^* = -1$, the roots are real with opposite signs, and a deviation grows along one direction and decays along the other, a saddle. For $\gamma = 0.2$ the hanging position has $\mu_\pm = -0.1 \pm 0.995\,\mathrm{i}$, so friction hardly changes the frequency and the amplitude falls by the factor e in $2/\gamma = 10$ time units. The upright position has $\mu_+ = 0.905$ and $\mu_- = -1.105$. The stable spiral is the attractor of Figure 11, and the saddle is the balanced pendulum that falls at the slightest disturbance.

Dissipation explains why the laser, which loses energy through its mirror, settles on an attractor. The kind of attractor that produces its irregular pulses remains open.

4 Three model systems

The notions of flow, attractor and dissipation become concrete on two models whose equations we can integrate and whose full state we know. They serve as references for the measured laser record, and each brings its own difficulty.

4.1 The Lorenz model of convection

The laser equations (1.5) are hard to test, since their parameters are known only roughly and only the intensity is measured. A reference model whose full state can be computed comes from fluid dynamics. A horizontal layer of fluid of height $\mathcal{D}$ lies between a lower plate at the temperature $\Theta_0 + \Delta\Theta$ and an upper plate at Θ_0, with $\Delta\Theta > 0$, a configuration called Rayleigh–Bénard convection. Fluid near the hot plate expands, becomes lighter than the fluid above it and tends to rise, while viscous friction and the diffusion of heat oppose the motion. For a small temperature difference the fluid stays at rest and heat crosses the layer by conduction alone. Above a threshold the fluid starts to turn in convection rolls, warm fluid rising in some places and cold fluid sinking in others.

The fluid is described by two fields, its velocity $\mathbf{v}(\mathbf{r}, t)$ and its temperature $\Theta(\mathbf{r}, t)$ at every position $\mathbf{r}$ and time t. Density changes caused by temperature are small, and the Boussinesq approximation keeps them only where they matter, in the buoyancy force. The fluid has the reference density ϱ_0, the pressure $\mathcal{P}(\mathbf{r}, t)$, the kinematic viscosity ν, the thermal diffusivity χ and the thermal expansion coefficient α. With the gravitational acceleration g_{gr} and the upward unit vector $\mathbf{e}_Z$, the fields obey

$$\partial_t \mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v} = -\frac{1}{\varrho_0}\nabla\mathcal{P} + \nu\nabla^2\mathbf{v} + g_{\mathrm{gr}}\,\alpha\,(\Theta - \Theta_0)\,\mathbf{e}_Z, \tag{4.1}$$

$$\nabla\cdot\mathbf{v} = 0, \tag{4.2}$$

$$\partial_t \Theta + (\mathbf{v}\cdot\nabla)\Theta = \chi\nabla^2\Theta. \tag{4.3}$$

Equation (4.1) is the Navier–Stokes equation, Newton's second law for a small parcel of fluid per unit mass. On the left, $\partial_t\mathbf{v}$ is the change of the velocity at a fixed point and $(\mathbf{v}\cdot\nabla)\mathbf{v}$ the change due to the parcel moving into a region of different velocity. Together they form the acceleration of the parcel. Of the forces on the right, the pressure gradient pushes fluid from high to low pressure, and the viscous term $\nu\nabla^2\mathbf{v}$ is the internal friction that smooths out velocity differences. The buoyancy

term pushes fluid upwards where it is warmer than the reference temperature and downwards where it is colder. Equation (4.2) states that the fluid is incompressible, so a volume of fluid keeps its size. Equation (4.3) is the heat equation with transport. The term $(\mathbf{v}\cdot\nabla)\Theta$ describes heat carried along by the moving fluid, and $\chi\nabla^2\Theta$ the diffusion of heat from warm to cold regions. The two advection terms, in which the fields transport themselves and each other, make the equations nonlinear. The state is a pair of fields with a value at every point of space, so the phase space has infinitely many dimensions.

Two dimensionless numbers control the problem. The Prandtl number $\sigma \coloneqq \nu/\chi$ compares the rates at which friction smooths velocity differences and diffusion smooths temperature differences, and it is a property of the fluid. The Rayleigh number

$$\mathrm{Ra} \coloneqq \frac{g_{\mathrm{gr}}\,\alpha\,\mathcal{D}^3\,\Delta\Theta}{\nu\,\chi} \tag{4.4}$$

compares the driving by buoyancy, proportional to $g_{\mathrm{gr}}\alpha\Delta\Theta$, with the damping by friction and diffusion, proportional to $\nu\chi$, for a layer of height $\mathcal{D}$. The conduction state, $\mathbf{v}=0$ with a temperature that falls linearly from the lower to the upper plate, solves equations (4.1) to (4.3) for every Ra. A parcel displaced upwards from it arrives in colder surroundings and is pushed further up by buoyancy, while friction slows it and diffusion equalises its temperature with the surroundings. As long as the damping dominates, the conduction state is stable. It becomes unstable at a critical Rayleigh number Ra_c, where buoyancy first overcomes dissipation, and convection sets in.

These equations reduce to ordinary differential equations in four steps [7, 8]. The motion is restricted to parallel rolls, so the flow is two-dimensional in the plane of a horizontal coordinate X and the height Z, with $0 \le Z \le \mathcal{D}$. Capital letters denote these positions, and the amplitudes of the modes are written in lower case. An incompressible flow in a plane can be written with a stream function $\varpi(X,Z,t)$ as $\mathbf{v} = (-\partial_Z\varpi,\ \partial_X\varpi)$, which satisfies equation (4.2) automatically. The fluid moves along the lines of constant ϖ. The temperature is then written as the conduction profile plus a deviation $\vartheta(X,Z,t)$ caused by the motion. The plates are idealised as free-slip boundaries, along which the fluid may slide, held at fixed temperatures. These conditions allow ϖ and ϑ to be expanded in sines and cosines that satisfy them. Inserting the expansion turns the partial differential equations into infinitely many ordinary differential equations for the time-dependent amplitudes of the modes, a Galerkin expansion. Finally, only three modes are kept,

$$\varpi \propto x(t)\,\sin\frac{\pi a X}{\mathcal{D}}\,\sin\frac{\pi Z}{\mathcal{D}}, \qquad \vartheta \propto y(t)\,\cos\frac{\pi a X}{\mathcal{D}}\,\sin\frac{\pi Z}{\mathcal{D}} - z(t)\,\sin\frac{2\pi Z}{\mathcal{D}}, \tag{4.5}$$

where a sets the aspect of the rolls, each roll being $\mathcal{D}/a$ wide. The amplitude x describes the circulation of one pair of rolls and y the temperature difference between the rising and the sinking fluid. The amplitude z describes the distortion of the mean vertical temperature profile away from the straight conduction profile (Figure 13). We insert the modes into equations (4.1) and (4.3), project each equation onto the three spatial patterns and discard the terms that generate other modes. Rescaling time and the amplitudes by constant factors yields the Lorenz-63 model with the state $\mathbf{x}=(x,y,z)$ and the law

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \qquad \mathbf{F}(\mathbf{x}) = \begin{pmatrix} \sigma(y-x) \\ x(\rho-z)-y \\ xy-\beta z \end{pmatrix}. \tag{4.6}$$

Time is measured in units of $\mathcal{D}^2/(\pi^2(1+a^2)\chi)$, the time in which diffusion alone would reduce the temperature pattern $\cos(\pi a X/\mathcal{D})\sin(\pi Z/\mathcal{D})$ of the rolls by the factor e. The parameter σ is the Prandtl number, and $\rho \coloneqq \mathrm{Ra}/\mathrm{Ra}_c$ is the heating in units of its critical value, with $\mathrm{Ra}_c = \pi^4(1+a^2)^3/a^2$ for the rolls of aspect a. The geometric factor is $\beta \coloneqq 4/(1+a^2)$. The critical Rayleigh number is smallest for $a^2 = 1/2$, with $\mathrm{Ra}_c = 27\pi^4/4 \approx 657.5$, and this aspect gives $\beta = 8/3$.

The first component of equation (4.6), $\sigma(y-x)$, describes the horizontal temperature difference y driving the circulation through buoyancy against viscous friction. By the second, the circulation x

carries warm fluid upwards through the background temperature gradient and creates the temperature difference y. It does so at a rate proportional to the heating, the term ρx, while diffusion removes the difference, the term $-y$. The third balances the distortion of the mean profile by the upward transport of heat, the term xy, against diffusion, the term $-\beta z$, which straightens the profile again. The two nonlinear terms, $-xz$ and xy, both come from the advection of heat by the flow, $(\mathbf{v}\cdot\nabla)\vartheta$ in equation (4.3). The term $-xz$ says that a distorted profile, with a weaker gradient in the interior, reduces the rate at which the flow creates new temperature differences. The advection of momentum, $(\mathbf{v}\cdot\nabla)\mathbf{v}$, contributes nothing, because for the single velocity mode of equation (4.5) the vorticity $\nabla^2\varpi$ is proportional to ϖ itself. A flow carries its own vorticity along its streamlines without change, so the first equation is linear.

Each approximation of the model limits what it says about a real fluid. The Boussinesq approximation keeps density changes only in the buoyancy term and treats the viscosity, the diffusivity and the expansion coefficient as constants. The flow is assumed two-dimensional, with parallel rolls of one wavelength, and the plates are free-slip boundaries, while real plates hold the fluid at rest. The truncation to three Fourier modes drops modes that are small close to the onset of convection. At the heating $\rho = 28$ used here they are not small, so the model does not describe real convection quantitatively. It captures nevertheless the onset of convection at $\rho = 1$, the steady rolls, their loss of stability and the irregular flipping between rolls of opposite sense. Even this three-variable reduction of fluid flow shows the irregular behaviour studied in these notes. We use the standard values $\sigma = 10$, $\rho = 28$ and $\beta = 8/3$.

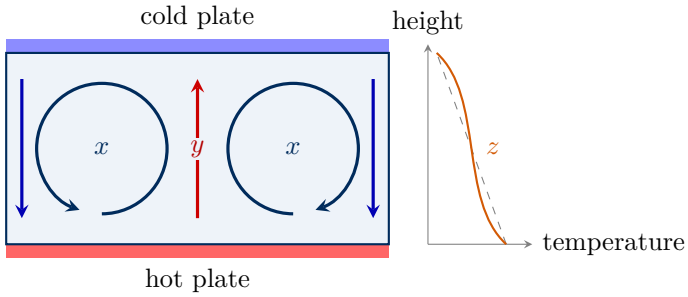


Figure 13: The convecting fluid layer of the Lorenz model. The amplitude x is the speed of the two convection rolls (navy arrows), and its sign gives the sense of rotation. The amplitude y is the temperature difference between the rising warm fluid (red) and the sinking cold fluid (blue). The amplitude z is the bending of the mean vertical temperature profile (orange) away from the straight profile of pure conduction (dashed).

The fixed points follow from setting the right-hand side of equation (4.6) to zero. The first component gives $x = y$, the third $z = x^2/\beta$, and the second then $x(\rho - 1 - z) = 0$. The origin, pure conduction, is one solution. For $\rho > 1$ the other two are the steady rolls

$$C_\pm = \Big(\pm\sqrt{\beta(\rho-1)},\ \pm\sqrt{\beta(\rho-1)},\ \rho-1 \Big) = (\pm 8.485,\ \pm 8.485,\ 27), \tag{4.7}$$

two steady convection patterns of equal strength that turn in opposite senses. The letter C here labels the rolls and is unrelated to the capacity $C(\tau)$. A small deviation $\boldsymbol{\delta}$ from a fixed point obeys $\dot{\boldsymbol{\delta}} = J\boldsymbol{\delta}$, with the Jacobian

$$J(x,y,z) = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix} \tag{4.8}$$

evaluated at the fixed point. The deviation is a sum of terms proportional to $e^{\mu t}$, where μ runs over the eigenvalues of J. A deviation along an eigenvector with $\mathrm{Re}\,\mu < 0$ decays, one with $\mathrm{Re}\,\mu > 0$ grows, and an imaginary part makes it rotate. At $\rho = 28$ the origin has the eigenvalues -22.828, -2.667 and 11.828. With one unstable direction, conduction is unstable and any small convective motion grows. The steady rolls have one strongly contracting direction, with the eigenvalue -13.855, and a complex pair $0.094 \pm 10.195\,i$ with a small positive real part. A trajectory therefore spirals

outwards around C_+ or C_-, at about 1.62 turns per time unit. When it comes close enough to the origin, it is thrown over to the other roll, where the spiralling starts again. The value at which the steady rolls lose their stability follows from the characteristic polynomial of J at $C_\pm$, $\mu^3 + (\sigma+\beta+1)\mu^2 + \beta(\sigma+\rho)\mu + 2\sigma\beta(\rho-1) = 0$. The rolls become unstable when the complex pair crosses the imaginary axis, $\mu = \pm i\omega$. Inserting $\mu = i\omega$ and separating real and imaginary parts gives $\omega^2 = \beta(\sigma+\rho)$ and $(\sigma+\beta+1)\,\beta(\sigma+\rho) = 2\sigma\beta(\rho-1)$, which is solved by

$$\rho_c = \frac{\sigma(\sigma+\beta+3)}{\sigma-\beta-1} = 24.74 \tag{4.9}$$

for $\sigma = 10$ and $\beta = 8/3$. The steady rolls are stable for $1 < \rho < \rho_c$ and lose their stability at $\rho = \rho_c$, so at $\rho = 28$ no stable fixed point remains. Figure 14 shows the computed motion, in which the rolls flip irregularly. The divergence of the Lorenz field is the trace of J,

$$\nabla \cdot \mathbf{F} = -(\sigma + 1 + \beta) = -41/3 \approx -13.667, \tag{4.10}$$

the same at every state. By equation (3.11) every volume shrinks as $V(t) = V(0)\,e^{-41t/3}$, by the factor $e^{-41/3} \approx 1.2 \times 10^{-6}$ per time unit, while the trajectory remains bounded and never repeats.

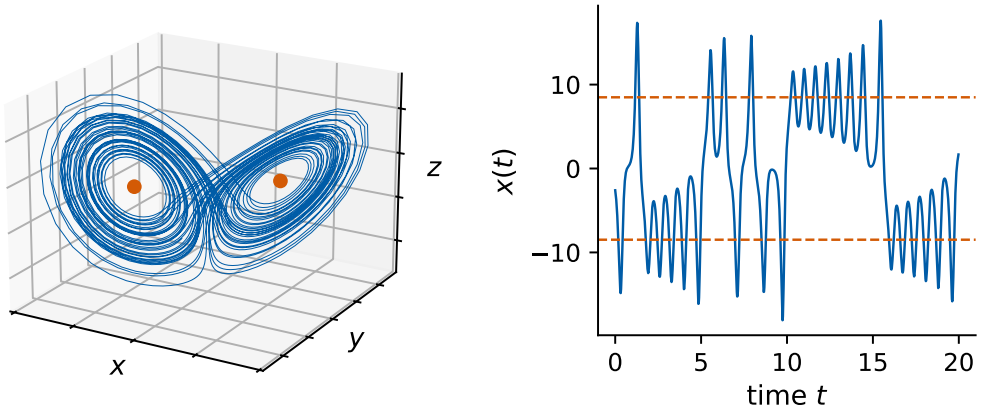


Figure 14: The Lorenz model at $\rho = 28$. Left, one long trajectory in the phase space (x, y, z), with the steady rolls $C_\pm$ as dots. Right, the coordinate $x(t)$ against time t, with the values of x at $C_\pm$ dashed. The signal oscillates around one of these values with growing amplitude and jumps irregularly to the other.

The record we use is the first coordinate, $s_k = x(k\Delta t)$, sampled every $\Delta t = 0.02$ time units, so the 4000 samples span 80 time units. The other two coordinates are hidden from the record, as P and W are hidden in the laser.

4.2 The Mackey–Glass delay equation

The Lorenz model has three state variables. Many systems respond to their own past with a delay, and a delay changes the size of the state. The Mackey–Glass equation models the concentration $c(t)$ of circulating blood cells [9]. New cells are produced in the bone marrow, and the production is controlled by the concentration in the blood. A new cell needs a maturation time τ_{MG} before it enters the blood, so the production at time t responds to the concentration a time τ_{MG} earlier (Figure 15). The law is

$$\dot{c}(t) = \underbrace{\frac{0.2\,c(t-\tau_{\mathrm{MG}})}{1 + c(t-\tau_{\mathrm{MG}})^{10}}}_{\text{production}} - \underbrace{0.1\,c(t)}_{\text{removal}}. \tag{4.11}$$

The production grows with the delayed concentration when it is small and is suppressed when it is large, and the cells are removed at the rate 0.1 per time unit. Continuing the solution from time t requires the whole history $c(t')$ for $t - \tau_{\mathrm{MG}} \le t' \le t$. The state of a delay equation is therefore a

function on this interval, and its phase space has infinitely many dimensions. As the delay grows, the motion changes from a steady state to a periodic oscillation and then to irregular motion. We use $\tau_{\mathrm{MG}} = 15$, at which the solution settles on a limit cycle, and $\tau_{\mathrm{MG}} = 25$ and 35, at which the orbit never closes. Each solution is sampled once per time unit, $s_k = c(k)$.

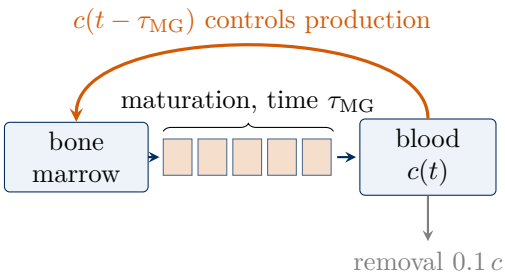


Figure 15: The Mackey–Glass model. Cells produced in the bone marrow mature during the time τ_{MG} before they enter the blood, where their concentration is $c(t)$ and they are removed at the rate 0.1. The production responds to the concentration $c(t-\tau_{\mathrm{MG}})$ of one maturation time earlier (orange feedback).

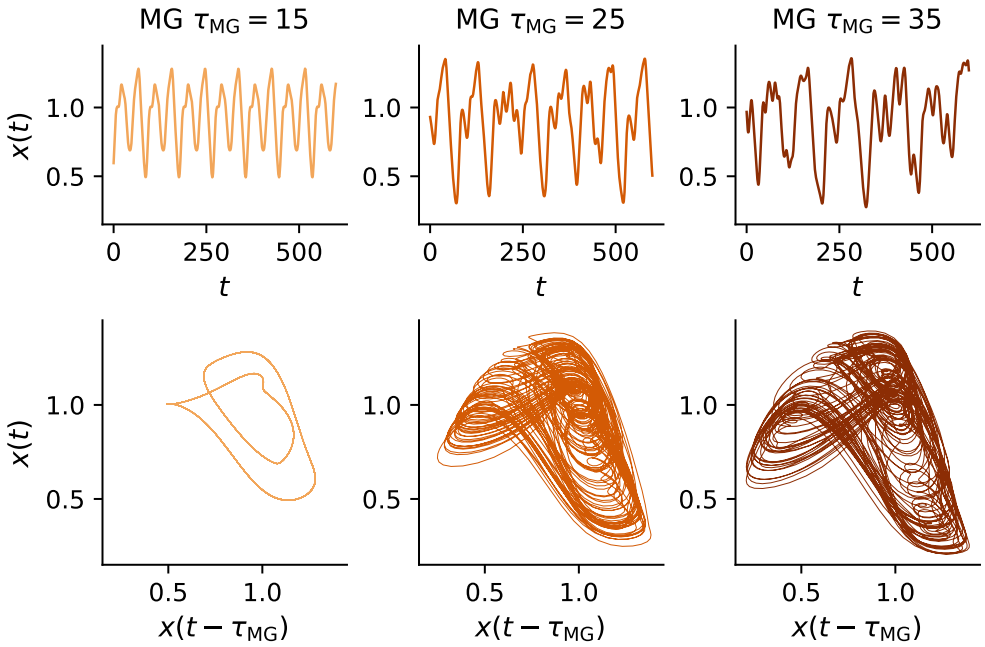


Figure 16: Solutions of the Mackey–Glass equation (4.11), with the concentration c written x on the axes. Top row, 600 time units of $c(t)$ for $\tau_{\mathrm{MG}} = 15$, 25 and 35. Bottom row, the value $c(t)$ against the delayed value $c(t-\tau_{\mathrm{MG}})$. For $\tau_{\mathrm{MG}} = 15$ the curve closes after two loops of different size, a limit cycle, and over one period $c(t)$ has three maxima, the lowest a small shoulder near $c = 1$. For $\tau_{\mathrm{MG}} = 25$ and 35 it never closes and fills a band that widens with the delay.

Although the phase space of the delay equation is infinite-dimensional, the three solutions of Figure 16 explore only a few of its directions. The periodic solution traces a closed curve, and the irregular ones settle on attractors that become more complicated as the delay grows [10].

4.3 The laser record revisited

With the Lorenz model at hand, the irregular pulses of the laser can be connected to a known mechanism. The Maxwell–Bloch equations (1.5) are the Lorenz equations in other variables [11]. Measuring time in units of the polarisation decay time, $t' := \gamma_\perp t$, and setting

$$x = \sqrt{\beta}\,E, \qquad y = \sqrt{\beta}\,g\,P, \qquad z = g\,(W_0 - W), \qquad \sigma = \frac{\gamma_c}{\gamma_\perp}, \qquad \beta = \frac{\gamma_\parallel}{\gamma_\perp}, \qquad \rho = g\,W_0, \quad (4.12)$$

turns equations (1.5) into equations (4.6) with the derivatives taken with respect to t'. The first equation is checked in one line, $dx/dt' = \sqrt{\beta}\,\dot{E}/\gamma_\perp = -\sigma\sqrt{\beta}\,(E - gP) = \sigma(y - x)$, and the other two follow in the same way. The field corresponds to the circulation, the polarisation to the temperature difference and the depletion of the inversion below its pumped value to the distortion of the temperature profile. The pump acts as the heating, and $\rho = 1$ is the lasing threshold, at which the state without light loses its stability. The steady lasing states, the analogues of the steady rolls $C_\pm$, lose their stability only if $\sigma > \beta + 1$, that is $\gamma_c > \gamma_\perp + \gamma_\parallel$. This condition describes a bad cavity, which loses light faster than the medium relaxes. Even then the steady states lose their stability only for a pump ρ above ρ_c, beyond which no stable steady state remains. Most lasers violate the bad-cavity condition, whereas the far-infrared ammonia laser can meet it. Intensity pulsations of the kind predicted by the Lorenz equations were observed in such a laser [27]. The detector measures $E^2 = x^2/\beta$, so the symmetry $(x, y, z) \mapsto (-x, -y, z)$ of the Lorenz equations becomes invisible in the record. In this language, a Lorenz orbit spirals outwards around one of the two convective fixed points, which the intensity of Figure 2 shows as pulses growing over many periods. The orbit then switches to the other lobe, which appears as a collapse after which the growth starts again, since both lobes give the same x^2. The analogy is qualitative, since the experiment is described by the Maxwell–Bloch equations only approximately and its parameters are known only approximately.

Figure 17 collects the five records used in these notes, the Lorenz coordinate, the three Mackey–Glass solutions and the laser intensity, each rescaled to $[0, 1]$. At first sight they look alike, irregular oscillations between zero and one. The periodic Mackey–Glass record repeats itself, the laser record contains rare collapses, and the Lorenz record jumps between two kinds of oscillation. So far these differences are impressions from looking at curves. Turning them into numbers requires quantities that can be computed from a record alone, without the law that produced it.

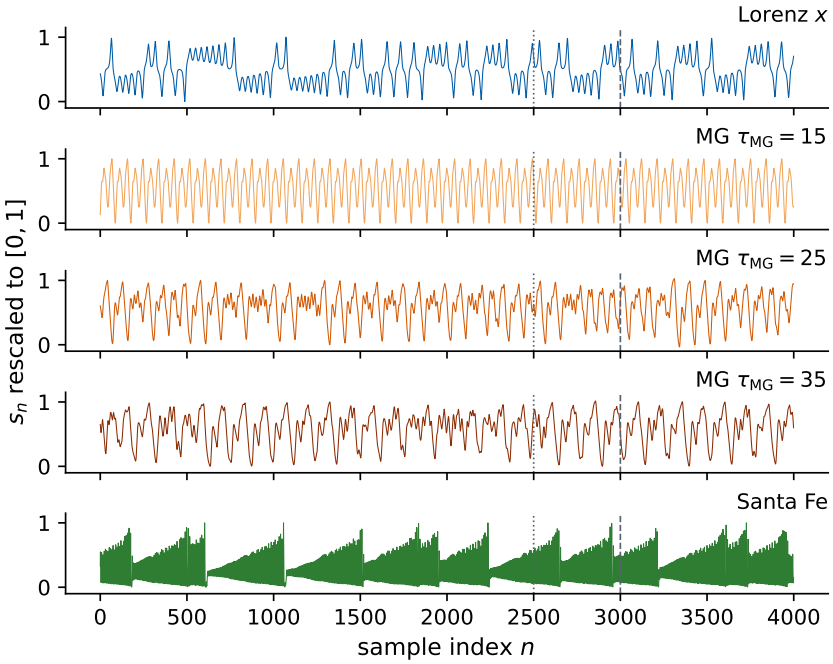


Figure 17: The five records, each with 4000 samples, rescaled to $[0, 1]$ with the minimum and maximum of samples 0 to 2499 and plotted against the sample index n. From top, the x coordinate of the Lorenz model, the Mackey–Glass solution at $\tau_{\mathrm{MG}} = 15$, 25 and 35, and the Santa Fe laser intensity. The dotted line marks the end of the training block (sample 2500) and the dashed line the start of the test block (sample 3000).

5 Regular signals and their statistics

The simplest records, whose forecast is known, show what quantities computed from a record alone must detect.

5.1 Signals that are forecast without error

Some signals can be forecast without error once their rule is recognised (Figure 18). A constant is forecast by persistence, $\tilde{s}_{k+\tau} = s_k$. A periodic signal with period T_p samples is forecast by the value one period earlier, $\tilde{s}_{k+\tau} = s_{k+\tau-T_\mathrm{p}}$ for $\tau \le T_\mathrm{p}$. A quasi-periodic signal, a sum of sine waves whose frequencies have an irrational ratio, never repeats and is nevertheless forecast without error by a linear rule. A single sampled sine wave $s_k = \sin(\omega k)$ obeys

$$s_k = 2\cos\omega \; s_{k-1} - s_{k-2}, \tag{5.1}$$

which follows from the identity $\sin(\omega k) + \sin(\omega(k-2)) = 2\cos\omega \, \sin(\omega(k-1))$. A sum of two sine waves with frequencies ω_1 and ω_2 therefore obeys a linear recurrence of order four, $s_k = \sum_{j=1}^{4} a_j s_{k-j}$. Its coefficients follow from the product of the two second-order rules. We fit the four coefficients by least squares on the past and iterate the recurrence into the future. For the quasi-periodic signal of Figure 18 the fitted coefficients are (3.8645, −5.7327, 3.8645, −1) to four decimals, and the largest forecast error over 40 steps is 7.4×10^{-12}. The periodic forecast errs by at most 7.2×10^{-15} and the constant one not at all. All three errors come from the rounding of floating-point numbers, amplified by the iterated recurrence in the quasi-periodic case.

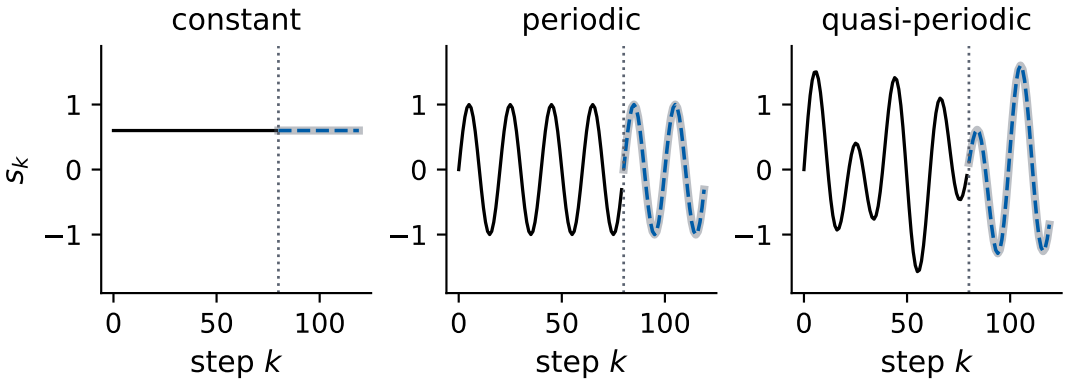


Figure 18: Three regular signals s_k against the step k. Black, the past up to $k = 80$ (dotted line). Grey, the true continuation. Dashed blue, the forecast. Left, a constant forecast by persistence. Middle, a periodic signal forecast by its value one period earlier. Right, a quasi-periodic signal, the sum of two sine waves with an irrational frequency ratio, forecast by iterating a fitted linear recurrence of order four.

Two other kinds of signals are hard to forecast, for different reasons. A stochastic signal contains in every new value a random part that no rule can know in advance. An irregular deterministic signal is produced by a law without any randomness and still resists forecasting.

5.2 The autocorrelation function

Telling a stochastic from an irregular deterministic record requires a measure of how strongly a record remembers its past. The simplest such measure is built from the Pearson coefficient (2.3). A stochastic process is a family of random variables s_k indexed by time, and a measured record is regarded as one realisation of it. The process is called stationary when its statistical properties do not depend on the origin of time. Its mean $\bar{s}$ and the covariance of two values n steps apart then depend on the separation n only, and the integer n is called the lag. For a stationary signal the autocovariance and the autocorrelation function (ACF) are

$$R(n) := \big\langle (s_k - \bar{s})(s_{k+n} - \bar{s}) \big\rangle, \qquad \rho(n) := \frac{R(n)}{R(0)}, \tag{5.2}$$

where the brackets denote the average over the random process. The argument n distinguishes $\rho(n)$ from the heating parameter ρ of the Lorenz model. By construction $\rho(0) = 1$ and $|\rho(n)| \le 1$. With

only one record of N samples, we replace the average over the process by an average over time. This replacement assumes that one long record visits the possible states with the same frequencies as many independent records would, a property called ergodicity. The estimate is

$$\tilde{\rho}(n) = \frac{\sum_{k=1}^{N-n} (s_k - \bar{s})(s_{k+n} - \bar{s})}{\sum_{k=1}^{N} (s_k - \bar{s})^2}, \tag{5.3}$$

with the record mean $\bar{s}$, the time average that estimates the mean of the process. It is the Pearson coefficient (2.3) of the pairs (s_k, s_{k+n}), except that the mean and the variance of the whole record are used for both members of a pair. The ACF measures how well s_k predicts s_{k+n} by a straight line.

Three signals whose ACF is known in advance show what it measures. A sine wave $s_k = \sin(2\pi k/T_\mathrm{p})$ has $\rho(n) = \cos(2\pi n/T_\mathrm{p})$, an oscillation that never decays. White noise, a sequence of independent random numbers ε_k with zero mean, has $\rho(n) = 0$ for every $n \geq 1$. The autoregressive process of order one, AR(1),

$$s_k = \phi\, s_{k-1} + \varepsilon_k, \qquad |\phi| < 1, \tag{5.4}$$

keeps the fraction ϕ of the previous value and adds a fresh random kick. Multiplying the equation at time $k+n$ by s_k and averaging gives $R(n) = \phi R(n-1)$, because the kick ε_{k+n} is independent of the earlier value, and so $\rho(n) = \phi^n$, an exponential decay. Figure 19 compares these three with the Lorenz, Mackey–Glass and laser records. The three irregular records forget their past, as the decay of their ACF shows, but more slowly than white noise. The Mackey–Glass and laser ACFs oscillate with the rhythm of their motion. The Lorenz ACF decays smoothly, with only a weak bump, a local maximum at a lag of 47 samples. The ACF of the Lorenz coordinate drops under $1/e$ after 15 samples and crosses zero first after 89 samples, or 1.78 time units. The laser ACF crosses zero already after 2 samples, since the intensity runs from a maximum to a minimum within a few samples.

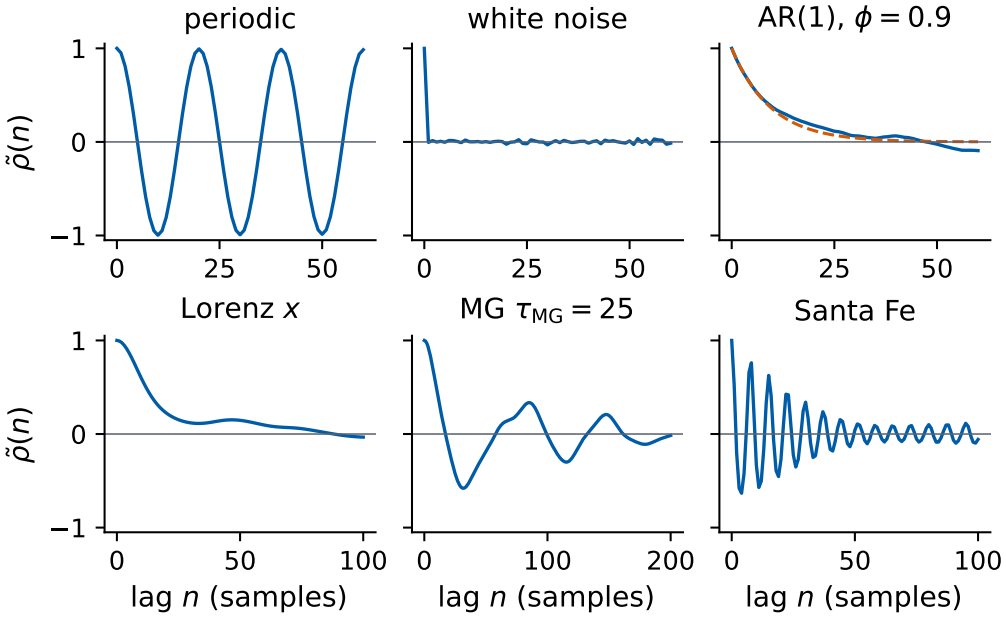


Figure 19: Estimated autocorrelation $\tilde{\rho}(n)$ of six signals against the lag n in samples, from records of 4000 samples. Top row, three signals of known type, a sine wave with period $T_\mathrm{p} = 20$, Gaussian white noise and the AR(1) process with $\phi = 0.9$, whose exact ACF 0.9^n is dashed. Bottom row, the Lorenz coordinate x, the Mackey–Glass record at $\tau_\mathrm{MG} = 25$ and the Santa Fe laser intensity.

5.3 The power spectrum

The ACF of the laser oscillates with the pulse period, a rhythm that is easier to read when the same correlations are arranged by frequency. The discrete Fourier transform of a record decomposes it

into oscillations with the frequencies $f_q = q/(N\Delta t)$, $q = 1, \ldots, N/2$, and the periodogram

$$\tilde{S}(f_q) := \frac{2\Delta t}{N} \left| \sum_{k=1}^{N} (s_k - \bar{s})\, e^{-2\pi i\, qk/N} \right|^2 \tag{5.5}$$

estimates how much of the variance of the record falls at each frequency. The power spectral density (PSD) is the limit of this estimate for an infinitely long record. A peak at f_q marks a component that repeats with the period $1/f_q$. A periodic signal gives sharp lines, white noise a flat spectrum, and AR(1) a smooth spectrum concentrated at low frequencies. By the Wiener–Khinchin theorem the PSD is the Fourier transform of the autocovariance [12],

$$S(f) = \Delta t \sum_{n=-\infty}^{\infty} R(n)\, e^{-2\pi i f n \Delta t}, \tag{5.6}$$

written here as the two-sided density over $-1/(2\Delta t) \le f \le 1/(2\Delta t)$, so the one-sided periodogram estimates $2S(f)$ for $f > 0$. The PSD carries the same information as the autocovariance $R(n)$. The ACF $\rho(n) = R(n)/R(0)$ lacks only the overall scale $R(0)$, the variance. A single periodogram scatters strongly from one frequency to the next, so we average the periodograms of overlapping segments of the record, at the price of a coarser frequency grid. Figure 20 shows the averaged spectra of the six signals. The laser has a strong peak at $f_\mathrm{a} \approx 0.138$ per sample, a period of 7.26 samples, which is the pulse period. A second peak lies at $f_\mathrm{b} \approx 0.262$ per sample, at $f_\mathrm{b}/f_\mathrm{a} = 1.90$, close to the harmonic $2f_\mathrm{a} = 0.275$ without coinciding with it. The Lorenz and Mackey–Glass records have broad spectra that decrease rapidly with frequency.

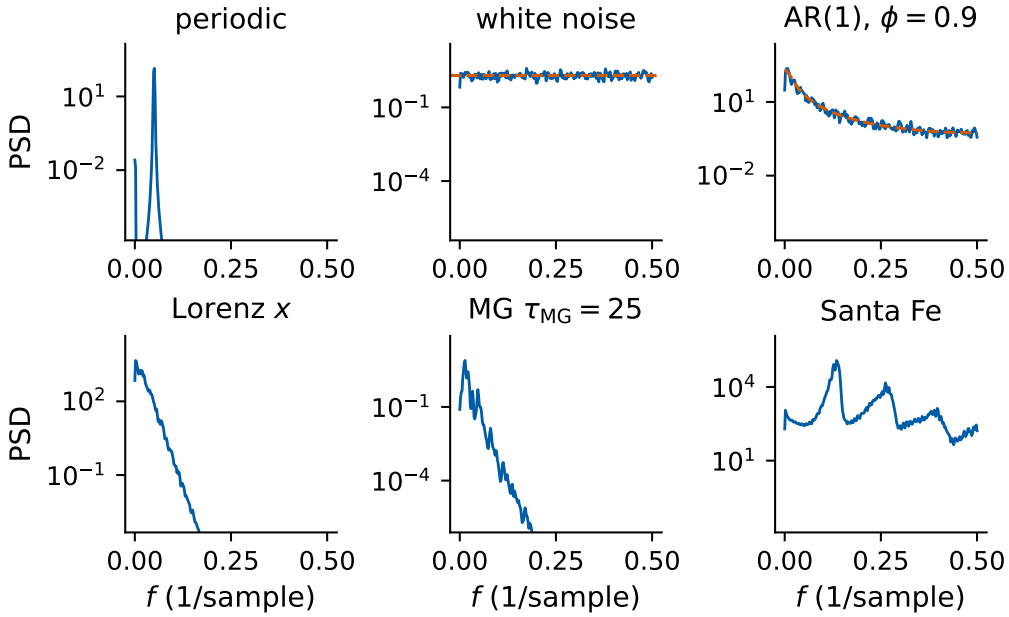


Figure 20: Power spectral density (PSD) of the six signals of Figure 19 on a logarithmic scale against the frequency f in cycles per sample. Each spectrum is estimated by averaging the periodograms of overlapping segments. The dashed curves in the top row are the exact spectra of white noise and of AR(1).

5.4 Persistence and the autocorrelation

The ACF also fixes the score of the simplest forecast, persistence. Its pairs of forecast and target are the pairs $(s_k, s_{k+\tau})$ of the record, whose Pearson correlation is $\tilde{\rho}(\tau)$ up to the slightly different averaging. The capacity of persistence is therefore $C(\tau) \approx \rho(\tau)^2$. A better forecast from s_k alone damps the deviation from the mean. For a stationary record with mean $\bar{s}$ and variance $R(0)$, a forecast $\tilde{s}_{k+\tau} = \bar{s} + b\,(s_k - \bar{s})$ has the mean squared error $R(0)\,(1 - 2b\rho(\tau) + b^2)$, which is smallest for $b = \rho(\tau)$,

with the value $R(0)\,(1-\rho(\tau)^2)$. Persistence, $b=1$, has the larger error $2R(0)\,(1-\rho(\tau))$. As affine functions of s_k, both forecasts have the capacity $\rho(\tau)^2$, while their errors differ by $R(0)\,(1-\rho(\tau))^2$. For $\rho(\tau)=0.5$ the damped forecast errs by $0.75\,R(0)$ and persistence by $R(0)$. The capacity is blind to this difference of level and scale, so it is read together with a scatter plot. Figure 21 confirms this on the test blocks of the five records. For the smooth model records persistence is a strong forecast at short horizons, with $C(1)$ between 0.976 and 0.991. For the laser it is weak already at one step, $C(1)=0.27$. Summed over 50 horizons, persistence reaches $C_{\mathrm{tot}}=8.4$ for the Lorenz record, 19.5, 13.3 and 11.8 for the three Mackey–Glass records and 5.5 for the laser. The oscillating curves show a blind spot of the capacity. At horizons where the ACF swings to negative values, persistence moves against the target, and the squared correlation counts this as skill. On the periodic Mackey–Glass record such horizons contribute 8.32 of its total of 19.5.

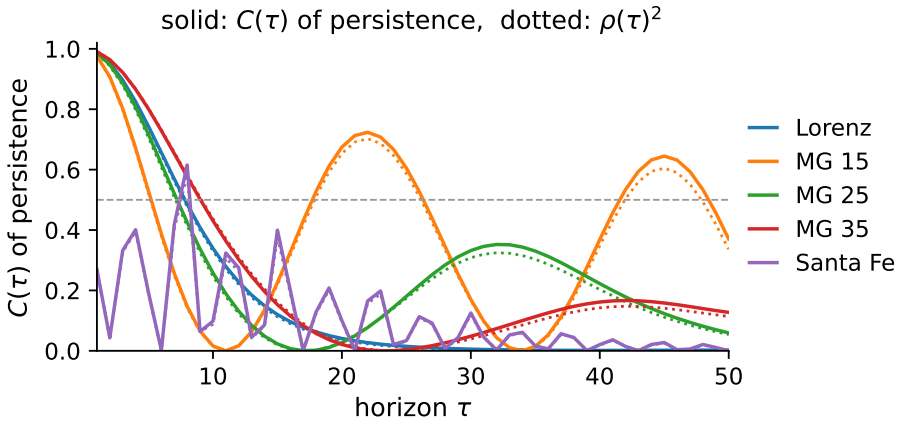


Figure 21: Capacity $C(\tau)$ of persistence on the 950 test origins of the five records against the horizon τ (solid) and the squared autocorrelation $\rho(\tau)^2$ of the test block (dotted). The dashed horizontal line marks $C=0.5$.

5.5 Limits of second-order statistics

Computed for deterministic and random signals alike, the ACF, the spectrum and the score of persistence leave open whether a record comes from a deterministic law. The ACF and the PSD are built from averages of products of two values, and they are called second-order statistics. They see only the part of the relation between two values that a straight line describes, and a deterministic relation can be invisible to them. Let a be a random number distributed symmetrically around zero and $b=a^2$. Then b is fixed by a, while the covariance of a and b is the average of a^3, which vanishes. The bottom row of Figure 19 shows that the deterministic Lorenz record has a smooth decaying ACF of the same kind as a random process with suitable correlations. A random linear process can be constructed with any prescribed PSD, and by the Wiener–Khinchin theorem it then has the same ACF as the record. Such a process can be generated from the record itself [25]. We compute the discrete Fourier transform of the record, keep the amplitude of every frequency and replace every phase by an independent random phase. Transforming the result back gives the surrogate. The phase-randomised surrogate has the spectrum of the record and with it its circular ACF. For the Lorenz record the periodograms of data and surrogate agree to round-off. Although the two records share their spectrum, the pairs (s_n, s_{n-10}), each value against the value ten samples earlier, look completely different (Figure 22). The data trace the folded structure of the attractor, and the surrogate a structureless cloud.

A short computation shows why the surrogate keeps the correlations of the record. With the samples numbered from 0, let $z_k := s_k - \bar{s}$, $k=0,\dots,N-1$, be the record with its mean removed and

$$X_q := \sum_{k=0}^{N-1} z_k\, e^{-2\pi i\, qk/N}, \qquad q=0,\dots,N-1, \tag{5.7}$$

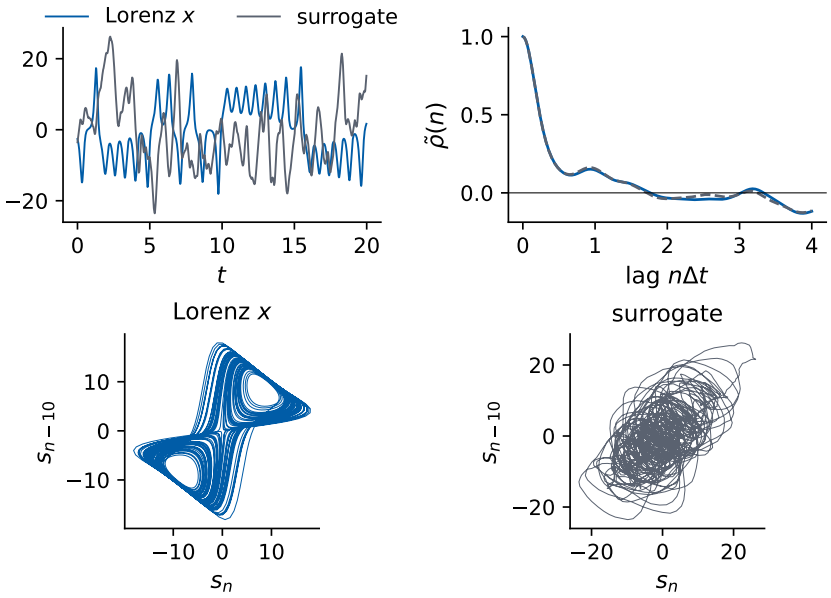


Figure 22: A record and its phase-randomised surrogate. Top left, the Lorenz coordinate x (blue) and one surrogate (grey) against time t. Top right, their estimated ACFs $\tilde{\rho}(n)$ against the lag $n\Delta t$ in time units, which coincide. Bottom, each value s_n against the value s_{n-10} ten samples earlier, for the data (left) and the surrogate (right).

its discrete Fourier transform. The circular autocovariance counts the index $k+n$ modulo N, so that every value has a partner n steps later,

$$R_{\mathrm{c}}(n) \coloneqq \frac{1}{N} \sum_{k=0}^{N-1} z_k \, z_{(k+n) \bmod N}. \tag{5.8}$$

Writing the squared modulus as a double sum and collecting the terms with $l = (k+n) \bmod N$, for which $e^{2\pi i\, q(l-k)/N} = e^{2\pi i\, qn/N}$, gives

$$\frac{1}{N}|X_q|^2 = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{l=0}^{N-1} z_k z_l \, e^{2\pi i\, q(l-k)/N} = \sum_{n=0}^{N-1} R_{\mathrm{c}}(n) \, e^{2\pi i\, qn/N}, \tag{5.9}$$

and the inverse transform turns this into the finite form of the Wiener–Khinchin theorem,

$$R_{\mathrm{c}}(n) = \frac{1}{N^2} \sum_{q=0}^{N-1} |X_q|^2 \, e^{-2\pi i\, qn/N}. \tag{5.10}$$

The circular autocovariance depends on the Fourier amplitudes $|X_q|$ alone and on none of the phases. A phase-randomised surrogate has the coefficients $\tilde{X}_q = |X_q| \, e^{i\theta_q}$ with independent random phases θ_q, chosen with $\theta_{N-q} = -\theta_q$, $\theta_0 = 0$ and, for even N, $\theta_{N/2} = 0$, so that the inverse transform is real. Its amplitudes are those of the record, $|\tilde{X}_q| = |X_q|$, so by equation (5.10) its circular autocovariance equals that of the record for every lag. The usual estimate (5.3) omits the n pairs that wrap around the end of the record, a fraction n/N of the sum. It therefore agrees between record and surrogate only up to these pairs. For the Lorenz record the estimates $\tilde{\rho}$ of data and surrogate agree to within 0.013 up to a lag of 100 samples. The circular ACFs agree to 2.2×10^{-16}, the round-off of the arithmetic.

Second-order statistics cannot separate an irregular deterministic record from noise with the same correlations, and the property that separates them is how the deterministic law treats small differences between states.

6 Growth of small errors

A deterministic law of motion fixes how a small error in the initial state develops, and the same development limits every forecast.

6.1 Two nearly equal starts

We assume that the law **F** is known without error and the initial state only up to a small error. In the Lorenz model of Figure 23, two solutions start at states that differ by 10^{-8} in the coordinate x, much finer than any measurement resolution. For a long time the two curves cannot be told apart. Then they separate, and their distance $|\boldsymbol{\delta}(t)|$ exceeds 1 at $t \approx 19.5$. The distance follows a straight line on the logarithmic scale of the right panel and grows exponentially, at the fitted rate 0.93 per time unit. The growth stops when the distance reaches the size of the attractor. Since the equations are deterministic and both solutions are computed to floating-point precision, the separation comes from the law of motion itself.

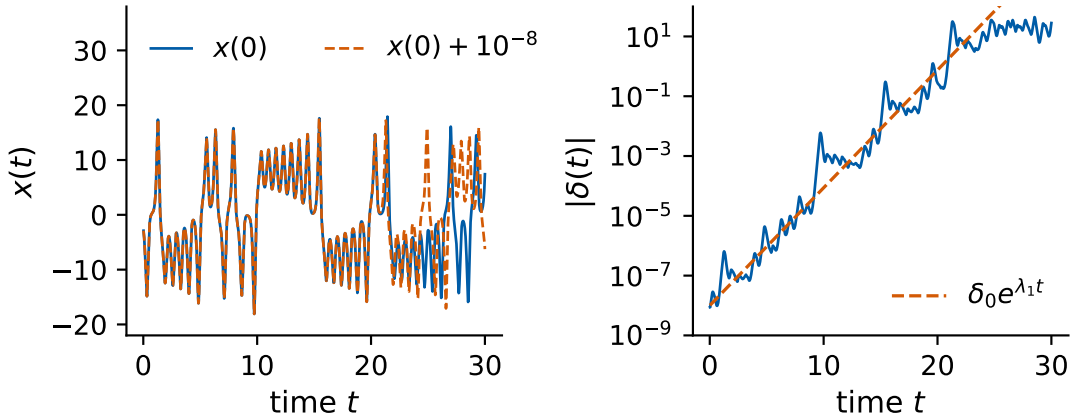


Figure 23: Two Lorenz solutions from nearly equal starts. Left, $x(t)$ against time t for the start $x(0)$ (solid) and for the start shifted by 10^{-8} in x (dashed). Right, the distance $|\boldsymbol{\delta}(t)|$ between the two states in the phase space (x, y, z) on a logarithmic scale, with the exponential $\delta_0 e^{0.907\,t}$ (dashed), $\delta_0 = 10^{-8}$, t in time units.

6.2 Stretching and folding

The mechanism by which a bounded motion separates nearby states without end appears already in a system with a single variable that evolves in steps. The logistic map $x_{n+1} = f(x_n) = 4x_n(1 - x_n)$, with $0 \le x_n \le 1$, is a model of a population with limited resources, in which x_n denotes the population. With the derivative $f'(x) = 4 - 8x$ of the map, two starts x_0 and $x_0 + \delta_0$ are separated after one step by about $|f'(x_0)|\,\delta_0$. After n steps the chain rule gives the separation

$$\delta_n \approx \delta_0 \prod_{j=0}^{n-1} |f'(x_j)| = \delta_0 \exp\Big(\sum_{j=0}^{n-1} \ln|f'(x_j)|\Big). \tag{6.1}$$

The average of $\ln|f'|$ along the orbit is the growth rate of small separations per step. Over 2×10^5 steps it is 0.6931, equal to $\ln 2$ to four decimals. Two orbits that start 10^{-10} apart stay closer than 0.01 for the first 28 steps and are unrelated a few steps later. Their separation grows at a fitted rate of 0.674 per step and reaches 0.1 after 31 steps, close to the estimate $\ln(0.1/10^{-10})/\ln 2 = 29.9$ (Figure 24). Where $|f'| > 1$ the map stretches distances, which produces the exponential growth. The interval $[0, 1]$ is nevertheless mapped onto itself, because the parabola folds the stretched interval back so that each value of x_{n+1} comes from two values of x_n. Stretching alone would send the orbit to infinity and folding alone would leave a stable orbit, while the two together keep the motion bounded and separate nearby states.

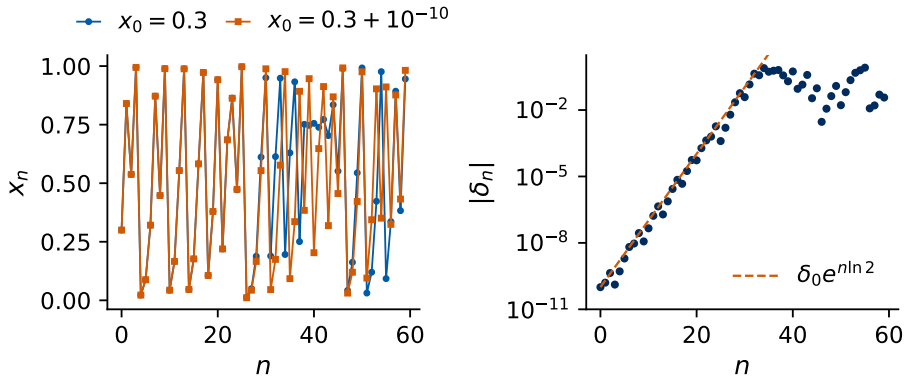


Figure 24: Sensitivity in the logistic map $x_{n+1} = 4x_n(1 - x_n)$. Left, two orbits x_n against the step n, started at $x_0 = 0.3$ and at $x_0 + 10^{-10}$. Right, their separation $|\delta_n|$ on a logarithmic scale, with the law $\delta_0 e^{n \ln 2}$ (dashed).

The Lorenz flow does the same in three dimensions, and a map of one variable can be read from it. We record only the successive maxima M_n of a coordinate and plot each maximum against the previous one, which gives a return map (Figure 25). The maxima are numbered by the integer n and are unrelated to the Jacobian matrix M_t of the flow map defined in equation (3.5). For the z coordinate of the Lorenz model, the 1063 maxima of a longer integration of 40000 samples never repeat. They lie on a thin curve shaped like a tent, so the next maximum is nearly a function of the previous one, as found in Ref. [8]. The slope of this map exceeds one on both branches, and it stretches and folds as the logistic map does. The 1336 maxima of the periodic Mackey–Glass solution at $\tau_{\mathrm{MG}} = 15$, again from 40000 samples, take three values in turn, about 1.01, 1.17 and 1.28. At $\tau_{\mathrm{MG}} = 35$ the maxima spread over a region of the plane, so one maximum does not fix the next.

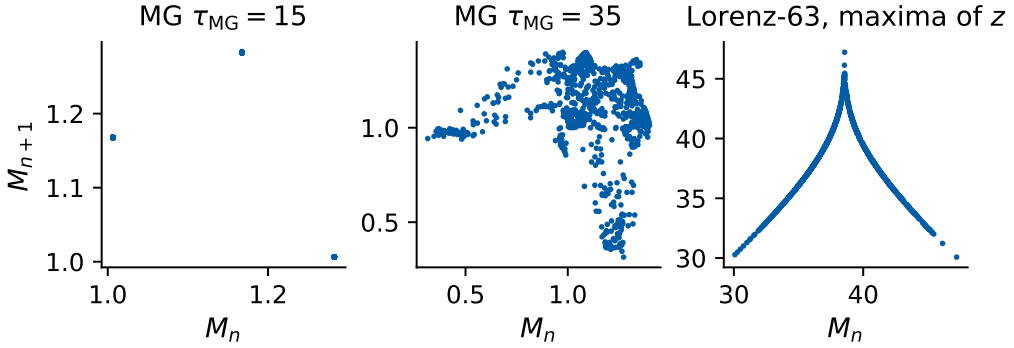


Figure 25: Return maps of successive maxima, each maximum M_{n+1} against the previous one M_n. Left, the Mackey–Glass record at $\tau_{\mathrm{MG}} = 15$, whose maxima repeat three values. Middle, the Mackey–Glass record at $\tau_{\mathrm{MG}} = 35$. Right, the maxima of z of the Lorenz model, which lie on a curve shaped like a tent.

6.3 Linearisation

To describe the exponential growth of the twin experiment quantitatively, let $\mathbf{x}(t)$ be the true trajectory and $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) + \boldsymbol{\delta}(t)$ a second one that starts at $\mathbf{x}(0) + \boldsymbol{\delta}_0$. Since both obey the law of motion, the separation obeys $\dot{\boldsymbol{\delta}} = \mathbf{F}(\mathbf{x} + \boldsymbol{\delta}) - \mathbf{F}(\mathbf{x})$. As long as the separation is small, a Taylor expansion of $\mathbf{F}$ to first order gives

$$\dot{\boldsymbol{\delta}} = J(\mathbf{x}(t))\,\boldsymbol{\delta}, \qquad J_{ij} = \frac{\partial F_i}{\partial x_j}, \tag{6.2}$$

with the Jacobian J of the vector field evaluated along the true trajectory. This equation is linear in $\boldsymbol{\delta}$ although the law itself is nonlinear, so the error at time t is a linear function of the initial error,

$$\boldsymbol{\delta}(t) = M_t\,\boldsymbol{\delta}_0, \qquad (M_t)_{ij} = \frac{\partial x_i(t)}{\partial x_j(0)}, \tag{6.3}$$

where M_t is the Jacobian matrix of the flow map defined in equation (3.5). It obeys the same linear equation (3.8), $\dot{M}_t = J(\mathbf{x}(t))M_t$ with $M_0 = I$, and it is computed by integrating these d^2 equations together with the trajectory. The matrix that measured the change of volumes also carries a small error from time 0 to time t.

6.4 Growth factors

The matrix M_t carries every small error forward, and the growth of an error depends on its direction. The squared length of the error at time t is

$$|\boldsymbol{\delta}(t)|^2 = \boldsymbol{\delta}(t)^\top \boldsymbol{\delta}(t) = \boldsymbol{\delta}_0^\top \tilde{M}_t \, \boldsymbol{\delta}_0, \qquad \tilde{M}_t \coloneqq M_t^\top M_t. \tag{6.4}$$

The matrix $\tilde{M}_t$ is symmetric with non-negative eigenvalues, so it has an orthonormal basis of eigenvectors,

$$\tilde{M}_t \, \mathbf{v}_i = \sigma_i(t)^2 \, \mathbf{v}_i, \qquad \mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}, \qquad \sigma_1(t) \geq \sigma_2(t) \geq \cdots \geq \sigma_d(t) \geq 0. \tag{6.5}$$

The numbers $\sigma_i(t)$ are the growth factors of the error, known in linear algebra as the singular values of M_t. The index i distinguishes them from the Prandtl number σ of the Lorenz model. An initial error $\varepsilon \mathbf{v}_i$ grows to the length $\varepsilon \, \sigma_i(t)$. A small ball of initial errors of radius ε becomes an ellipsoid with semi-axes $\varepsilon\sigma_1(t), \ldots, \varepsilon\sigma_d(t)$ (Figure 26). Expanding a generic initial error as $\boldsymbol{\delta}_0 = \sum_i c_i \mathbf{v}_i$ gives $|\boldsymbol{\delta}(t)|^2 = \sum_i c_i^2 \sigma_i(t)^2$. Once the growth factors have separated, the first term dominates the sum, and the largest growth factor sets the growth of a typical error.

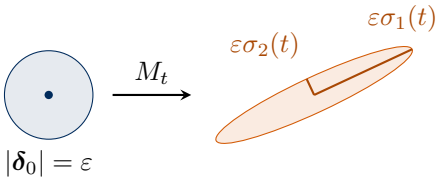


Figure 26: A ball of initial errors of radius ε around a state is mapped by M_t into an ellipsoid. Its semi-axes are ε times the growth factors $\sigma_i(t)$, along the directions $M_t\mathbf{v}_i$. The state itself stays bounded, and only the error grows or shrinks.

6.5 Lyapunov exponents

The growth factors change in time, and a forecast over long times depends only on their long-time behaviour. On the Lorenz attractor they grow or decay exponentially, at rates that settle to fixed values over long times.

Definition 6.1 (Lyapunov exponents)**.** The Lyapunov exponents of a trajectory are the long-time growth rates of the growth factors,

$$\lambda_i \coloneqq \lim_{t\to\infty} \frac{1}{t} \ln \sigma_i(t), \qquad i = 1, \ldots, d, \tag{6.6}$$

so that $\sigma_i(t) \sim e^{\lambda_i t}$ for long times, with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$.

The multiplicative ergodic theorem of Oseledets guarantees that these limits exist for almost every initial state on an attractor [13, 14]. When the motion is ergodic, so that time averages along a trajectory equal averages over the attractor, the limits take one set of values for almost all initial states. The exponents are then properties of the system, shared by almost every trajectory on the attractor. A positive exponent means exponential growth of errors in the corresponding direction

and a negative one exponential decay. A bounded motion that does not repeat and has $\lambda_1 > 0$ is called chaotic. The largest exponent governs the growth of a typical small error,

$$|\boldsymbol{\delta}(t)| \approx |\boldsymbol{\delta}_0|\, e^{\lambda_1 t}, \qquad \lambda_1 = \lim_{t\to\infty} \lim_{|\boldsymbol{\delta}_0|\to 0} \frac{1}{t} \ln \frac{|\boldsymbol{\delta}(t)|}{|\boldsymbol{\delta}_0|}. \tag{6.7}$$

A flow that keeps moving has one exponent equal to zero. An initial error along the direction of motion, $\boldsymbol{\delta}_0 = \epsilon\, \mathbf{F}(\mathbf{x}(0))$ with a small ϵ, only shifts the state along its own trajectory by the time ϵ, so it evolves as $\boldsymbol{\delta}(t) = \epsilon\, \mathbf{F}(\mathbf{x}(t))$. For a bounded trajectory that does not approach a fixed point, the speed $|\mathbf{F}(\mathbf{x}(t))|$ stays bounded and returns again and again to values of the same order. The error along the direction of motion neither grows nor decays exponentially, and its exponent is zero.

The matrix M_t cannot be followed for long times, because its columns all turn into the most expanding direction and its entries exceed the range of floating-point numbers. The standard algorithm re-orthonormalises at regular intervals [15]. Starting from an orthonormal matrix Q_0, we integrate the tangent equation over one interval Δt, equal here to the sampling interval. The result is factorised into an orthonormal matrix and an upper triangular one,

$$M_{\Delta t}(\mathbf{x}_{n-1})\, Q_{n-1} = Q_n\, T_n, \qquad \lambda_i \approx \frac{1}{n\Delta t} \sum_{l=1}^{n} \ln |(T_l)_{ii}|, \tag{6.8}$$

where $M_{\Delta t}(\mathbf{x}_{n-1})$ carries errors from the state $\mathbf{x}_{n-1}$ over one interval. The factorisation is the Gram–Schmidt procedure. The first column of Q_n follows the most expanding direction, and $(T_n)_{11}$ is its stretching in this interval. The second column is the part of the second vector orthogonal to the first, so $(T_n)_{11}(T_n)_{22}$ is the growth of the area spanned by the first two vectors. In general the product of the first ℓ diagonal entries is the growth of an ℓ-dimensional volume, so the method returns the exponents in decreasing order. Since all logarithms add up to $\ln|\det M_{\Delta t}|$, the computed exponents can be checked against the time average of the divergence. The running averages converge as shown in Figure 27.

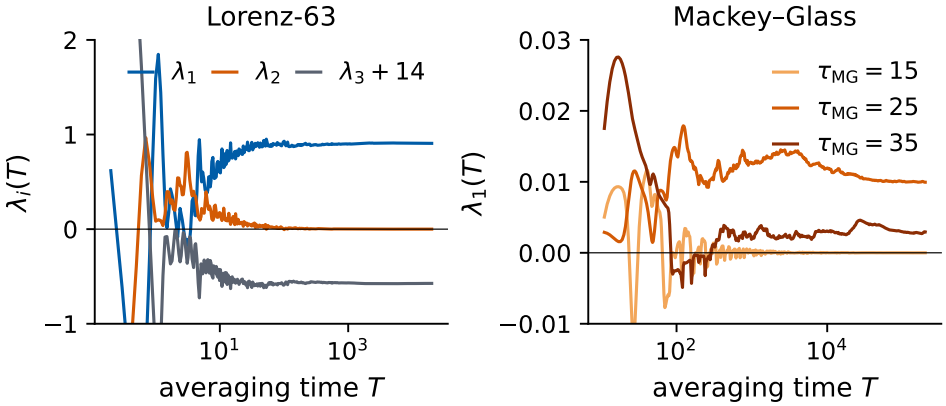


Figure 27: Convergence of the Lyapunov exponents computed by equation (6.8). Left, the running averages $\lambda_i(t)$ of the Lorenz model against the averaging time t (written T in the drawing) on a logarithmic axis. The third exponent is shifted by 14 to fit the scale. Right, the leading exponent $\lambda_1(t)$ of the Mackey–Glass equation for the three delays. The Lorenz values settle within about 100 time units and the Mackey–Glass values within about 10^4.

Table 1 shows the finite-time values $\frac{1}{t}\ln\sigma_i(t)$ of the Lorenz model, computed from the matrix M_t itself, together with the long-time result of this procedure. The finite-time values drift slowly towards the limits, which is why the definition (6.6) takes the long-time limit. The long-time spectrum is $\lambda_1 = 0.907$, $\lambda_2 = 0.0001$ and $\lambda_3 = -14.573$ per time unit, with statistical errors of about 0.001. One direction stretches, the direction of motion is neutral, and one direction contracts strongly.

Table 1: Finite-time exponents $\frac{1}{t}\ln\sigma_i(t)$ of the Lorenz model at $\rho = 28$ for four times t. They are computed from the Jacobian M_t of the flow map and averaged over 100 starting points along the attractor. The last row gives the long-time Lyapunov exponents from the orthonormalisation procedure, and the last column the sum of the three values. Since the smallest singular value of M_t falls below the round-off error of the largest one, the third value is computed as $\frac{1}{t}\ln|\det M_t|$ minus the first two. The last column therefore equals $\frac{1}{t}\ln|\det M_t|$.

time t	$i=1$	$i=2$	$i=3$	sum
1	1.935	−0.794	−14.807	−13.667
2	1.375	−0.362	−14.679	−13.667
5	1.083	−0.127	−14.623	−13.667
10	0.998	−0.062	−14.602	−13.667
long time	0.907	0.0001	−14.573	−13.667

6.6 The sum of the exponents

The last column of Table 1 does not change with time, a consequence of the change of phase-space volumes. The determinant of a matrix is the product of its singular values up to a sign, since $\det\tilde{M}_t = (\det M_t)^2 = \prod_i \sigma_i(t)^2$, so

$$|\det M_t| = \sigma_1(t)\,\sigma_2(t)\cdots\sigma_d(t). \tag{6.9}$$

Taking the logarithm, dividing by t, inserting equation (3.11) for $\det M_t$ and letting $t\to\infty$ gives the sum rule

$$\sum_{i=1}^{d}\lambda_i = \lim_{t\to\infty}\frac{1}{t}\ln|\det M_t| = \lim_{t\to\infty}\frac{1}{t}\int_0^t (\nabla\cdot\mathbf{F})(\mathbf{x}(s))\,ds =: \langle\nabla\cdot\mathbf{F}\rangle. \tag{6.10}$$

The sum of the Lyapunov exponents is the time average of the divergence along the trajectory, the average rate at which the flow changes phase-space volumes. For a conservative system the divergence vanishes and the exponents sum to zero. Hamilton's equations, written as $\dot{\mathbf{x}} = \Omega\,\nabla H$ with the state $\mathbf{x} = (\mathbf{q},\mathbf{p})$ and the matrix $\Omega = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$, have the Jacobian $J = \Omega\,\mathcal{H}$, where $\mathcal{H}$ is the symmetric matrix of second derivatives of H. Since $\Omega^\top = -\Omega$ and $\Omega^2 = -I$, one finds $J^\top\Omega + \Omega J = \mathcal{H}\Omega^\top\Omega + \Omega^2\mathcal{H} = 0$, and with $\dot{M}_t = JM_t$,

$$\frac{d}{dt}(M_t^\top\Omega M_t) = M_t^\top(J^\top\Omega + \Omega J)M_t = 0, \qquad M_t^\top\Omega M_t = \Omega. \tag{6.11}$$

A matrix with this property is called symplectic. Its transpose and the product $\tilde{M}_t = M_t^\top M_t$ are symplectic as well. A symplectic matrix Y satisfies $Y^{-1} = \Omega^{-1}Y^\top\Omega$, so Y^{-1} has the same eigenvalues as Y. The eigenvalues $\sigma_i(t)^2$ of $\tilde{M}_t$ therefore come in pairs σ^2 and $1/\sigma^2$, and the Lyapunov exponents in pairs $\pm\lambda$. Every expanding direction of a Hamiltonian system is matched by a contracting one. For a dissipative system the sum is negative, volumes shrink, and the motion approaches an attractor of zero volume, on which errors can nevertheless grow in some directions. For the Lorenz model the divergence is the constant $-41/3$, so the finite-time sums $\frac{1}{t}\ln|\det M_t|$ equal $-41/3 \approx -13.667$ at every time, and the long-time exponents add up to -13.667. A bounded set can thus stretch errors without end and still shrink volumes. A small ball of initial states is drawn out into a long thin filament along the expanding direction and flattened along the contracting one. The nonlinearity of the field folds the filament back onto the attractor.

6.7 The prediction horizon

An initial error of size δ_0 grows on average as $\delta_0 e^{\lambda_1 t}$. A forecast remains useful until the error reaches a tolerance Δ, which happens after the prediction horizon

$$T_{\text{pred}} = \frac{1}{\lambda_1}\ln\frac{\Delta}{\delta_0}. \tag{6.12}$$

This time is measured in units of the Lyapunov time $T_\lambda \coloneqq 1/\lambda_1$, over which an error grows by the factor e. For the Lorenz model $T_\lambda = 1.1$ time units, or 55 samples at $\Delta t = 0.02$. We test equation (6.12) on 200 pairs of Lorenz solutions that start on the attractor 10^{-8} apart in random directions (Figure 28). Individual pairs scatter because the local stretching rate varies over the attractor. The average of $\ln|\boldsymbol{\delta}(t)|$ over the pairs grows at 0.897 per time unit, close to $\lambda_1 = 0.907$. With the tolerance $\Delta = 1$, small compared with the size of the attractor, the average reaches Δ at $t = 20.4$. Equation (6.12) gives $T_{\text{pred}} = 20.3$ time units, or about 1015 samples. Beyond this time the linearisation no longer applies, and the distance saturates at the size of the attractor, as for two unrelated points of the attractor.

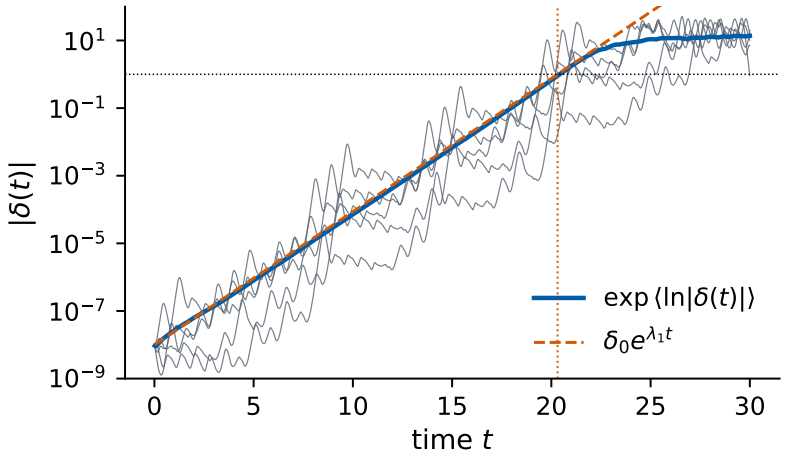


Figure 28: Growth of small errors in the Lorenz model. Grey, the distance $|\boldsymbol{\delta}(t)|$ against time t for five of 200 pairs of solutions that start on the attractor 10^{-8} apart in random directions. Blue, the average $\exp\langle\ln|\boldsymbol{\delta}(t)|\rangle$ over all 200 pairs. Orange dashed, the law $\delta_0 e^{\lambda_1 t}$ with $\lambda_1 = 0.907$. The horizontal dotted line marks the tolerance $\Delta = 1$ and the vertical one the prediction horizon T_{pred} of equation (6.12).

Because of the logarithm in equation (6.12), a thousand times smaller initial error extends the horizon by only $\ln(1000)\,T_\lambda \approx 6.9\,T_\lambda$. In an experiment such a gain in precision may require a new instrument. Even the exact law of a chaotic system forecasts a trajectory only over a finite horizon. A model learned from data adds its own errors to those of the initial state and cannot do better. We therefore state the quality of every forecast as a function of the horizon, measured in Lyapunov times. The limit applies to individual trajectories, whereas averages over the attractor, such as the mean and the variance of x, remain predictable over arbitrarily long times. Fresh measurements reset the error.

6.8 The Kaplan–Yorke dimension

Besides the growth of errors, the Lyapunov exponents determine the number of coordinates needed to locate a state on the attractor. A dissipative flow shrinks every volume to zero, the Lorenz flow by the factor $e^{-41/3}$ per time unit. Its motion still does not collapse onto a point or a curve, since errors along the expanding direction keep growing. The attractor is therefore an object of zero volume that is more than a curve. Its dimension, which need not be an integer, sets how many coordinates a reconstruction of the state from a record needs.

Dimension by counting boxes

A smooth curve of length ℓ is covered by about $N_\varepsilon = \ell/\varepsilon$ boxes of side ε, and a smooth surface of area $\mathcal{A}$ by about $\mathcal{A}/\varepsilon^2$ boxes. The exponent of $1/\varepsilon$ in the number of boxes is the dimension, which defines the box-counting dimension of a bounded set,

$$D_B \coloneqq \lim_{\varepsilon \to 0} \frac{\ln N_\varepsilon}{\ln(1/\varepsilon)}, \qquad \text{so that} \quad N_\varepsilon \propto \varepsilon^{-D_B} \text{ for small } \varepsilon, \tag{6.13}$$

where N_ε is the smallest number of boxes of side ε that cover the set. The definition gives 1 for a curve and 2 for a surface, and for some sets a value between integers, the simplest being the middle-thirds Cantor set (Figure 29). Its construction starts from the interval $[0,1]$ and removes the open middle third, which leaves two intervals of length $1/3$. The middle third of each remaining interval is then removed, and so on without end. After n steps 2^n intervals of length 3^{-n} remain, so boxes of side $\varepsilon = 3^{-n}$ give $N_\varepsilon = 2^n$ and

$$D_B = \frac{\ln 2^n}{\ln 3^n} = \frac{\ln 2}{\ln 3} \approx 0.631 \tag{6.14}$$

for every n. The set has zero length, since its total length $(2/3)^n$ vanishes as $n \to \infty$, and yet it contains uncountably many points. Its dimension lies between that of a point and that of a line. A strange attractor is a set of the same kind, built by stretching and folding.



Figure 29: The first steps of the construction of the middle-thirds Cantor set. After n steps $N = 2^n$ intervals of length 3^{-n} remain, so the number of boxes of side 3^{-n} grows as 2^n and the box-counting dimension is $\ln 2/\ln 3 \approx 0.631$.

Dimension from growth rates

The product $\sigma_1(t)\cdots\sigma_\ell(t)$ is the factor by which a small ℓ-dimensional volume spanned by the ℓ most expanding directions grows, so its growth rate is the partial sum

$$\Sigma_\ell \coloneqq \lambda_1 + \cdots + \lambda_\ell. \tag{6.15}$$

For the Lorenz model lengths grow since $\Sigma_1 > 0$, areas do not shrink since $\Sigma_2 \approx \lambda_1 > 0$, and three-dimensional volumes shrink to zero since $\Sigma_3 = -41/3 < 0$. On a bounded attractor that keeps its size, the volume measured in its own dimension D can neither grow nor shrink on average. Pieces of lower dimension are stretched without limit and pieces of higher dimension are squeezed to nothing. We place the dimension of the attractor where the growth rate changes sign. Since the growth rate is known only at the integers ℓ, we join the values at j and $j+1$ by a straight line [16],

$$\Sigma(D) \coloneqq \Sigma_j + (D - j)\,\lambda_{j+1}, \qquad j \le D \le j+1, \tag{6.16}$$

with j the last index at which $\Sigma_j \ge 0$, and define the dimension as the point where this line crosses zero (Figure 30).

Definition 6.2 (Kaplan–Yorke dimension)**.** With $j \coloneqq \max\{\ell\colon \Sigma_\ell \ge 0\}$, the Kaplan–Yorke dimension is

$$D_{\mathrm{KY}} \coloneqq j + \frac{\Sigma_j}{|\lambda_{j+1}|}. \tag{6.17}$$

The integer part j counts the directions in which the motion is not contracted on balance. The fractional part $\Sigma_j/|\lambda_{j+1}|$ measures how strongly the net expansion of these directions beats the contraction of the next one. Stronger stretching or weaker contraction gives a larger dimension. A stable fixed point has only negative exponents, so $j = 0$ and $D_{\mathrm{KY}} = 0$. A stable limit cycle has $\lambda_1 = 0$, the direction along the cycle, and all other exponents negative, so $j = 1$ and $D_{\mathrm{KY}} = 1$, a curve.

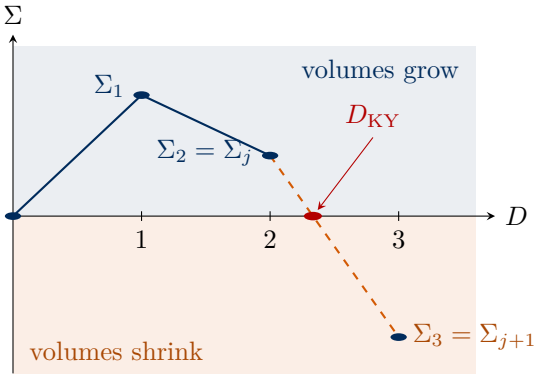


Figure 30: Sketch of the Kaplan–Yorke construction for $j = 2$, with invented values of the exponents. Dots, the partial sums Σ_ℓ, the growth rates of ℓ-dimensional volumes, defined only at whole ℓ. Dashed, the straight line $\Sigma(D)$ between Σ_j and Σ_{j+1}. The dimension D_{KY} is the point where this line crosses zero. For dimensions smaller than D_{KY} volumes grow, for larger ones they shrink.

Dissipative baker's map

The straight line in the definition is a choice, which we test on an example where the dimension can be computed in two independent ways. The dissipative baker's map acts on the unit square of points (q, w) with a constant $0 < c < \frac{1}{2}$ [24],

$$q_{n+1} = 2q_n \bmod 1, \qquad w_{n+1} = \begin{cases} c\,w_n, & q_n < \frac{1}{2}, \\ c\,w_n + \frac{1}{2}, & q_n \geq \frac{1}{2}. \end{cases} \tag{6.18}$$

The left half of the square is stretched horizontally by the factor 2 and squeezed vertically by the factor c into the strip $0 \leq w < c$. The right half goes in the same way into the strip $\frac{1}{2} \leq w < \frac{1}{2} + c$. Each step multiplies areas by $2c < 1$, so the map is dissipative. Its Jacobian matrix is $\mathrm{diag}(2, c)$ at every point, so the growth factors after n steps are 2^n and c^n and the Lyapunov exponents are $\lambda_1 = \ln 2$ and $\lambda_2 = \ln c$ for every orbit. With $\Sigma_1 = \ln 2 > 0$ and $\Sigma_2 = \ln(2c) < 0$, equation (6.17) gives $j = 1$ and

$$D_{\mathrm{KY}} = 1 + \frac{\ln 2}{\ln(1/c)}. \tag{6.19}$$

After n steps the square has become 2^n horizontal strips of full width and height c^n, placed vertically like the intervals of a Cantor set. Covering them with boxes of side $\varepsilon = c^n$ needs c^{-n} boxes along each strip, so $N_\varepsilon = 2^n c^{-n}$ and equation (6.13) gives

$$D_B = \lim_{n\to\infty} \frac{\ln(2^n c^{-n})}{\ln(c^{-n})} = 1 + \frac{\ln 2}{\ln(1/c)}, \tag{6.20}$$

the same value.

The box-counting dimension counts every occupied box once, however rarely the motion visits it. A third dimension weights the boxes by their share of the motion. Let $p_i(\varepsilon)$ be the fraction of time that a long trajectory spends in box i of side ε.

Definition 6.3 (Information dimension)**.** The information dimension of an attractor is

$$D_1 := \lim_{\varepsilon\to 0} \frac{-\sum_i p_i(\varepsilon) \ln p_i(\varepsilon)}{\ln(1/\varepsilon)}. \tag{6.21}$$

The numerator is the Shannon entropy of the visiting frequencies, the information gained on average when one learns which box the state occupies. If all N_ε occupied boxes are visited equally often, $p_i = 1/N_\varepsilon$, the entropy equals $\ln N_\varepsilon$ and $D_1 = D_B$. The entropy of N_ε frequencies never exceeds $\ln N_\varepsilon$, so $D_1 \leq D_B$, and boxes that the motion visits rarely make D_1 smaller than D_B. For the

baker's map the doubling of q visits both halves of the square equally often, so every strip carries an equal share of the motion and $D_1 = D_B$.

We check the three dimensions numerically for $c = 1/3$. The Jacobian of the map, computed by automatic differentiation at points of the attractor, gives the exponents $0.693 = \ln 2$ and $-1.099 = \ln(1/3)$, and with them $D_{\mathrm{KY}} = 1.631$. Counting boxes needs points spread over the attractor with the frequencies of a long trajectory, and a direct iteration of the map in floating-point arithmetic cannot provide them. Doubling q shifts its binary digits one place to the left, so a double-precision number between $\frac{1}{2}$ and 1 becomes 0 after 50 to 53 steps. The computed orbit then ends at the fixed point $(0, 0)$. The map selects its branch by the leading binary digit of q, and for almost every starting q these digits behave like independent tosses of a fair coin. Drawing the branches at random therefore generates a typical trajectory, and 2×10^6 points obtained in this way give the counts of Figure 31. The coordinate q holds the digits not used yet, so it is uniform on $[0, 1)$ and independent of w. We therefore count boxes along w, and the interval in q adds 1 to every dimension. Both the number of occupied boxes and the entropy of the visiting frequencies grow with the slope $\ln 2/\ln 3$ against $\ln(1/\varepsilon)$, and the fits give $D_B = 1.631$ and $D_1 = 1.631$. For $c = 1/3$ the attractor is a line times a Cantor set of dimension 0.631, and $D_{\mathrm{KY}} = D_B = D_1 = 1 + \ln 2/\ln 3$. In this example the straight-line interpolation of equation (6.17) gives the correct dimension.

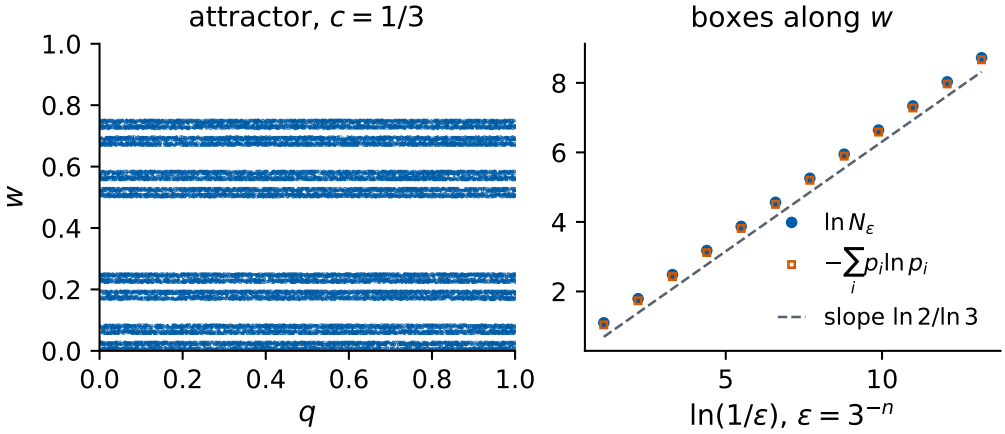


Figure 31: The dissipative baker's map with $c = 1/3$. Left, 2×10^4 points of the attractor in the plane of q and w, horizontal lines arranged in w like a Cantor set. Right, the logarithm $\ln N_\varepsilon$ of the number of occupied boxes (dots) and the entropy $-\sum_i p_i \ln p_i$ of the visiting frequencies (squares) against $\ln(1/\varepsilon)$, for boxes of side $\varepsilon = 3^{-n}$ along w, $n = 1, \ldots, 12$. The dashed line has slope $\ln 2/\ln 3$. The two slopes are the dimension of the Cantor set in w, and the interval in q adds 1 to both.

The Lorenz attractor

For the Lorenz model the exponents of Table 2 give $\Sigma_1 = 0.907$, $\Sigma_2 = 0.907$ and $\Sigma_3 = -13.667$, so $j = 2$ and

$$D_{\mathrm{KY}} = 2 + \frac{\Sigma_2}{|\lambda_3|} = 2.06. \tag{6.22}$$

The dimension is only slightly above two, because the strong contraction along the third direction, with $|\lambda_3|$ about sixteen times λ_1, nearly compensates the stretching. The attractor is almost a surface. Since it is bounded, the stretched sheet is folded onto itself again and again into thin layers, and the fraction 0.06 beyond two describes this fine layering (Figure 32). As a heuristic, about $\lceil D_{\mathrm{KY}} \rceil = 3$ independent numbers locate a state on the Lorenz attractor, where $\lceil \cdot \rceil$ rounds up to the next integer.

Table 2: Kaplan–Yorke dimension of the Lorenz attractor. Lyapunov exponents λ_ℓ per time unit and their partial sums Σ_ℓ.

ℓ	λ_ℓ	$\Sigma_\ell = \lambda_1 + \cdots + \lambda_\ell$
1	0.907	0.907
2	0.0001	0.907
3	−14.573	−13.667



point, 0 curve, 1 surface, 2 Lorenz, 2.06

Figure 32: Sets of increasing dimension, from left to right a point, a curve, a surface and a schematic cross-section of the Lorenz attractor. The attractor is a sheet folded into many thin layers, with a dimension slightly above two. The layers are drawn for illustration and do not show computed positions.

Dimension and the degree of chaos

To follow how the Mackey–Glass dimension changes with the dynamics, we store the history of the delay equation on a grid of step 0.1. The equation then becomes a map with 151, 251 and 351 variables for the three delays, and the same orthonormalisation procedure gives the leading exponents (Table 3). Exponents whose magnitude is below three standard errors are set to zero before equation (6.17) is applied, since their sign is not determined. At $\tau_{\mathrm{MG}} = 15$ the leading exponent is zero and all others are negative, so the attractor is a limit cycle with $D_{\mathrm{KY}} = 1.00$. At $\tau_{\mathrm{MG}} = 25$ one exponent is positive, one is zero and the next two are small and negative, so $j = 3$, $\Sigma_3 = 0.00434$ and $D_{\mathrm{KY}} = 3 + 0.00434/0.0275 = 3.16$. At $\tau_{\mathrm{MG}} = 35$ the positive exponent is smaller and two further exponents are zero within their errors. The first clearly negative exponent, $\lambda_4 = -0.00726$, is weaker, so $D_{\mathrm{KY}} = 3 + 0.00294/0.00726 = 3.41$. As the delay grows, the motion fills more dimensions, and at $\tau_{\mathrm{MG}} = 35$ this happens through weaker contraction while the leading exponent decreases. Dimension and separation rate measure different aspects of a chaotic motion. The Lorenz attractor, with one positive exponent and very strong contraction, lies just above two.

Table 3: Leading Lyapunov exponents per time unit and Kaplan–Yorke dimension of the Mackey–Glass attractors. Entries marked 0 are zero within three standard errors and are set to zero in equation (6.17).

delay	λ_1	λ_2	λ_3	λ_4	D_{KY}
$\tau_{\mathrm{MG}} = 15$	0	−0.0127	−0.0269	−0.0798	1.00
$\tau_{\mathrm{MG}} = 25$	0.00995	0	−0.00561	−0.0275	3.16
$\tau_{\mathrm{MG}} = 35$	0.00294	0	0	−0.00726	3.41

A higher dimension requires more numbers to locate the state, and a reconstruction from a record then needs more delay coordinates and a longer window. A larger leading exponent shortens the Lyapunov time and with it the prediction horizon.

Limits of the Kaplan–Yorke dimension

The Kaplan–Yorke conjecture states that D_{KY} equals the information dimension of the attractor. The conjecture holds for the baker's map and is confirmed for many other systems, without being proven in general [14]. Both dimensions refer to the frequencies with which a typical trajectory visits the attractor. Exceptional trajectories visit it differently, for example the fixed point $(0, 0)$ of the baker's map, which stays in a single box. The dimension also depends on the contracting exponents, in particular λ_{j+1}. These are the hardest to estimate from a measured record, since a finite record contains few pairs of states that are close along a strongly contracting direction. A Kaplan–Yorke

dimension computed from a record carries an uncertainty that the exponents of the equations do not have.

7 The state from one measured variable

The exponents of the laser cannot be computed from equations, and a forecast of its record requires its state. The record provides one number per sample, and one coordinate does not determine a three-dimensional state.

7.1 One value does not fix the state

Figure 33 shows the problem on the Lorenz record. The coordinate x takes the value 4.62 at sample 266 and the value 4.64 at sample 325. The two values are nearly equal, and yet in the first case x rises by 0.63 in the next step and in the second case it falls by 1.24. A rule that receives only the present value s_k has to give the same forecast in both cases and is wrong in at least one of them. By equation (1.4), the value of x is one coordinate of the three-dimensional state. One value of x belongs to states with different y and z that move in different directions.

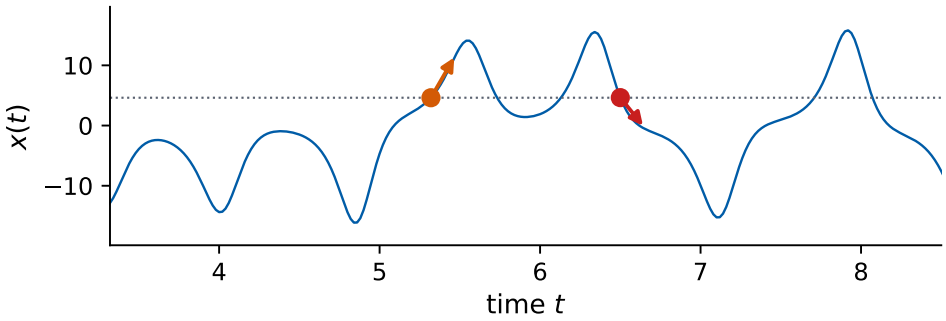


Figure 33: One value, two futures. A stretch of the Lorenz coordinate $x(t)$ against time t. The dotted line marks one value of x, which the signal reaches twice, once rising (orange dot) and once falling (red dot). The arrows show the course of the signal over the next eight samples.

7.2 Past values carry the hidden coordinates

The missing information is contained in the past of the record. Consider a harmonic oscillator with the state (x, v), position and velocity, of which we measure only the position $x(t) = \cos t$. The difference of two consecutive samples divided by the sampling interval approximates the velocity,

$$v(k\Delta t) \approx \frac{x_k - x_{k-1}}{\Delta t}, \tag{7.1}$$

with an error of at most 0.025 for $\Delta t = 0.05$. The pair (x_k, x_{k-1}) therefore carries the same information as the state. So does almost any pair $(x_k, x_{k-\kappa})$ with a fixed lag κ, except for lags of a multiple of half a period, at which the two values are equal or opposite. Plotted in the plane, the pairs trace a deformed copy of the circle that the true state traces (Figure 34).

For the Lorenz model, the first component of equation (4.6) gives $y = x + \dot{x}/\sigma$. Differentiating this relation and using the second component gives $z = \rho - (\dot{y} + y)/x$ with $\dot{y} = \dot{x} + \ddot{x}/\sigma$. The coordinate x together with its first two time derivatives therefore determines the full state (x, y, z) wherever $x \neq 0$. Time derivatives can be approximated by differences of past samples, and in practice we use the past samples themselves, which are less sensitive to noise than their differences. The hidden coordinates are thus contained in the successive values of the measured one.

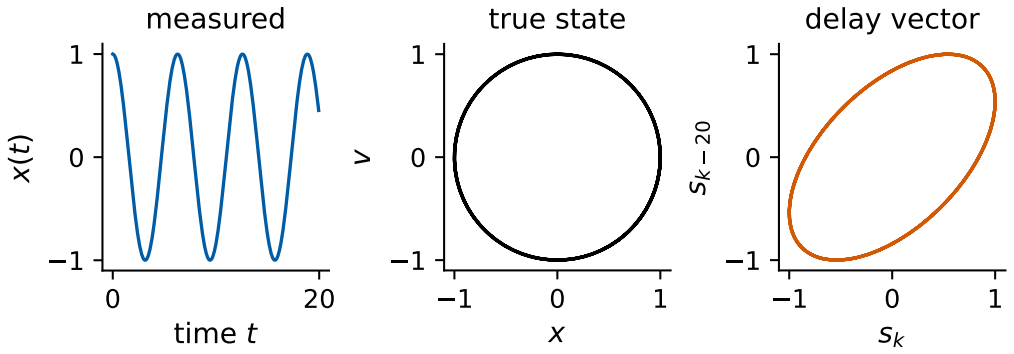


Figure 34: Phase space from one coordinate of a harmonic oscillator sampled at $\Delta t = 0.05$. Left, the measured position $x(t)$ against time. Middle, the true state, velocity v against position x. Right, the pairs (s_k, s_{k-20}) of the measured samples $s_k = x(k\Delta t)$, a lag of 20 samples, which trace a deformed copy of the same closed curve.

7.3 Delay vectors and the delay map

When the law is unknown, so is the combination of past values that recovers each coordinate, and we collect several past values of the record into one vector.

Definition 7.1 (Delay vector)**.** For a record s_k, an integer delay $\kappa \geq 1$ and an integer embedding dimension m, the delay vector at step k is

$$\mathbf{u}_k := (s_k,\, s_{k-\kappa},\, s_{k-2\kappa},\, \ldots,\, s_{k-(m-1)\kappa}) \in \mathbb{R}^m. \tag{7.2}$$

The delay vector reuses the symbol $\mathbf{u}_k$ of the window (2.2), which is the special case $\kappa = 1$, $m = L$. For the Lorenz model with the measured coordinate x and $m = 3$,

$$\mathbf{x}_k = (x(k\Delta t),\, y(k\Delta t),\, z(k\Delta t)), \qquad \mathbf{u}_k = (x(k\Delta t),\, x((k-\kappa)\Delta t),\, x((k-2\kappa)\Delta t)). \tag{7.3}$$

The state contains three different quantities at one time, and the delay vector contains one quantity at three times. Both can carry the same information. Figure 35 shows that the two lobes of the Lorenz attractor reappear in the delay vectors built from x alone, deformed but recognisable.

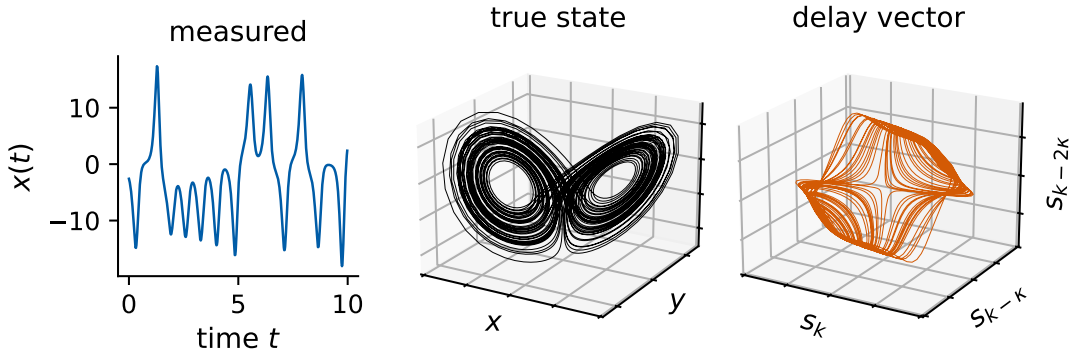


Figure 35: Phase space from one coordinate of the Lorenz model. Left, the measured coordinate $x(t)$ against time t. Middle, the true state (x, y, z). Right, the delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$ of the record $s_k = x(k\Delta t)$ with $\kappa = 10$ samples, which show the two lobes in deformed coordinates.

Each entry of the delay vector is a measurement of an earlier state. Every earlier state is determined by the present one through the flow map, $\mathbf{x}_{k-j\kappa} = \varphi_{-j\kappa\Delta t}(\mathbf{x}_k)$ by equation (3.2). For noise-free data the j-th entry is therefore

$$s_{k-j\kappa} = h(\mathbf{x}_{k-j\kappa}) = h(\varphi_{-j\kappa\Delta t}(\mathbf{x}_k)), \qquad j = 0, 1, \ldots, m-1, \tag{7.4}$$

a function of the single present state $\mathbf{x}_k$. The whole delay vector is the value of one map of phase space into $\mathbb{R}^m$.

Definition 7.2 (Delay map)**.** Let A be the attractor on which the states lie. The delay map $\mathcal{E}\colon A \to \mathbb{R}^m$ assigns to every state the delay vector that a noise-free record would produce,

$$\mathcal{E}(\mathbf{x}) \coloneqq \Big(h(\mathbf{x}),\, h(\varphi_{-\kappa\Delta t}(\mathbf{x})),\, \ldots,\, h(\varphi_{-(m-1)\kappa\Delta t}(\mathbf{x}))\Big), \qquad \mathbf{u}_k = \mathcal{E}(\mathbf{x}_k). \tag{7.5}$$

Its image $\mathcal{E}(A) \subset \mathbb{R}^m$, the cloud of delay vectors, is the reconstructed attractor.

The delay map contains the unknown measurement function and the unknown flow, so it is never evaluated (Figure 36). The record supplies its values $\mathbf{u}_k$, and the mathematical question concerns its properties.

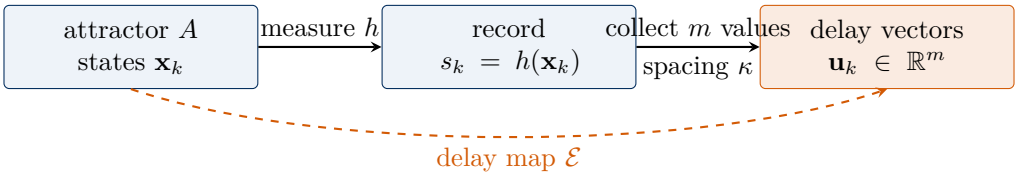


Figure 36: From the hidden state to the delay vector. The measurement h turns each state into one stored number, and collecting m stored numbers spaced by κ samples gives the delay vector. The composite map from the state to the delay vector is the delay map $\mathcal{E}$ of equation (7.5).

7.4 Embedding and Takens' theorem

A forecast from past values is possible when the delay vectors can replace the state, which requires that $\mathcal{E}$ loses no information about it. A map $\mathcal{E}$ of the attractor into $\mathbb{R}^m$ is called an embedding when it is one-to-one and continuous with a continuous inverse on its image. Different states then give different delay vectors,

$$\mathbf{x} \neq \mathbf{x}' \quad \Longrightarrow \quad \mathcal{E}(\mathbf{x}) \neq \mathcal{E}(\mathbf{x}'), \qquad \mathbf{x}, \mathbf{x}' \in A, \tag{7.6}$$

nearby delay vectors belong to nearby states, and the state is a function of the delay vector,

$$\mathbf{x}_k = \mathcal{E}^{-1}(\mathbf{u}_k). \tag{7.7}$$

The image $\mathcal{E}(A)$ is then a faithful copy of the attractor. Distances and angles are distorted, while the lobes and holes of the attractor survive and neighbouring states remain neighbouring delay vectors. A single measured value, $m = 1$, violates the condition, as the two nearly equal values with different futures in Figure 33 show. The condition for a sufficient number of coordinates involves the box-counting dimension D_B of the attractor, defined in equation (6.13).

Theorem 7.3 (Takens' embedding theorem and its extension)**.** Let the motion take place on a bounded and closed set A of phase space. Let the delay map (7.5) be built from the smooth and invertible flow map $\varphi_{\kappa\Delta t}$ and a smooth measurement function h.

(i) If A lies on a smooth surface of integer dimension D_{M}, then for generic pairs of flow map and measurement function the delay map with $m \geq 2D_{\mathrm{M}} + 1$ coordinates is an embedding [17].

(ii) If A has the box-counting dimension D_B and $m > 2D_B$, then for almost every smooth measurement function the delay map is one-to-one on A [18]. This statement requires in addition mild conditions on the periodic orbits of low period.

In statement (i) the word generic means that the property holds for an open and dense set of pairs. In statement (ii) almost every is meant in the sense of prevalence, a notion of almost every for spaces of functions introduced for this purpose in Ref. [18]. Both exclude exceptional cases that an arbitrarily small change destroys. Examples are a constant measurement function, which cannot distinguish any states, and a delay equal to the period of an orbit, which makes all coordinates equal on that orbit. The bound $m > 2D_B$ has a geometric origin, since two pieces of a set of dimension D

placed generically in $\mathbb{R}^m$ intersect in a set of dimension $2D - m$. They therefore meet when $2D \geq m$ and miss each other when $m > 2D$. A closed curve, $D = 1$, drawn in the plane, $m = 2$, can cross itself, while in three dimensions a small perturbation removes any crossing. Figure 37 shows this for the periodic signal

$$s(t) = \cos t + 0.6\cos(2t + 0.7), \tag{7.8}$$

sampled with 400 samples per period and embedded with a delay of a quarter period, $\kappa = 100$ samples. Its attractor is a closed curve, with $D_B = 1$. In two delay coordinates the curve crosses itself, and at the crossing two different states, samples 205 and 294 of the period, have delay vectors only 0.013 apart. There $\mathcal{E}^{-1}$ does not exist, and the delay vector does not determine on which branch the signal will continue. In three coordinates, $m = 3 > 2D_B$, the same two states are 1.819 apart, every delay vector belongs to one state, and the future is unique. As in Figure 3, the crossing signals missing coordinates.

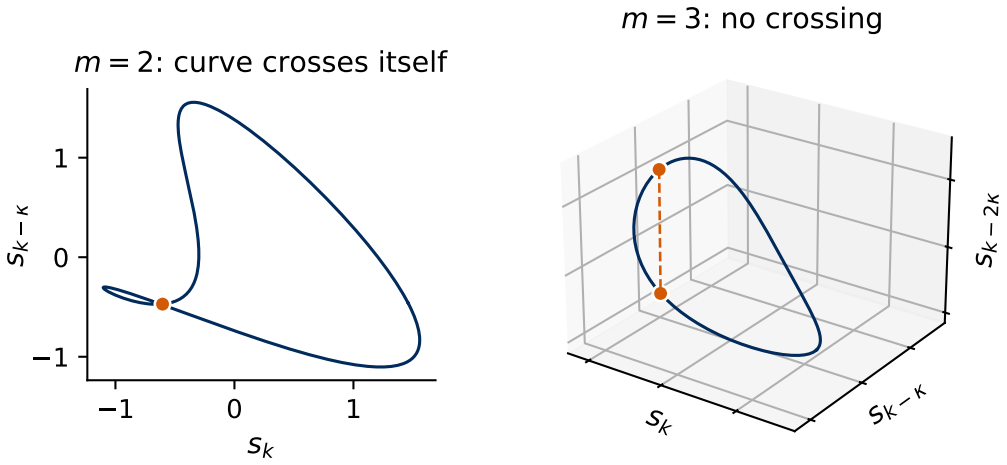


Figure 37: Delay vectors of the periodic signal $s(t) = \cos t + 0.6\cos(2t + 0.7)$ with a delay of a quarter period. Left, $m = 2$, the curve $(s_k, s_{k-\kappa})$ crosses itself (orange dot), so two different states have the same delay vector. Right, $m = 3$, the curve $(s_k, s_{k-\kappa}, s_{k-2\kappa})$ has no crossing, and the dashed orange segment joins the two states that collided in the plane.

Since the box-counting dimension is hard to estimate reliably from a finite record, the Kaplan–Yorke dimension serves as a practical estimate of it. Substituting D_{KY} for D_B gives a guide, without the guarantee of the theorem. For the Lorenz attractor $2D_{\mathrm{KY}} = 4.12$, so $m = 5$ would be sufficient if the two dimensions agreed. The bound is sufficient and often larger than necessary. Since $\lceil D_{\mathrm{KY}} \rceil$ numbers are needed to locate a state on the attractor, the dimension that a particular record requires often lies between $\lceil D_{\mathrm{KY}} \rceil$ and $\lfloor 2D_{\mathrm{KY}} \rfloor + 1$, where $\lfloor \cdot \rfloor$ rounds down. This range is a heuristic without a theorem behind it, and the dimension has to be found from the record itself.

7.5 The forecast map

An embedding recovers the state, and the state fixes the future, so the forecasting task of equation (2.2) has a firm basis. When $\mathcal{E}$ is an embedding, the flow induces a dynamics of the delay vectors. Starting from a delay vector, the inverse map gives the state, the flow map advances it by one sample, and the delay map returns to delay coordinates,

$$\mathbf{u}_{k+1} = \mathcal{E}(\varphi_{\Delta t}(\mathbf{x}_k)) = (\mathcal{E} \circ \varphi_{\Delta t} \circ \mathcal{E}^{-1})(\mathbf{u}_k). \tag{7.9}$$

The delay vector spans

$$L = (m - 1)\kappa + 1 \tag{7.10}$$

consecutive samples, so a window of at least this length contains all its entries. Followed by a measurement, this chain of maps leads from a delay vector to a future measured value. It defines the object that every forecasting model of a deterministic record estimates.

Definition 7.4 (Forecast map)**.** Let A be an invariant set of states, a set that the flow carries into itself, $\varphi_t(A) = A$ for every t, such as the attractor. Let h be the measurement function, φ_t the deterministic flow map and $\mathcal{E}\colon A \to \mathbb{R}^m$ the noise-free delay map of equation (7.5), assumed one-to-one on A. Let the horizon $\tau \geq 1$ be an integer number of samples. The forecast map for the horizon τ is

$$G_\tau\colon \mathcal{E}(A) \to \mathbb{R}, \qquad G_\tau \coloneqq h \circ \varphi_{\tau\Delta t} \circ \mathcal{E}^{-1}. \tag{7.11}$$

The composition is read from right to left. The inverse delay map $\mathcal{E}^{-1}$ takes a delay vector to the state that produced it, and the flow map $\varphi_{\tau\Delta t}$ advances this state by τ samples. The measurement function h then turns the advanced state into the value that the instrument records. For a noise-free record, $\mathbf{u}_k = \mathcal{E}(\mathbf{x}_k)$, and

$$G_\tau(\mathbf{u}_k) = h\Big(\varphi_{\tau\Delta t}(\mathcal{E}^{-1}(\mathbf{u}_k))\Big) = h(\varphi_{\tau\Delta t}(\mathbf{x}_k)) = h(\mathbf{x}_{k+\tau}) = s_{k+\tau}. \tag{7.12}$$

The forecast map gives the measured value τ samples ahead from the present delay vector alone (Figure 38).

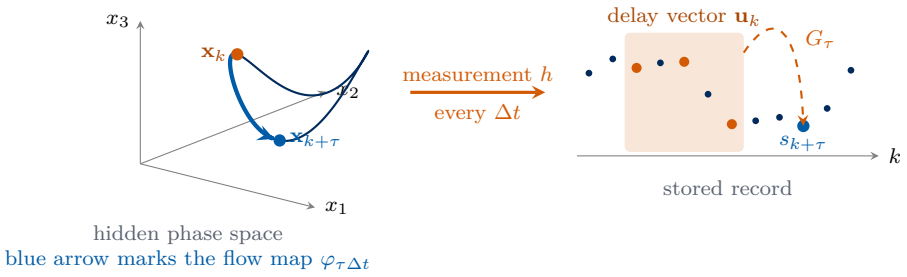


Figure 38: The forecast map. The hidden state $\mathbf{x}_k$ is carried by the flow map to $\mathbf{x}_{k+\tau}$ (blue arrow), and the measurement turns the states into the stored record. The forecast map G_τ of equation (7.11) goes directly from the delay vector $\mathbf{u}_k$ (orange samples, here $m = 3$) to the future value $s_{k+\tau}$ (dashed). It does not pass through the hidden phase space.

The map G_τ is defined only on admissible delay vectors, the points of $\mathcal{E}(A)$ that noise-free states of the invariant set produce. A measured delay vector contains noise and in general lies slightly off this set, where G_τ is not defined. A forecast from noisy inputs needs an estimate of G_τ extended to a neighbourhood of $\mathcal{E}(A)$, and the noise sets the error of this estimate. The inverse $\mathcal{E}^{-1}$ exists only where $\mathcal{E}$ is one-to-one. When the assumption that $\mathcal{E}$ is one-to-one fails, as at the crossing of Figure 37, the composition (7.11) is undefined at the shared delay vectors. A function from delay vectors to future values exists there only if both states lead to the same future measured value.

The forecast map is unknown, because h, φ and $\mathcal{E}$ are unknown, and it can be estimated from the pairs $(\mathbf{u}_k, s_{k+\tau})$ that the record itself provides. Forecasting a deterministic system from one measured variable becomes a regression problem with a finite-dimensional input, whose window must reach at least the L samples of equation (7.10) back. Such a window contains the delay vector as every κ-th entry. The model $\tilde{G}_\tau$ of equation (2.2) estimates the forecast map G_τ applied to the delay vector selected from the window.

The theorem says nothing about the form of G_τ, which is nonlinear even for simple systems, and nothing about noise, which limits how finely nearby delay vectors can be distinguished. Its genericity assumption can also fail for a physical measurement. In the Maxwell–Bloch description the laser intensity $h \propto E^2$ takes the same value on the two states related by the symmetry $(E, P, W) \mapsto (-E, -P, W)$. The intensity is an exceptional measurement function in the sense of the theorem, and no delay vector of intensities recovers the sign of the field. The delay map built from the intensity is not one-to-one, and the inverse $\mathcal{E}^{-1}$ in equation (7.11) does not exist. A reconstruction from the intensity can at best embed the system with each pair of symmetric states identified as one point. The laser equations are unchanged by the symmetry, so the two states of a pair remain symmetric partners at all later times and produce the same future intensity. A map from intensity

delay vectors to the future intensity still exists and assigns to each delay vector the common future intensity of the two states behind it.

The contracting exponents, and with them D_{KY}, remain poorly determined by a reconstruction, even for a chaotic attractor. A finite record contains few pairs of states close along a strongly contracting direction.

The theorem guarantees an embedding for generic measurement functions and a sufficient number of coordinates, with exceptional delays such as those resonant with periodic orbits excluded. For a finite and noisy record some delays work much better than others, and the sufficient number of coordinates often exceeds what is needed.

8 Delay and dimension from the record

The embedding theorem promises a reconstruction for almost every delay and a sufficient number of coordinates. For the laser and Lorenz records it names neither the delay nor the number of coordinates, and both have to be read from the record.

8.1 Effect of the delay

We choose the delay first, because the number of coordinates is tested with a fixed delay. Figure 39 shows what the delay does to the reconstruction of the Lorenz attractor. A very short delay makes successive coordinates nearly equal, $s_{k-\kappa} \approx s_k$, so all delay vectors lie close to the diagonal and noise dominates their differences. A very long delay makes the coordinates nearly unrelated on a chaotic attractor, since the dynamics has lost the memory of the earlier value, and the reconstruction becomes a tangle. With $\kappa = 10$ samples the two lobes reappear, smoothly deformed. With $\kappa = 89$, the first zero of the ACF, the geometry is lost. The ACF measures only linear dependence, and its first zero, the classical choice of the delay, comes far too late for this record. A good delay makes the coordinates as different as possible while the dynamics still relates them. Finding it requires a measure of dependence that sees every kind of relation, linear or nonlinear.

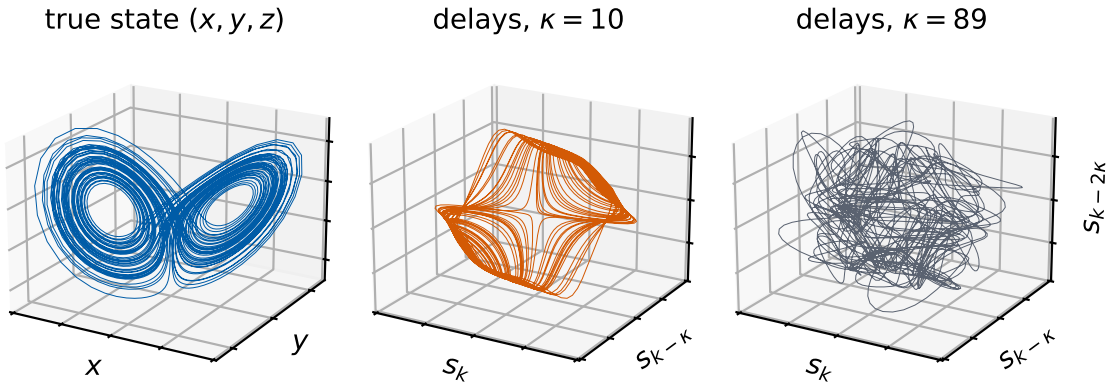


Figure 39: The Lorenz attractor and two reconstructions from the record $s_k = x(k\Delta t)$. Left, the true state (x, y, z). Middle, the delay vectors $(s_k, s_{k-\kappa}, s_{k-2\kappa})$ with $\kappa = 10$ samples. Right, the same with $\kappa = 89$ samples, the first zero of the autocorrelation.

8.2 Mutual information

In probability theory the uncertainty of a random variable a that takes the values i with the probabilities p_i is measured by its entropy,

$$H(a) := -\sum_i p_i \ln p_i, \tag{8.1}$$

with the convention $0 \cdot \ln 0 = 0$, so that impossible values contribute nothing. The symbol H here denotes the entropy and is unrelated to the Hamiltonian. With the natural logarithm it is measured in nats. A variable with n equally likely values has $H = \ln n$, the largest entropy possible for n values, and a variable with a certain value has $H = 0$. Consider a second variable b with the values j, joint probabilities p_{ij} and single probabilities $p_i = \sum_j p_{ij}$ and $p_j = \sum_i p_{ij}$. The uncertainty of a that remains once b is known is the conditional entropy $H(a|b) := -\sum_{i,j} p_{ij} \ln(p_{ij}/p_j)$. The mutual information (MI) is the reduction of uncertainty,

$$\mathrm{MI} := H(a) - H(a|b) = \sum_{i,j} p_{ij} \ln \frac{p_{ij}}{p_i\, p_j}, \tag{8.2}$$

where the second form follows by inserting the definitions, and the convention $0 \cdot \ln 0 = 0$ again removes empty cells. The second form is symmetric in a and b and never negative, by Gibbs' inequality $\sum_{i,j} p_{ij} \ln(p_{ij}/q_{ij}) \geq 0$ for any two probability distributions p and q, applied with $q_{ij} = p_i p_j$. It vanishes if and only if $p_{ij} = p_i p_j$ for all cells, that is, if and only if the two variables are independent. It is large when one value nearly determines the other, through a straight line, a parabola such as $b = a^2$, or any other curve. For $b = a$ the conditional entropy vanishes and $\mathrm{MI} = H(a)$.

For a record we take the two values a delay κ apart, $a = s_{k-\kappa}$ and $b = s_k$, and estimate the probabilities from a histogram. The range of the record is cut into 16 intervals of equal width. For every pair $(s_{k-\kappa}, s_k)$ we note the interval i of the first value and the interval j of the second. The fraction of pairs that fall into the cell (i, j) estimates $p_{ij}(\kappa)$, and equation (8.2) then gives one number $\mathrm{MI}(\kappa)$ for each delay. To compare all delays on an equal footing, we compute the curve $\mathrm{MI}(\kappa)$ for $\kappa = 0, \ldots, 100$ from the same number of pairs at every delay. These are the 2400 pairs that the first 2500 samples provide at the largest delay. With these 2400 pairs, 16 intervals per value give 256 cells with about 9 pairs each on average. With so few counts per cell the estimate is biased upwards, because random fluctuations of the counts look like dependence, and the bias is larger for finer grids. A coarser grid reduces the bias but blurs the relation between the two values, so the number of intervals is a compromise. Absolute values of $\mathrm{MI}(\kappa)$ depend on the grid, while the choice of the delay uses only the shape of the curve against κ.

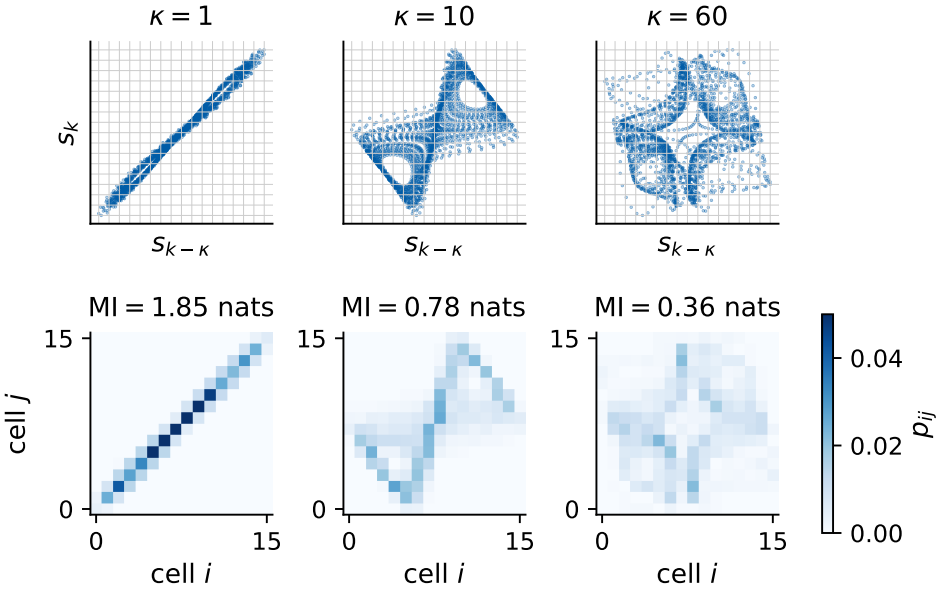


Figure 40: Mutual information of the Lorenz record from a histogram. Top row, the pairs $(s_{k-\kappa}, s_k)$ for the delays $\kappa = 1$, 10 and 60 samples, with the grid of 16×16 cells. Bottom row, the fraction p_{ij} of pairs in each cell (i, j) as a shade of blue, with $\mathrm{MI}(\kappa)$ computed from it above each panel.

At $\kappa = 1$ the pairs of the Lorenz record lie close to the diagonal (Figure 40), since x hardly changes within one sample, and the mutual information is large, 1.85 nats. A second coordinate of this kind repeats the first. At $\kappa = 10$ the pairs spread over a curved band, with 0.78 nats. The two values still

depend on each other, so the reconstruction stays smooth, while each adds information that the other lacks. At $\kappa = 60$ the histogram has spread further, with 0.36 nats, although the two values are still dependent. At $\kappa = 0$ the two values coincide, so MI(0) in Figure 41 is the entropy of the binned record, 2.62 nats. It lies close to the largest possible value $\ln 16 = 2.77$ nats for 16 intervals. The curve falls fast over the first few samples and has a shallow first local minimum of 0.77 nats at $\kappa = 10$ samples, or 0.2 time units. It then rises slightly and keeps falling at larger delays, with further shallow minima (left panel of Figure 43). The minimum differs slightly from the value of the histogram of Figure 40 at the same delay, because the curve uses the same 2400 pairs at every delay. All of this happens long before the ACF crosses zero.

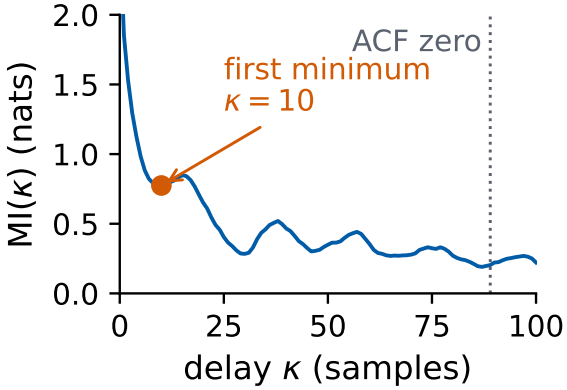


Figure 41: Mutual information MI(κ) between $s_{k-\kappa}$ and s_k of the Lorenz record, in nats, against the delay κ in samples. The dot marks the first local minimum, $\kappa = 10$, and the dotted line the first zero of the ACF, at 89 samples.

The delay is taken at the first local minimum of MI(κ) [19]. At small delays $s_{k-\kappa}$ nearly fixes s_k, and as the delay grows the second value adds more and more information that the first lacks, so MI(κ) falls. For an oscillating signal the fall does not continue smoothly. The crossing example of Figure 37 used a delay of a quarter period, at which the two coordinates of a sine wave are a sine and a cosine. These separate the rising and the falling branch best, while at half a period the second value is again fixed by the first. In the Lorenz record the trajectory turns around a lobe at about 1.62 turns per time unit, the rate given by the linearisation at $C_\pm$. The period is about 31 samples, so a quarter turn takes about 8 samples, and the first dip of MI(κ) near $\kappa = 10$ lies close to this quarter turn. The first local minimum is thus the earliest delay at which the second coordinate is least redundant while the turning motion still ties it to the first. At much larger delays the stretching and folding of the chaotic flow spread the values over the whole attractor and lower MI(κ) further. This low dependence is of no use, since finite data cannot resolve the folded structure.

For the Lorenz record the first dip is shallow, and the curve in Figure 41 is almost flat around $\kappa = 10$. Its location depends on details of the estimator, such as the number of intervals and the number of pairs. The rule is a heuristic, and a delay plot drawn for a few delays, as in Figure 39, checks that the choice serves its purpose.

8.3 False nearest neighbours

With the delay fixed, we choose the number of coordinates next. If m is too small, the reconstruction folds the attractor onto itself as in the left panel of Figure 37. Points from distant parts of the attractor then appear next to each other, and adding one coordinate separates them. In two delay coordinates the Lorenz orbit (Figure 42) passes through the same region from two different lobes. Two points of different lobes lie there at the distance 0.20. With the third coordinate added they are 17.8 apart. The two points are different states with different futures, and a forecast that relied on the neighbour would be wrong. Such a pair is called a pair of false nearest neighbours (FNN), and counting them gives a criterion for m [20].

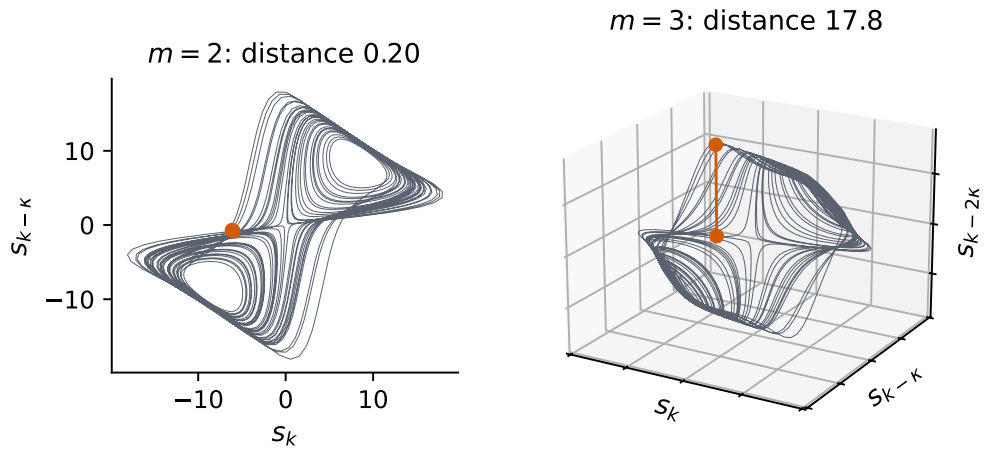


Figure 42: A pair of false nearest neighbours of the Lorenz record with $\kappa = 10$. Left, the delay vectors $(s_k, s_{k-\kappa})$ in $m = 2$, where the two orange points are nearest neighbours at the distance 0.20. Right, the same points with the third coordinate $s_{k-2\kappa}$ added, where they lie on different lobes, 17.8 apart.

We run the procedure over $m = 1, 2, 3, \ldots$ with κ fixed by the mutual information. For a trial m we form all delay vectors $\mathbf{u}_k^{(m)}$ of equation (7.2). For every k we find the nearest neighbour $\mathbf{u}_j^{(m)}$, the delay vector at the smallest Euclidean distance

$$r_k = \|\mathbf{u}_k^{(m)} - \mathbf{u}_j^{(m)}\| = \Big(\sum_{i=0}^{m-1} \big(s_{k-i\kappa} - s_{j-i\kappa}\big)^2 \Big)^{1/2}, \tag{8.3}$$

searched only among $|k - j| > 50$, so that j belongs to a different pass of the trajectory than k. Here r_k is a distance, unrelated to the Pearson coefficient r. Adding one coordinate gives the pair the new entries $s_{k-m\kappa}$ and $s_{j-m\kappa}$ and the new difference $\Delta_k = |s_{k-m\kappa} - s_{j-m\kappa}|$. The neighbour is counted as false when the new coordinate alone pulls the pair apart far beyond its previous distance,

$$\Delta_k > 15\, r_k. \tag{8.4}$$

We compute the fraction FNN(m) of indices k with a false neighbour and increase m until this fraction drops under 5%. The first such m is the estimated embedding dimension. The factor 15 and the threshold of 5% are heuristic choices. The method of Ref. [20] includes a second, absolute test for neighbours that are far apart from the start. It counts a pair as false also when $\sqrt{r_k^2 + \Delta_k^2}$ exceeds twice the standard deviation of the record. The absolute test matters for records that contain noise.

8.4 Delay and dimension of the five records

Mutual information and false nearest neighbours were developed on the Lorenz record, whose dimension is known from its equations. The other records test them where the answer is less clear, and Figure 43 applies both criteria to all five. Since both choose the length of the input window of a forecasting model, we compute them on the training block alone, the first 2500 samples. The validation and test blocks never enter them. The first minimum of the mutual information gives $\kappa = 10$ samples for the Lorenz record and 2 for the laser. For the Mackey–Glass records it gives 8, 15 and 21 at $\tau_{\mathrm{MG}} = 15$, 25 and 35. The ratio test drops under 5% at $m = 3$ for the laser and at $m = 4$ for the Lorenz record and the two chaotic Mackey–Glass records. For the periodic Mackey–Glass record it drops at $m = 2$, since a closed curve drawn in the plane crosses itself only at isolated points. For the Lorenz record the fraction drops from 0.950 at $m = 1$ to 0.173, 0.064 and 0.035 at $m = 2$, 3 and 4. The chaotic Mackey–Glass records need only four coordinates although the state of a delay equation is infinite-dimensional, since only the dimension of the attractor matters for the reconstruction. The value of m need not equal the number of state variables, since the Lorenz model has three variables and its record needs $m = 4$ here. For the laser the value $m = 3$ says nothing by itself about the three Maxwell–Bloch variables. With the absolute test included, the Lorenz fraction

stays between 2.8% and 6.7% for $m = 4$ to 10 and grows again at the largest m. The renewed growth is a finite-record effect of points in sparsely visited regions, whose nearest neighbours are far away.

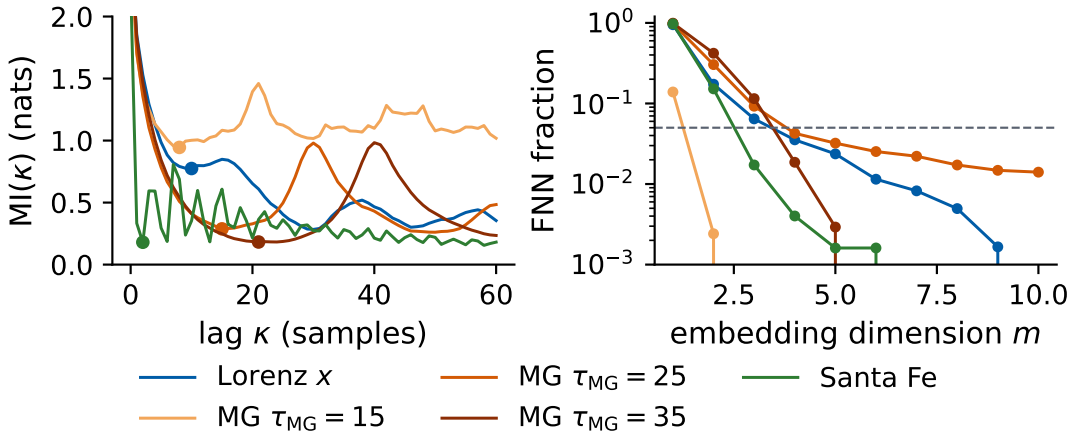


Figure 43: Delay and dimension of the five records. Left, the mutual information MI(κ) in nats against the delay κ in samples, with dots at the first local minimum. Right, the fraction of false nearest neighbours by the ratio test (8.4) against the embedding dimension m, on a logarithmic scale, with the threshold of 5% dashed.

9 Determinism in a measured record

The delay and the dimension give the laser record a reconstructed state, and with it the second question about the record can be answered with numbers. A law with a few variables and a linear process driven by many unobserved influences can both produce an irregular record, and the autocorrelation function cannot tell them apart. The phase-randomised surrogate of Figure 22 is coloured noise with the spectrum of the Lorenz record. We compute every statistic below for the deterministic records and for such noise side by side. Two linear random processes with a known answer serve as controls, the AR(1) process of Section 5, $s_k = \phi\, s_{k-1} + \varepsilon_k$ with $\phi = 0.8$, and the noisy oscillator $s_k = 1.6\, s_{k-1} - 0.9\, s_{k-2} + \varepsilon_k$. The latter is an autoregressive process of order two, AR(2), and both have independent Gaussian kicks ε_k of unit variance. The two roots of $z^2 - 1.6z + 0.9$ have the modulus $\sqrt{0.9} \approx 0.949$, which makes the AR(2) process stationary, and they are complex, which makes its motion oscillate. Since these statistics describe the records and choose no setting of a forecasting model, they use all 4000 samples. Only the false nearest neighbours, like the choice of the delay and the dimension, use the first 2500 samples.

9.1 False neighbours of chaos and noise

A deterministic record unfolds once the delay vector has enough coordinates, and the fraction of false nearest neighbours falls at that dimension. The fraction of a noise record could be expected to stay high, since a random new coordinate separates any two neighbours. Figure 44 tests this expectation on the first 2500 samples of the Lorenz and laser records and of a phase-randomised surrogate of each, embedded with the delay of its data. The AR(1) process with $\kappa = 1$ serves as the third noise record.

The expectation holds only at small dimensions (Table 4). At $m = 2$ the two deterministic records have about a sixth of their neighbours false, and the three noise records more than half. By $m = 4$ the ratio test falls under 5% for all five series of the figure. Since a noise record does not unfold, its fall has another cause. Each added coordinate adds a random amount to the distance r_k between neighbours, so the typical distance grows with m and the ratio Δ_k / r_k of equation (8.4) shrinks with it. The absolute test was added for this reason, and it keeps the surrogate of the Lorenz record above 10% at every $m \geq 3$. For the surrogate of the laser and for AR(1) it falls to 0.008 and 0.032 at

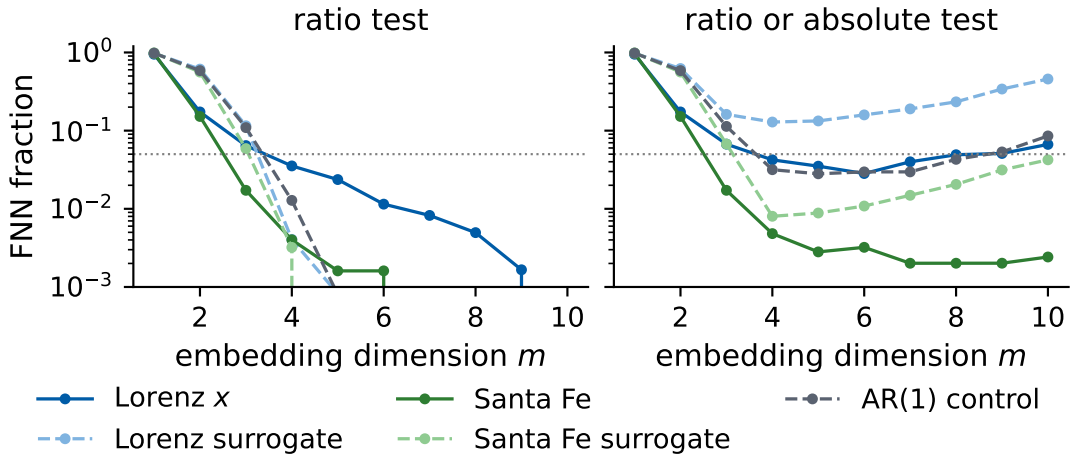


Figure 44: Fraction of false nearest neighbours against the embedding dimension m, on a logarithmic scale, for the Lorenz and laser records (solid), their phase-randomised surrogates and the AR(1) process (dashed). Left, the ratio test (8.4) alone. Right, a neighbour counted as false when it fails the ratio test or the absolute test, $\sqrt{r_k^2 + \Delta_k^2} > 2\sigma_s$ with σ_s the standard deviation of the record. The dotted line marks 5%, and fractions equal to zero lie below the axis.

Table 4: False nearest neighbours of chaos and noise, from the first 2500 samples. The ratio test (8.4) at $m = 2$ and $m = 4$, the combined test at $m = 4$, and the smallest combined fraction for $m = 3, \ldots, 10$.

record	ratio, $m = 2$	ratio, $m = 4$	combined, $m = 4$	combined, $m \geq 3$, smallest
Lorenz x	0.17	0.035	0.042	0.028
Lorenz surrogate	0.61	0.004	0.129	0.129
Santa Fe laser	0.15	0.004	0.005	0.002
laser surrogate	0.56	0.003	0.008	0.008
AR(1)	0.59	0.013	0.032	0.028

$m = 4$, as low as the 0.042 of the Lorenz record. With a few thousand samples, the fraction of false neighbours gives the number of delays that unfold a record known to be deterministic. Determinism itself has to be decided by a statistic that compares the record with noise directly.

9.2 Forecast by the nearest neighbour

In a deterministic system, states that are close now remain close for a while, and the method of analogues [28] is the simplest forecast that uses this property. For the delay vector $\mathbf{u}_k$ of equation (7.2) we find its nearest neighbour $\mathbf{u}_j$ among the delay vectors with $|k - j| > 50$. The value that followed the neighbour is the forecast,

$$\tilde{s}_{k+\tau} = s_{j+\tau}, \qquad \mathrm{NE}(\tau) := \frac{\left\langle (s_{k+\tau} - \tilde{s}_{k+\tau})^2 \right\rangle^{1/2}}{\sigma_s}, \tag{9.1}$$

where the average runs over all delay vectors of the record, $\tau \geq 1$ is the horizon and σ_s is the standard deviation of the targets. The normalised error $\mathrm{NE}(\tau)$ has two reference values. The forecast by the mean of the record has $\mathrm{NE} = 1$, and the forecast by a value unrelated to the target has $\mathrm{NE} = \sqrt{2}$. The second value follows since two independent values a and b with the same mean and the same variance σ_s^2 have $\langle (a - b)^2 \rangle = 2\sigma_s^2$. For a deterministic record the neighbour lies on the same part of the attractor as $\mathbf{u}_k$. Its future differs from the true future by an error that starts at the distance of the neighbours and grows with the Lyapunov exponent. $\mathrm{NE}(\tau)$ therefore starts small and grows with τ. For a linear random process the neighbour carries its own random kicks. For AR(1) with a neighbour that coincides with $\mathbf{u}_k$, $s_j = s_k$, the two next values are $\phi s_k + \varepsilon_{k+1}$ and $\phi s_k + \varepsilon_{j+1}$. Their

difference $\varepsilon_{k+1} - \varepsilon_{j+1}$ has the variance 2, the record has the variance $1/(1-\phi^2)$, and

$$\mathrm{NE}(1) = \sqrt{2\,(1-\phi^2)} = 0.85 \quad \text{for } \phi = 0.8. \tag{9.2}$$

The analogue forecast of AR(1) adds the kick of the neighbour to the kick of the target. It is therefore worse than the best linear forecast ϕs_k, whose normalised error is $\sqrt{1-\phi^2}$.

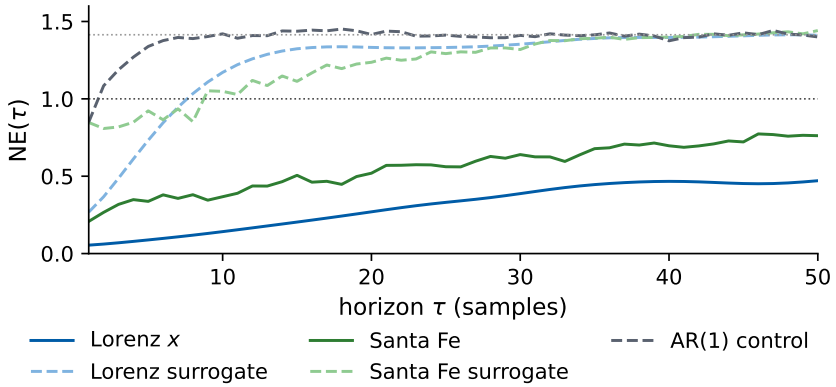


Figure 45: Normalised error NE(τ) of the nearest-neighbour forecast (9.1) against the horizon τ in samples, for the Lorenz and laser records (solid), their phase-randomised surrogates and AR(1) (dashed). The dotted lines mark the forecast by the mean of the record, NE = 1, and by an unrelated value, NE = $\sqrt{2}$.

In Figure 45, one step ahead the Lorenz record has NE(1) = 0.054 and its surrogate 0.27, the laser 0.208 and its surrogate 0.85, and AR(1) 0.850, the value of equation (9.2). The error of the deterministic records grows smoothly with the horizon, to 0.47 and 0.76 at $\tau = 50$. The error of the noise records passes the level of the mean within ten steps, reaching 1.17, 1.05 and 1.42 at $\tau = 10$, and approaches $\sqrt{2}$. Because the surrogate of the Lorenz record is a smooth signal, its next value lies close to its present one and its one-step error is moderate. This error then grows much faster than that of the data. Judging the large margin between the curves of a record and its surrogate requires a scale.

9.3 Surrogate data

The scale comes from a hypothesis test whose null hypothesis H_0, the hypothesis to be tested, states that the record is a stationary linear Gaussian process,

$$s_k = \mu + \sum_{l \geq 0} \psi_l \, \varepsilon_{k-l}, \tag{9.3}$$

with a mean μ, real coefficients ψ_l and independent Gaussian kicks ε_k. All statistical properties of such a process follow from its mean and its autocovariance, so a record that obeys H_0 is described completely by its spectrum. A test compares the record with many other records of its process, and phase randomisation provides them from the record itself [25]. By equation (5.10) the circular autocovariance depends on the Fourier amplitudes $|X_q|$ alone, so the surrogates keep the circular autocovariance and the estimate $\tilde{\rho}(n)$ up to the wrap-around pairs. Under H_0 the Fourier phases of a long record are independent and uniformly distributed, so a surrogate with new random phases is a new realisation of the same process. Each surrogate then gives one sample of the statistic under H_0.

The phase-randomised surrogate has a nearly Gaussian distribution of values, since every value is a sum of many oscillations with random phases. The values of the Lorenz record have a two-humped distribution, and a statistic could reject H_0 because of the distribution alone. A static nonlinear measuring device, a detector whose reading is a monotone function $\tilde{h}$ of the intensity, turns a Gaussian linear process into a non-Gaussian one without adding any dynamics. The iterative amplitude-adjusted Fourier transform (IAAFT) realises this wider hypothesis, a linear Gaussian

process seen through a fixed monotone function $\tilde{h}$ [26]. It starts from a random permutation of the record and repeats two operations. The first replaces the Fourier amplitudes of the current series by those of the record and keeps the current phases. The second sorts the values of the record and gives the i-th smallest value of the series the i-th smallest value of the record. The first operation fixes the spectrum and spoils the distribution, while the second fixes the distribution and spoils the spectrum a little. After a few hundred repetitions the series has the values of the record in a new order and approximately its spectrum. We apply it to the Lorenz, Mackey–Glass ($\tau_{\mathrm{MG}} = 25$) and laser records and to the two autoregressive controls. The Fourier amplitudes of 19 IAAFT surrogates of each deviate from those of the record by at most 2.7% of the largest amplitude.

9.4 Rank test

We compute a test statistic Q for the record, Q_0, and for M surrogates, $Q_1, \ldots, Q_M$, where the number M is unrelated to the matrix M_t. If H_0 holds, the record and the surrogates are realisations of the same process. The $M + 1$ values are then exchangeable, i.e. their joint distribution does not change when they are relabelled, and every ordering of the $M + 1$ values is equally likely. When ties have zero probability, the rank of Q_0, its position in the sorted list, is uniformly distributed over $1, \ldots, M + 1$. A statistic such as the forecast error NE(1) is expected to be small for a deterministic record, and the test rejects H_0 when Q_0 is the smallest of all values. The probability of a false rejection is

$$P(\text{rank of } Q_0 = 1 \mid H_0) = \frac{1}{M+1}, \tag{9.4}$$

which is the significance level α of the test. With $M = 19$ surrogates $\alpha = 1/20 = 0.05$, and $\alpha = 0.05$ is also the finest level that 19 surrogates can resolve. A two-sided test, which rejects when Q_0 is the smallest or the largest value, has $\alpha = 2/(M + 1)$ and needs $M = 39$ for the same level. The level is independent of the distribution of Q, which need not be known or Gaussian. It assumes only that the surrogates realise H_0, so the approximate spectrum of the IAAFT surrogates makes the level approximate as well.

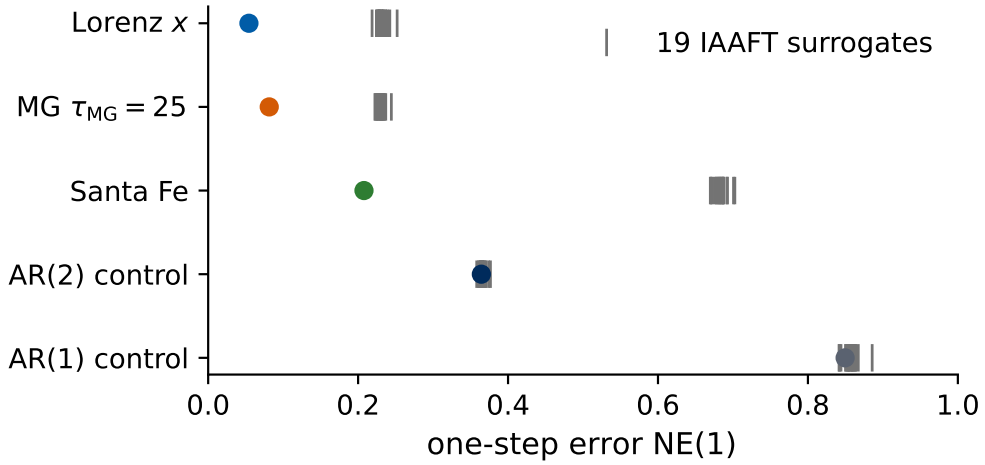


Figure 46: Rank test with the one-step error NE(1) of the nearest-neighbour forecast (9.1) as the statistic. For each record the dot is the error of the record and the vertical ticks are the errors of its 19 IAAFT surrogates. The null hypothesis is rejected at the level 0.05 when the dot lies to the left of all ticks.

The three deterministic records have errors far below those of all their surrogates (Table 5 and Figure 46), and H_0 is rejected for each of them. The two controls, for which H_0 holds, fall among their surrogates, and H_0 is kept. A different statistic, the skewness of the increments, $\langle (s_{k+1} - s_k)^3 \rangle / \langle (s_{k+1} - s_k)^2 \rangle^{3/2}$, measures whether a record rises and falls in the same way. A linear Gaussian process has zero skewness, since it looks the same when played backwards. With 39 phase-randomised surrogates and a two-sided test this statistic rejects H_0 for the Mackey–Glass record and keeps it for the Lorenz and laser records. The laser rises slowly and collapses fast, and its

Table 5: Rank test of the linear Gaussian null hypothesis with the statistic NE(1) and 19 IAAFT surrogates per record. Rank 1 is the smallest error of the 20 values and rejects the hypothesis at the level 0.05.

record	NE(1)	surrogates, smallest	surrogates, largest	rank of 20
Lorenz x	0.054	0.219	0.252	1
MG, $\tau_{\mathrm{MG}} = 25$	0.081	0.222	0.244	1
Santa Fe laser	0.208	0.670	0.703	1
AR(2)	0.364	0.358	0.377	8
AR(1)	0.850	0.842	0.886	4

asymmetry appears on the time scale of a pulse. The score z is the distance of the record value from the surrogate mean in units of the surrogate standard deviation. Increments over 7 and 8 samples, one pulsation period, give z-scores of -8.5 and -17.1 against the surrogates, while increments over one sample give -0.5. These lags were chosen after looking at the record, so they illustrate where the asymmetry lies and are no valid test at the level 0.05. For the Lorenz record the reason is the symmetry $(x, y, z) \mapsto (-x, -y, z)$ of equation (4.6), which makes x and $-x$ equally likely and every odd moment of its increments zero. A kept hypothesis means that the chosen statistic sees no difference between the record and its surrogates, and the linearity of the record remains open.

9.5 Leading exponent from the record

A record that is deterministic has Lyapunov exponents, and the leading one can be estimated without equations. The Rosenstein algorithm takes every delay vector $\mathbf{u}_i$, finds its nearest neighbour $\mathbf{u}_{j(i)}$ with $|i - j(i)| > 50$, and follows how far apart the two are after n further steps [21]. Each pair of neighbours plays the role of the two nearby starts of Figure 23, and the average of the logarithm of the distance over the pairs is

$$\langle \ln \delta(n) \rangle \coloneqq \frac{1}{N_{\mathrm{p}}} \sum_i \ln \left\| \mathbf{u}_{i+n} - \mathbf{u}_{j(i)+n} \right\|, \tag{9.5}$$

with N_{p} the number of pairs.

To relate this curve to λ_1, we linearise the flow around one pair. Let $\boldsymbol{\delta}_0$ be its initial separation, with components $c_1, c_2, \ldots$ along the directions that the flow stretches by the factors $\sigma_1(t) \geq \sigma_2(t) \geq \ldots$ of equation (6.5), evaluated at $t = n\Delta t$. While the separation is small, its length after n samples is

$$\delta(n) = \Big(\sum_l c_l^2 \, \sigma_l(n\Delta t)^2 \Big)^{1/2} = |c_1| \, \sigma_1(n\Delta t) \Big(1 + \sum_{l \geq 2} \frac{c_l^2}{c_1^2} \, \frac{\sigma_l(n\Delta t)^2}{\sigma_1(n\Delta t)^2} \Big)^{1/2}, \tag{9.6}$$

where the components c_l refer to the stretching directions of $M_{n\Delta t}$ and change slowly with n. Its logarithm is

$$\ln \delta(n) = \ln |c_1| + \ln \sigma_1(n\Delta t) + \tfrac{1}{2} \ln \Big(1 + \sum_{l \geq 2} \frac{c_l^2}{c_1^2} \, \frac{\sigma_l(n\Delta t)^2}{\sigma_1(n\Delta t)^2} \Big), \tag{9.7}$$

and averaged over many pairs each term has its own time dependence. The last term decays to zero as the separation turns into the most expanding direction, at the rate $2(\lambda_1 - \lambda_2)$. For a flow with one positive exponent $\lambda_2 = 0$, so this term falls by the factor e within half a Lyapunov time, and while it decays it lowers the slope. The middle term grows on average at a rate that approaches $\lambda_1 \Delta t$ per sample from above. The stretching over a time $t + t'$ is at most the product of the stretchings over t and over t', so the average of $\ln \sigma_1(t)$ grows at most additively in time. The finite-time exponent $\langle \ln \sigma_1(t) \rangle / t$ is therefore never smaller than its limit λ_1. Table 1 shows it falling from 1.935 at $t = 1$ to 0.998 at $t = 10$ for the Lorenz model. The linearisation holds while $\delta(n)$ stays small compared with the size ℓ_A of the attractor. The distance saturates at about ℓ_A, which lowers the slope again. The curve (9.5) therefore has the slope $\lambda_1 \Delta t$ only over a range of horizons after the finite-time excess has

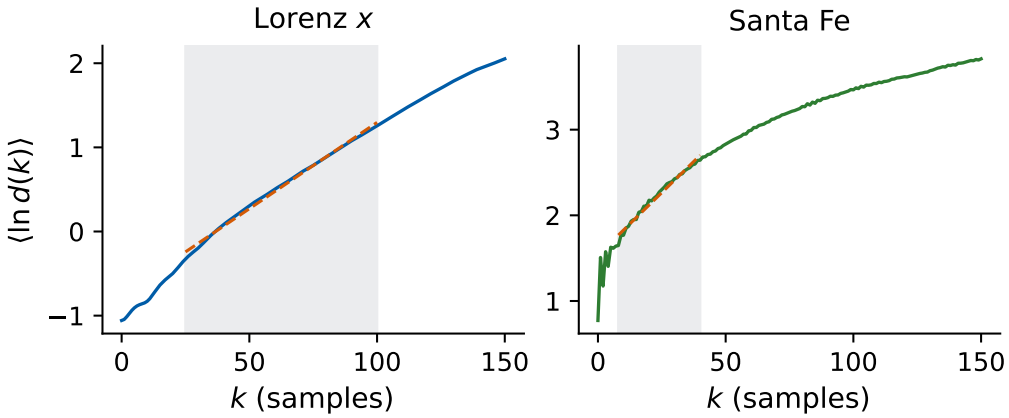


Figure 47: The largest Lyapunov exponent from a record. The average logarithmic distance $\langle \ln \delta(n) \rangle$ of equation (9.5) of pairs of neighbouring delay vectors against the number of steps n. Left, the Lorenz record with $m = 4$, $\kappa = 10$, and right, the laser with $m = 3$, $\kappa = 2$. On the axes these quantities are written $\langle \ln d(k) \rangle$ and k. The shaded range is used for the straight-line fit (dashed).

died out and before the saturation. This range is long only when $\ln(\ell_A/\delta_0)$ is large, that is, when neighbours start very close compared with the size of the attractor.

For the Lorenz record, with $m = 4$ and $\kappa = 10$, the fit over 25 to 100 samples in Figure 47 gives 1.03 per time unit, against 0.907 from the equations. The curve has no range of constant slope. Its local slope falls from 1.55 per time unit at $n = 25$ to 0.91 at $n = 100$, from above towards λ_1 as the finite-time excess dies out. For the laser, with $m = 3$ and $\kappa = 2$, the fit over 8 to 40 samples gives $\lambda_1 = 0.029$ per sample. The laser curve has no such range either. Its local slope, averaged over one pulsation period, falls from 0.072 per sample at $n = 4$ to 0.030 at $n = 20$ and 0.020 at $n = 40$ (left panel of Figure 48). The mean distance of the neighbours starts at 2.2 intensity units, about two of the 256 levels of the digitiser. It grows over the first 150 samples by the factor 21 before it saturates, which leaves about 3.1 factors of e of growth. The finite-time excess and the saturation overlap, and the fitted slope depends on the chosen range.

The Sano–Sawada method fits, around every delay vector $\mathbf{u}_i$, a linear map $\tilde{B}_i$ to the evolution of its K nearest neighbours [22],

$$\tilde{B}_i = \arg\min_{B} \sum_{j \in \mathcal{N}_i} \left\| (\mathbf{u}_{j+1} - \mathbf{u}_{i+1}) - B\,(\mathbf{u}_j - \mathbf{u}_i) \right\|^2, \tag{9.8}$$

with $\mathcal{N}_i$ the set of the K neighbours, and applies the orthonormalisation of equation (6.8) to the product of the matrices $\tilde{B}_i$ along the record. It measures the growth over one step at the scale of a whole neighbourhood, while Rosenstein's curve follows single pairs over many steps. The least-squares fit determines $\tilde{B}_i$ only along the directions in which the displacements $\mathbf{u}_j - \mathbf{u}_i$ are spread. Their root-mean-square (rms) extents $w_1 \geq w_2 \geq \ldots$ along the principal directions, the singular values of the $K \times m$ matrix of displacements divided by $\sqrt{K}$, show how much of each direction the fit sees. A thin attractor, of dimension close to two in a space of $m = 4$ delays, fills its neighbourhoods along two directions and leaves the others almost empty. For the laser with $K = 63$ the median extents are 0.189, 0.164, 0.047 and 0.017 standard deviations of the record (right panel of Figure 48). One level of the digitiser is 0.021 standard deviations. In the two thin directions the displacements extend over about two levels and less than one level of the digitiser. The fitted entries of $\tilde{B}_i$ there describe mostly the rounding, and the exponents of these directions describe the fit. Over $K = 16$ to 256 the third exponent ranges from -0.42 to -0.29 and the fourth from -1.56 to -1.37 per sample. For the Lorenz record, whose contracting exponent is -14.573 per time unit, the same method returns -2.2 for the third exponent. A direction contracted by the factor $e^{-14.6}$ per time unit holds almost no spread of neighbours, an extent of only 0.008 standard deviations. The leading exponent lies in the well-filled directions and stays between 0.064 and 0.076 per sample over the same range of K. This value is close to the early local slope of Rosenstein's curve.

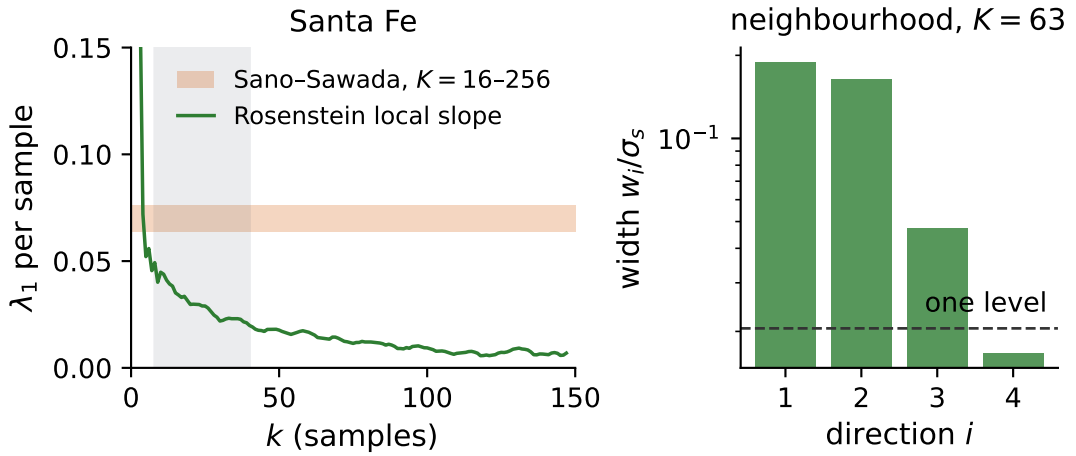


Figure 48: Two estimators of the leading exponent of the laser. Left, the local slope of Rosenstein's curve (9.5), averaged over 7 samples, against the number of steps k. The fit range of Figure 47 is shaded, and the orange band marks the range of the Sano–Sawada estimate for $K = 16$ to 256 neighbours. Right, the median rms extents w_i of neighbourhoods of $K = 63$ delay vectors with $m = 4$, $\kappa = 2$ along their principal directions i, in units of the standard deviation σ_s of the record. The scale is logarithmic, and the dashed line marks one level of the digitiser. The steep start of the local slope at $k < 4$ lies above the plotted range.

The two estimates of the laser exponent are 0.029 per sample from Rosenstein's curve and 0.065 from the local maps with $m = 4$. On a short, rounded record of a thin attractor they measure different things. The curve of single pairs has no range of constant slope, and its fitted slope falls with the fit range. The local maps see the growth over one step in the filled directions and nothing in the empty ones. Neither gives the exponent of the laser with a known error, so we state every number with its method. For the same reason the Kaplan–Yorke dimension of the laser from the local maps, 2.05 with $m = 3$ and 2.16 with $m = 4$, inherits the uncertainty of the contracting exponents. Setting the small second exponent at $m = 4$ to zero gives 2.17.

9.6 Correlation dimension

A dimension computed from distances alone, without exponents, gives a second view of the attractor and a second test against noise. A deterministic attractor has a dimension that does not change once the delay space is large enough to hold it, while noise fills every dimension it is given. The Grassberger–Procaccia algorithm measures this with the correlation sum [29], the fraction of pairs of delay vectors closer than a radius r (unrelated to the Pearson coefficient),

$$C_m(r) \coloneqq \frac{1}{N_c} \sum_{\substack{i<j \\ j-i>50}} \mathbf{1}\big(\|\mathbf{u}_i^{(m)} - \mathbf{u}_j^{(m)}\| < r\big), \tag{9.9}$$

where $\mathbf{1}(\cdot)$ is 1 when its condition holds and 0 otherwise and N_c is the number of pairs in the sum. Pairs closer in time than 50 samples are left out, because they are close only since the trajectory is continuous [30]. The sum $C_m(r)$ estimates the probability that two points of the attractor, drawn with the frequencies of the motion, lie closer than r. On a set of dimension D_2 this probability scales as a power at small r,

$$C_m(r) \propto r^{D_2}, \qquad D_2 = \lim_{r \to 0} \frac{\ln C_m(r)}{\ln r}, \tag{9.10}$$

and D_2 is called the correlation dimension. A point drawn on a curve has a neighbour within r with a probability proportional to r, and a point on a surface with a probability proportional to r^2. For these sets D_2 is therefore 1 and 2. The correlation dimension weights every box of side r by the square of its visiting frequency, the probability that both points of a pair fall into it. It obeys $D_2 \le D_1 \le D_B$, and for an attractor whose boxes are visited nearly equally often, as for the baker's map, the three are equal. We fit the slope of $\ln C_m$ against $\ln r$ over the radii where

$10^{-4} < C_m(r) < 10^{-2}$, small enough for the power law and large enough for a few hundred pairs. Every record is first standardised to zero mean and unit variance. We also include Lorenz x with added independent Gaussian measurement noise of 5% of its standard deviation.

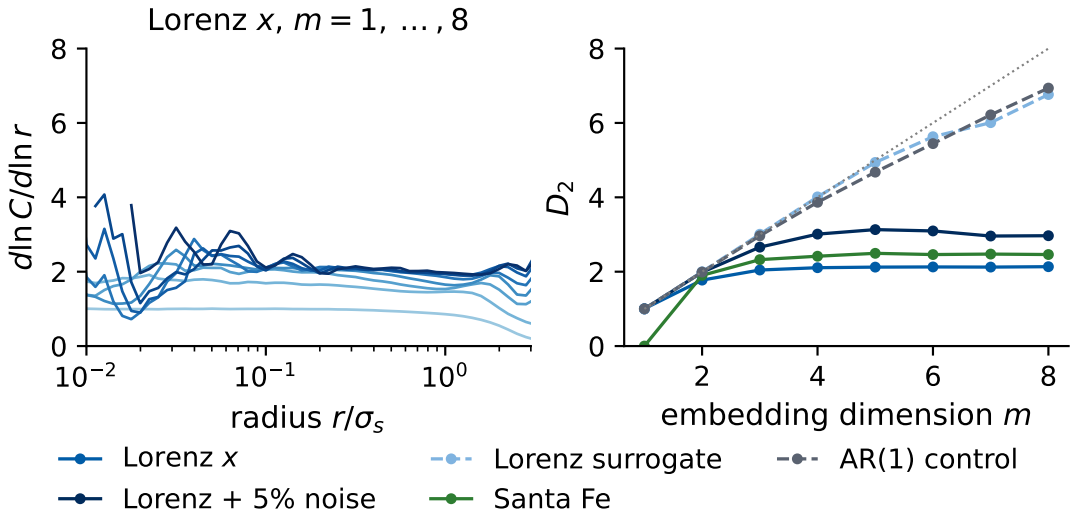


Figure 49: Correlation dimension. Left, the local slope $d \ln C_m / d \ln r$ of the correlation sum (9.9) of the Lorenz record against the radius r in units of the standard deviation σ_s, for $m = 1, \ldots, 8$ from light to dark. Right, the dimension D_2 fitted where $10^{-4} < C_m < 10^{-2}$ against the embedding dimension m. Solid lines show the Lorenz record, the Lorenz record with 5% measurement noise and the laser record, and dashed lines the surrogate of the Lorenz record and AR(1). The dotted line is $D_2 = m$.

Table 6: Correlation dimension D_2 at the embedding dimensions $m = 4$ and $m = 8$.

record	D_2, $m = 4$	D_2, $m = 8$
Lorenz x	2.11	2.13
Lorenz x with 5% noise	3.01	2.97
Santa Fe laser	2.42	2.46
surrogate of Lorenz x	4.01	6.76
AR(1)	3.87	6.94

In the left panel of Figure 49 the local slopes of the Lorenz record rise with m at first and then collect on a common plateau. The fitted dimension saturates accordingly (Table 6). From $m = 4$ on, the Lorenz record has $D_2 \approx 2.13$, close to $D_{\mathrm{KY}} = 2.06$ from the exponents, and the laser saturates at $D_2 \approx 2.46$. For the surrogate and AR(1), D_2 grows with m nearly as m itself, to 6.76 and 6.94 at $m = 8$. The saturation separates the deterministic records from the noise, and its value estimates the dimension of the attractor, 2.13 for the Lorenz record and 2.46 for the laser. The Lorenz record with measurement noise also reaches a plateau, at a higher value.

9.7 Limits of the diagnostics

Measurement noise, the null hypothesis and the length of the record limit every diagnostic of this section. Noise of size η moves every delay vector by about η in all m directions. At radii below a few η the record therefore looks like noise and fills all dimensions. Measurement noise of 5% raises the plateau of the fitted dimension of the Lorenz record from 2.13 to 2.97 at $m = 8$. This happens because the fit range lies at radii where the noise already matters. This noise also sets the distance of the nearest neighbours, which shortens the scaling range of Rosenstein's curve. It fills the thin directions of the local maps of the Sano–Sawada method.

The surrogate test rejects one specific null hypothesis, a stationary linear Gaussian process seen

through a static monotone measurement. A nonlinear random process, a record whose statistics drift in time, or a measuring device with memory can produce a rejection as well. A rejection shows structure beyond linear correlations and leaves determinism as one explanation among others.

A correlation sum over N points resolves power laws only over a limited range of radii. Estimates of a dimension from N points cannot exceed about $2\log_{10} N$ [31], here 7.2. The values of D_2 near 7 for the noise records at $m = 8$ already approach this bound, while the plateaus of the deterministic records near two lie far below it. Within these limits all diagnostics agree for the Lorenz and laser records. Their nearest-neighbour forecasts are far better than those of noise with the same spectrum, and the rank test rejects the linear Gaussian hypothesis. Their correlation dimension saturates at a small value, and the leading exponent is positive by both estimators.

9.8 Characteristics of the five records

Table 7 collects the quantities of these notes for all five records, which together fix what a forecast of each record can achieve. The delay κ comes from the mutual information and the dimension m from the false nearest neighbours, both computed on the training block. The window length L follows from equation (7.10). The Kaplan–Yorke dimension and the leading exponent come from the equations for the model records and from the record for the laser. The leading exponent is given per sample, $\lambda_1 \Delta t$, and the Lyapunov time $T_\lambda = 1/\lambda_1$ in samples, the unit in which forecast horizons are counted.

Table 7: Characteristics of the five records. κ, L and T_λ are in samples and λ_1 is per sample. For the laser λ_1 is the Rosenstein estimate and D_{KY} the Sano–Sawada estimate with $m = 3$ and, after the slash, with $m = 4$. The periodic record has $\lambda_1 = 0$ within its error, so its Lyapunov time is infinite.

record	κ	m	$L = (m-1)\kappa + 1$	D_{KY}	λ_1 per sample	T_λ
Lorenz x	10	4	31	2.06	0.0181	55
Mackey–Glass, $\tau_{\mathrm{MG}} = 15$	8	2	9	1.00	≈ 0	∞
Mackey–Glass, $\tau_{\mathrm{MG}} = 25$	15	4	46	3.16	0.0099	101
Mackey–Glass, $\tau_{\mathrm{MG}} = 35$	21	4	64	3.41	0.0029	340
Santa Fe laser	2	3	5	2.05 / 2.16	0.029	34

Every record satisfies $\lceil D_{\mathrm{KY}} \rceil \le m$, so the window holds at least as many numbers as are needed to locate a state on the attractor. For the chaotic records m is smaller than the sufficient bound $2D_{\mathrm{KY}}$, consistent with the statement that the bound is often larger than needed. The windows are short, from 5 samples for the laser to 64 samples for the Mackey–Glass record at $\tau_{\mathrm{MG}} = 35$, whose window must reach back over the delay of the equation. The Lyapunov times set very different limits for horizons of 1 to 50 samples. For the Lorenz record the longest horizon is about one Lyapunov time, over which an initial error grows on average by the factor $e^{50/T_\lambda} = 2.48$. For the chaotic Mackey–Glass records all horizons lie well within one Lyapunov time, with growth factors 1.64 and 1.16 over 50 samples. The periodic record has no finite Lyapunov time. The laser has the shortest Lyapunov time, 34 samples from the Rosenstein estimate and 15 samples from the Sano–Sawada estimate. Over 50 samples a small initial forecast error grows by a factor between 4.3 and 26. The two difficulties named with the Kaplan–Yorke dimension appear side by side across the records. The chaotic Mackey–Glass records have the largest dimensions and need the longest windows, while their Lyapunov times are long. The laser and the Lorenz record have dimensions between two and three and short windows, while their Lyapunov times are the shortest.

A window of L past values contains the delay vector, which we assume determines the present state. The embedding theorem does not guarantee this, since the values of m found here lie below the sufficient bound $2D_{\mathrm{KY}}$. The assumption rests on the small fraction of false nearest neighbours at these m. With it the forecast map G_τ of equation (7.11) exists, and chaos limits its useful range to horizons of the order of the Lyapunov time. The map itself is unknown and nonlinear, and it has to

be learned from the pairs of windows and future values that the training block provides.

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